

Mass Exit in the Market for Governance

A Behavioral Foundation for Theories of State Development*

Shusuke Ioku[†]

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ABSTRACT. State development theories posit that political fragmentation lets subjects exit to other authorities, spurring jurisdictional competition that drives institutional development, yet behavioral evidence remains scarce. I develop a theory of exit through alternative authority and test its implications in Tokugawa Japan. Across nearly 300 domains, villages sought concessions from home rulers by fleeing en masse or appealing through neighboring lords or to the central authority. I exploit domain conversions to central rule, which brought central authority closer to neighboring villages, as a natural experiment. Difference-in-differences analysis estimates a 4.3 p.p. increase after conversions in the probability of exit among nearby villages under other lords. The analysis further shows that proximity to neighboring capitals predicted more exit and that domains with easier access to neighboring rulers tended to set lower tax rates, especially with economically attractive destinations. These findings strengthen the behavioral foundations of theories linking political fragmentation to state development.

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[†]Ph.D. Candidate, Department of Political Science, University of Rochester

1. INTRODUCTION

Many influential theories of state development rest on a common premise: political fragmentation gives subjects access to alternative authorities, and that access disciplines extraction and shapes institutions. This premise carries the central causal weight of leading explanations for the Great Divergence (Cox 2017; Fernández-Villaverde et al. 2023; Jones 2003; Mokyr 2017); for accounts linking Europe’s polycentric structure to constrained taxation (Abramson 2017; Dincecco and Wang 2018; Sng and Moriguchi 2014; Stasavage 2016); for fiscal-federalism arguments that competing jurisdictions tame Leviathan (Qian and Weingast 1997; Rodden 2003; Tiebout 1956; Weingast 1995); and for recent theories arguing that the modern territorial state system cartelized the *market for governance* by restricting subject mobility (Acharya and Lee 2018; 2023). Each case shares the same basic mechanism: fragmentation offers subjects credible exit options, and the threat or use of exit forces rulers to compete or cooperate. Yet the behavioral foundation of this chain has yet to be directly demonstrated: whether access to authority outside the home jurisdiction shaped how subjects resisted. I test this link.

Mass exit has served as a form of resistance across many historical contexts. In medieval Europe, migration to towns offered serfs a way to resist their lords (Adas 1981, 217). In precolonial Java and Burma, cultivators escaped excessive demands by transferring their allegiance to rival patrons (Adas 1981, 229–231). Gandhi described an Indian village whose inhabitants began leaving in protest against their prince, prompting him to withdraw his command (Gandhi and Parel 1997, 95). In Tokugawa Japan, villages fled collectively into neighboring domains and petitioned neighboring rulers for permission to remain (Moore 1993, 255).

A common setting for such resistance combined horizontal divisions among neighboring local rulers with a vertical hierarchy under a common superior, as in Languedoc’s lordships and towns under the early modern French crown (Gerring et al. 2011, 405) and county jurisdictions within Qing China’s provincial and imperial hierarchy (Sng and Moriguchi 2014, 446–447).^[3] I develop a theory of how communities use these alternative

^[3]Comparable arrangements appeared in territories of the Holy Roman Empire under the emperor (Cantoni, Mohr, and Weigand 2024), princely states under the British Raj (Iyer 2010), and emirates

authorities to challenge extraction. Neighboring protection can sustain villagers’ absence and place their grievance beyond the home ruler’s exclusive control. When the common superior accepts that grievance, its intervention can reinforce the pressure to concede. These routes can make temporary departure effective even when villagers wish to return to their land. Easier access to another authority should lead more communities to choose exit. When neighboring jurisdictions offer sufficiently attractive livelihoods, rulers should lower taxes to retain cultivators. These tax concessions should reduce, but not eliminate, the increase in exit.^[4]

Tokugawa Japan (1615–1868) combined these two dimensions of authority within a common political order. Nearly 300 feudal domains exercised local autonomy without sovereignty, each with its own legal code, tax system, and border controls, under a central authority, the Tokugawa shogunate. The shogunate retained supra-domain arbitration, attainer power, direct peasant-appeal authority, and power to assume direct rule (Moore 1993; Ravina 1999; White 1995). I use *exit through alternative authority* for collective flight into a neighboring domain (*chōsan*), a petition routed through a neighboring lord, or an appeal over the home lord’s head to the shogunate (*osso*). These routes created different forms of leverage, but each moved villagers or their claim beyond the home lord’s exclusive control.^[5]

This setting helps overcome two obstacles to testing the argument. First, open rebellion leaves abundant documentary traces, whereas exit is often subtle and covert. Limited records have therefore led most studies to infer exit indirectly rather than observe it at the community level (Bailey 2014, 76). Newly digitized records of peasant resistance compiled by Aoki (1975) document both direct confrontation and exit through

of northern Nigeria under colonial supervision (Gerring et al. 2011, 399). These settings varied in local autonomy and subjects’ access to other authorities.

^[4]A model of tax competition formalizes the economic outside-option mechanism in [Supporting Information §A](#).

^[5]Tokugawa mass exit operated as active resistance rather than permanent relocation: peasants presented demands to home-domain officials while seeking permission from neighboring rulers to cross the border, with some documented cases recording return once concessions were granted. Moore (1993, p. 255), drawing on Borton’s catalog, recorded 106 such mass desertions across the Tokugawa period as evidence of “the solidarity of the Japanese village”; my catalog substantially extends this count and covers the full archipelago.

alternative authority at the village level across the entire Tokugawa period. Second, comparisons between entire regions, such as Europe and China, conflate differences along many dimensions and provide little within-case variation for examining subjects' choices. Within Tokugawa's common political order, villages varied widely in access to rival rulers, while intense inter-domain competition for agricultural labor made mass migration a credible strategy. The post-1615 peace also sharply weakened the recurrent war-finance pressures that Rosenthal and Wong (2011) identify as overwhelming the exit-discipline mechanism in early modern Europe, reducing a major competing pressure on extraction (Moore 1993).

The empirical analysis begins with two behavioral tests and then examines a downstream fiscal trace. First, I employ a difference-in-differences analysis exploiting domain conversions to direct shogunal rule (*tenryo*), which brought shogunal authority closer to surrounding villages. After direct rule began, nearby villages governed by other lords recorded 4.3 percentage points more exit but no detectable change in direct confrontation, and event studies show no detectable pre-conversion differences. A grid-year panel tracking changes as nearby capitals were established, abolished, or relocated finds the same pattern: easier access to another active capital predicts more exit but no more direct confrontation.

Second, among villages that resisted, greater geographic exposure was associated with choosing exit over direct confrontation. I measure each village's exposure as the ratio of its distance to the home-domain capital to its distance to the nearest active non-home capital. This ratio is a spatial proxy for access to alternative authority, not a direct measure of competition, bargaining, or another ruler's willingness to receive migrants. I use *governance conflict* for the condition in which villages under one ruler can credibly move themselves or their claims under another. Its two components are access to alternative authority and an outside option worth the costs of invoking it. Geographic exposure proxies for access; in the fiscal analysis, neighboring domains' capacity to absorb labor captures the attractiveness of migration as an outside option. Across all recorded resistance, this exit-*share* result holds robustly under home-neighbor dyad fixed effects. Decomposing the ratio into its two absolute distances shows that foreign-capital

proximity is the larger and more robust component. Home-capital remoteness is positive but less precise under spatial inference. Distance to the nearest domain border does not significantly predict exit choice, supporting the importance of access to an authority able to protect villagers or mediate their demands.

As a downstream fiscal trace, a late-period domain cross-section associates greater geographic exposure with lower reported tax rates. The estimates are more negative where neighboring domains could absorb migrants, though the difference between groups is imprecise.

The paper makes three core contributions. First, it provides direct, micro-level evidence for a central behavioral premise running through four traditions in political economy, each resting on the same link between access to outside authority and exit: the Great Divergence argument that ruler competition for mobile subjects checked extraction and fueled European growth (Jones 2003; Mokyr 2017); state-formation accounts in which exit forced ruler bargains and built representation (Dincecco and Wang 2018; Stasavage 2016); fiscal-federalism claims that competition among jurisdictions tames Leviathan through taxpayer and labor mobility (Rodden 2003; Tiebout 1956; Weingast 1995); and the cartel theory of the territorial state, in which modern borders are erected specifically to suppress subject exit (Acharya and Lee 2018; 2023). The evidence links better access to outside authority with communities' use of exit, supplying behavioral evidence for the first link in this shared causal chain.

Second, the paper brings exit from the background to the center of the study of state development. In Hirschman's (1970) terms, the dominant analytical traditions are built around voice: revolution and rebellion drive regime change (Acemoglu and Robinson 2001; Skocpol et al. 1982), collective mobilization makes states (Hintze 1975; Tilly 1978), and the threat of revolt disciplines extraction (Besley and Persson 2009; Fearon 2011). When exit appears, it shapes incentives (Besley and Persson 2009; Fearon 2011) or long-run sorting (Jones 2003) without being directly observed. Historians of everyday resistance have long documented otherwise: exit often predominated over revolt (Adas 1981; Bailey 2014; Camp 2004; Diouf 2014; Kolchin 1990; Scott 1976; 1985; Sreenivasan 2004). This paper documents exit directly as a strategy of resistance alternative to open rebellion.

Third, the paper explains how collective exit can constrain extraction within a common political hierarchy. Studies of horizontal exit emphasize individual preference-sorting (Peterson 1992; Tiebout 1956), flight beyond state reach (Carneiro 1970; Herbst 2000; Pruett 2024; Scott 2009), or indirect evidence from jurisdictional havens, movement restrictions, and tax-base responses (Guardado and Franco-Vivanco 2025; Kleven et al. 2020; Matranga and Natkhov 2025). Other strands show how exit can spur voice (Hirschman 1993; Pfaff and Kim 2003; Schoppa 2022) or how governments induce emigration (Lueders 2025; Michel, Miller, and Peters 2023). I distinguish the pressure to retain cultivators from the threat of superior intervention. Neighboring shelter sustains absence, while the threat of superior sanction can strengthen villagers’ demands even when they cannot resettle permanently. The Tokugawa evidence links access to outside authority with community resistance; historical cases show how shelter and hierarchical oversight could support that resistance.^[6]

2. EXIT THROUGH ALTERNATIVE AUTHORITY

Theories of state development identify war and exit as two forces that constrain rulers, but direct tests have focused on war. Financing armies required rulers to extract men, money, and supplies from holders of capital, land, and labor. The resulting bargains produced tax bureaus, representative assemblies, and eventually citizenship (Tilly 1985; 1992). Research now specifies when war built states rather than destroyed them (Dincecco and Wang 2018; Spruyt 2017). Exit operates through a different constraint: divided authority gives subjects alternatives, so excessive extraction can cost rulers their tax base or expose them to outside intervention. This mechanism underpins arguments about the Great Divergence, fiscal federalism, and the territorial state system.

Tilly’s account recognizes mobility as a constraint but confines its treatment to elites and regional trajectories. Merchants and financiers in Europe’s capital-intensive zones could “escape with their capital to another business site if the ruler’s demands became unbearable,” forcing rulers in Genoa, Amsterdam, and the Hanseatic cities to

^[6]Moore (1993, p. 230) characterized the late Tokugawa order as marked by “vertical and horizontal fissures... perhaps equally important,” foregrounding the horizontal axis without operationalizing it; this paper supplies the community-level measurement.

offer favorable terms (Tilly 1992, 53). In coercion-intensive zones, rulers and landlords instead fixed peasants in place. Russia’s Assembly Code of 1649 and Prussian serfdom bound cultivators to the land because departure would have undermined extraction (Tilly 1992, 22–23, 141). Tilly thus treats mobility as a difference between regional trajectories and peasant exit as an option rulers foreclosed, without analyzing horizontal exit as a strategy of popular resistance (Tilly 1992, 187). The question here is how communities attached to land can use outside authority to resist extraction despite restrictions on movement.

2.1. Actors, Authority, and Choices

The setting contains local jurisdictions that govern their own subjects within a wider hierarchy. Local rulers control taxation and administration, while a superior authority can arbitrate disputes and sanction them. Subjects can approach another local ruler horizontally or the superior authority vertically. The jurisdictions need not be sovereign or military rivals. Horizontal protection requires a reachable ruler able and willing to receive subjects; hierarchical enforcement additionally requires a superior willing to act on their claim.

The central relationship is between a ruler who depends on cultivators for revenue and village communities that bear extraction collectively. Raising taxes increases receipts from those who comply but can induce others to resist. Villagers seek to protect their livelihoods and secure concessions from the authority that taxes them. For landholding peasants, departure puts a valuable local asset at risk: their cultivation rights are attached to household holdings or membership in a landholding village (Brown 1993, 94–95, 107–108; Nakabayashi et al. 2020, 175–176). These fiscal obligations differ from tenants’ rental obligations to village landlords. Distance, forgone income, and continued access to land make exit costly even when another authority is nearby. Regional differences in tenure change the form of that cost, without eliminating it.

Rulers set taxes knowing that villagers can compare demands at home with alternatives elsewhere. A village can first petition its own ruler for a concession. If its demand remains unresolved, villagers can comply, confront the home ruler directly, or invoke an

outside authority. *Direct confrontation* presses demands against the home ruler through coercive petition, property destruction, or uprising. *Exit through alternative authority* moves villagers or their claim beyond the home ruler's exclusive control. This distinction adapts Hirschman's contrast between withdrawing from an authority and pressing it to change (Hirschman 1970, 30). Studies of peasant resistance likewise distinguish flight and other forms of avoidance from direct confrontation (Adas 1981, 217–219; Scott 1976, 65). Compliance preserves income after taxes without the immediate costs of resistance. Confrontation offers concessions through collective pressure at home; exit offers outside protection or intervention at the cost of reaching and using another authority. Ordinary petitions to the home ruler remain within his jurisdiction and are distinct from exit.

Exit through alternative authority creates leverage through lost revenue, neighboring protection, and superior intervention. Collective departure deprives the home ruler of cultivators and tax payments. A neighboring ruler can shelter villagers, mediate their demands, and publicize misgovernment beyond the home ruler's control. An appeal to the superior invokes its power to investigate or sanction the home ruler. These sources of leverage can operate together: neighboring shelter sustains villagers while they pursue a superior's intervention, and a neighboring ruler can transmit or support their appeal. Access to a neighboring capital can therefore strengthen an appeal to the superior as well as make collective departure more feasible.

The cost of reaching another authority and the value of its support are separate conditions. Geography determines how far subjects must travel and how many authorities they can approach. Cultural and legal distance determines what they give up by leaving; the receiving authority's decisions determine the livelihood, protection, or intervention they can obtain. Crossing a border can reduce the home ruler's immediate coercive reach; approaching another authority can add recognition, mediation, and a path to superior intervention. Access to outside authority is therefore distinct from remoteness from the home ruler, and proximity alone does not guarantee protection.

2.2. How Exit Constrains Extraction

Receiving rulers can benefit from sheltering or absorbing cultivators instead of returning them automatically. Absorption increases their revenue, while surrendering villagers restores a rival's tax base (Miyazaki 1995). Reporting misgovernment can also expose the home ruler to superior sanctions. These incentives can obstruct a ruler cartel that closes subjects' alternatives (Acharya and Lee 2018). They do not eliminate cooperation: shelter costs, obligations to other rulers, and interdomain agreements for the reciprocal return of fugitives can favor repatriation (Miyazaki 1990, 38–39). The home ruler must therefore consider both villagers' responses and the support available elsewhere. Access to a neighboring jurisdiction has little value when its ruler refuses protection.

Superior intervention adds leverage when the center treats a grievance as grounds to investigate or sanction the home ruler. The center can use petitions to learn about crop failures and misconduct that local officials would otherwise conceal, but hearing a complaint does not commit it to accepting villagers' demands (Vlastos 1986, 42–43, 53). It can investigate misconduct while defending fiscal demands or enforcing rules against unauthorized appeals. A neighboring ruler may bring villagers' grievances before the superior while providing shelter or mediation. Villagers may also petition the superior directly, but a neighboring ruler's backing can strengthen their bargaining position against the home ruler. These routes expose local administration to scrutiny that the home ruler cannot control.

Outside involvement changes the pressure on a ruler after an ordinary petition has failed. Neighboring shelter can place villagers beyond his immediate coercive reach while they press their demands. Lost cultivation and revenue give him reason to settle. If a superior is willing to sanction him, temporary shelter can also give villagers time to obtain intervention without establishing a permanent livelihood elsewhere. Without that prospect, leverage rests more heavily on sustaining absence. If outside authorities enforce return or reject the grievance, proximity offers less protection. These responses to a particular dispute are distinct from lowering general tax demands in anticipation of resistance.

A credible threat of continued absence can nevertheless end in return. Temporary shelter sustains villagers while they bargain. When pressure rests on lost cultivation and revenue, continued absence must be sustainable through shelter or a livelihood elsewhere; permanent resettlement additionally requires feasible land or work. Absorbing many cultivators may also require investment in settlement infrastructure. Villagers meanwhile lose income, incur shelter costs, and risk weakening access to household- or village-held land. Sufficient relief makes return attractive by restoring those benefits, while the ruler recovers cultivation and receipts.

2.3. Exit and Confrontation

Exit and confrontation can compete for participation and organizational effort. Departure can weaken simultaneous confrontation by removing participants, raising the opportunity cost of mobilization, or lowering confidence that collective action will attract enough support (Sellars 2019). Exit should therefore substitute for contemporaneous confrontation when collective departure withdraws participants and organizational capacity needed at home, and outside support does not offset that loss. This condition applies most directly to mass departure—the focus of this paper—and mass confrontation.

Exit can also support confrontation. Migrants may transmit resources, information, or organizational support (Pfaff and Kim 2003); departure can increase the leverage of those who remain (Karadja and Prawitz n.d.); and an outside option can back further demands (Clark, Golder, and Golder 2017; Hirschman 1993). Outside recognition, shelter, and political pressure can strengthen action at home. Neighboring prosperity could also intensify grievances about the home ruler’s performance, while departures could weaken his resources and make confrontation more promising.

In Tokugawa Japan, the village was a unit of collective resistance. At a given stage of a dispute, mobilizing villagers for confrontation at home and organizing mass departure generally offered alternative uses of the same participants and organizational capacity. The substitution logic therefore applies to the village-level analyses of tactical choice that follow. Complementarity can also operate across villages, although these dynamics

lie beyond the scope of this paper. One village's departure may weaken a shared ruler or attract outside support, strengthening confrontation by villages that remain.

2.4. Observable Implications

Better access to outside authority should encourage exit and, when neighboring livelihoods are sufficiently attractive, restrain taxation. [Supporting Information \(SI\) §A](#) formalizes the economic outside-option mechanism under common tax rates.

First, holding taxes fixed, easier access or stronger protection should make exit more attractive relative to compliance. Direct confrontation should not increase if grievances, resources, and coordination incentives remain unchanged. The related fixed-tax result concerns higher neighboring income or lower neighboring taxes, which improve the material outside option without changing confrontation's payoff ([SI Proposition 1](#)).

Second, better access should continue to encourage exit when rulers respond. Tax cuts retain potential leavers but sacrifice revenue from villages that would have complied. This cost limits concessions, so tax adjustment does not fully offset increased exit. Under the stated equilibrium conditions, [SI Proposition 2](#) establishes increased exit incidence regardless of the tax response and a higher exit share among resisting villages when the home tax falls.

The fiscal implication depends on the attractiveness of neighboring livelihoods after taxes. Easier exit gives rulers reason to retain cultivators, but can also reduce the pool available for confrontation. When neighboring net income is sufficiently attractive, retention dominates and rulers lower extraction; otherwise, relief from confrontation can permit higher taxes ([SI Proposition 3](#)). More exit therefore need not imply lower taxation; that depends on how attractive neighboring livelihoods are.

3. EMPIRICAL EVIDENCE

The empirical analysis examines exit occurrence using domain conversions and grid-year variation, resistance composition using village choices, and the fiscal consequence using a domain cross-section. The historical cases establish how outside authority operated in Tokugawa Japan; the quantitative tests examine behavioral implications and whether

geographic exposure is associated with lower taxation where neighboring domains can absorb migrants. They do not separately identify the interaction of horizontal and vertical authority or establish long-run institutional development.

3.1. Setting: Tokugawa Japan

During the Tokugawa period (1603–1868), nearly 300 domains, each held by a lord (*daimyō*), governed an archipelago smaller than France. [Figure 1\(a\)](#) maps the 264 capitals active in 1868, and [Figure 1\(b\)](#) tracks the number active as domains were created, merged, transferred, or absorbed into shogunal land. These domains originated in territories controlled by local warriors during the Warring States period and, earlier, in private estates known as *shoen*. Although subordinate to the shogunate, each domain maintained its own government and military forces and set its own tax rates ([Ravina 1999, 1000](#)).

Domains held territory at the shogunate’s pleasure and varied in size, wealth, and villages’ proximity to rival rulers. A common central authority, language, and broader legal and religious order kept vertical authority and the costs of changing jurisdictions comparatively stable while villages differed in the authorities they could approach. Cross-domain migration was prohibited, and domains could not leave the hierarchy. The prohibition made collective physical departure costly even where another ruler was nearby; appeals by representatives did not require whole communities to abandon their land.

Capital-centered fiscal administration links domain fragmentation to village choices. Domains shared an administrative structure centered on the castle town ([Hall 1955](#)), while the village contract system (*murauke*) made villages responsible to the *daimyō* for tax collection ([Saxonhouse 1995; Walthall 1997, 89; White 1995](#)). Domain capitals housed the officials who received petitions, decided on shelter and mediation, and kept records that could reach Edo ([Hall 1955; Ravina 1999](#)). Crossing a border placed villagers beyond their lord’s immediate reach; reaching an administrative center made their grievance visible to another authority and could open a path to Edo. A neighboring domain governed from a distant castle town offered less access than one governed from nearby.

The village contract made extraction a collective obligation administered through identifiable offices. When a tax dispute arose, the village addressed the same institutional system that collected its dues: home officials received its demands, an outside capital could decide shelter or mediation, and Edo could receive the resulting record. The relevant distinction was therefore not simply whether individual households moved. It was whether the village kept its dispute within the home domain or placed it before an outside authority.

The post-1615 settlement prevented recurrent inter-domain warfare for most of the next two and a half centuries. Domains fought no major wars against one another until the shogunate’s collapse in the 1860s.^[7] Domains still maintained military forces and responded to foreign threats, and the shogunate fought limited domestic campaigns. Those demands did not reproduce sustained competition among sovereign states. Domain rivalry operated principally through competition for cultivators, shelter of fugitives, and shogunal resolution of interjurisdictional disputes, enabling the “everyday forms of resistance” (Scott 1985) rather than the recurrent warfare central to conventional state-building accounts (Hintze 1975; Tilly 1978; Weber 1919).

Feudal, imperial, federal, and other layered systems combine local autonomy and superior authority in different ways (Gerring et al. 2011, 377–382, 405–407). Early modern Europe contained both sovereign states without a common superior and layered arrangements in which principalities, cities, and estates remained under imperial or royal authority. Movement could entail changes in language, religion, or law. Qing administration placed local jurisdictions more uniformly within a single sovereign hierarchy. Tokugawa Japan combined local autonomy with a common shogunal order and comparatively low cultural and legal distance. These arrangements offered different combinations of horizontal alternatives and vertical oversight.

^[7]The Shimabara Rebellion of 1637–38 was an anti-Christian and anti-Matsukura uprising suppressed by a shogunal coalition, not unrestricted inter-*daimyō* war. The Chōshū War (1864–1866) similarly pitted the shogunate against Chōshū, and the Boshin War (1868–1869) accompanied regime collapse. Institutional peace held between Shimabara and the 1860s. The tax data cover the period’s end, but tax policies remained stable even in the final years (Smith 1958).

European fragmentation also supplied rulers’ military rivals alongside subjects’ exit options. Exit could moderate taxes in peaceful conditions, but endemic warfare overwhelmed this restraint (Rosenthal and Wong 2011, 169, 184). Cross-regional comparisons also conflate differences in law, kinship, religion, bureaucratic recruitment, and geography between Europe and China. Europe’s fragmentation itself arose partly from medieval ecclesiastical competition (Grzymala-Busse 2023). These comparisons cannot by themselves isolate the exit channel. Within Tokugawa Japan, rival centers varied in number and proximity across villages and years while the hierarchy, broader institutional order, and exposure to recurrent warfare remained comparatively stable. This variation permits examination of subjects’ responses to alternative authority apart from recurrent war finance; it does not identify the long-run institutional consequences of exit.

3.2. Observed Forms of Exit and Direct Confrontation

The empirical analysis maps the distinction in §2.1 onto Aoki’s (1975) five recorded tactics: coercive appeals (*gōso*), destruction (*uchikowashi*), rebellion (*hōki*), desertion (*chōsan*), and end-run appeals (*osso*).^[8] The first three constitute *direct confrontation*; the last two constitute *exit through alternative authority*.^[9] Figure 1(c) maps these incidents, and Figure 1(d) plots their frequency over time.

The grouped exit outcome combines desertion (*chōsan*) and end-run appeals (*osso*), both of which place villagers or their claims beyond the home ruler’s exclusive control. Neighboring authorities can support either tactic through shelter, mediation, or backing for an appeal to the shogunate. The authority ultimately addressed and the neighboring support used to reach it are distinct features of the same action. Ordinary petitions to the home ruler are excluded and are not part of the *osso* category used here.

The full destination audit in SI §B.4 reports destinations and coverage (Table A.9). The desertion-only analyses isolate horizontal exit by observed action rather than inferred destination, and the findings persist under this restriction (SI §C.5, §D.2.8, and §D.2.9).

^[8]Later studies follow this classification. White (1995) analyzes protest repertoires and their evolution; Vlastos (1986) treats end-run appeals as a distinct form that dominated the mid-Tokugawa period; Walthall (1997) translates primary documents from all five categories.

^[9]“Direct confrontation” is more accurate than “fight” because coercive appeals were ritualized and often only implicitly violent.

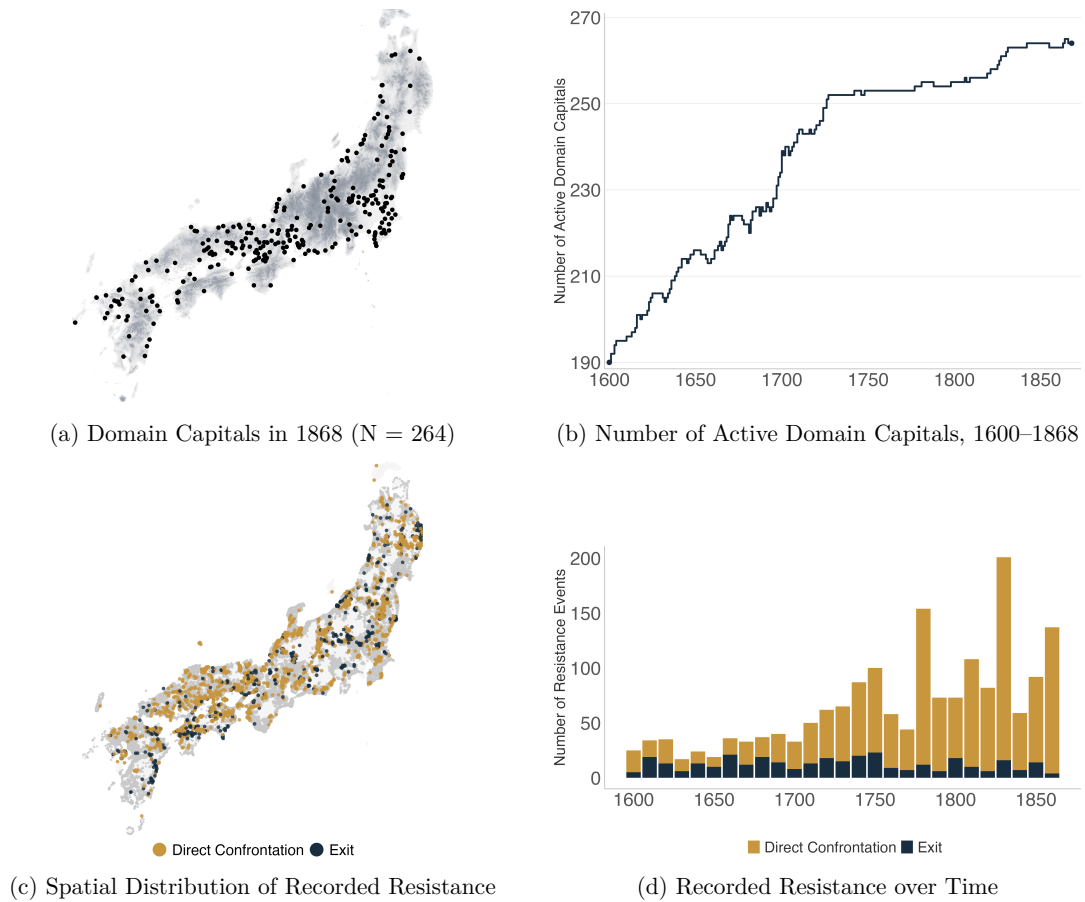


Figure 1: The Tokugawa Domain System and the Record of Peasant Resistance

Note: Tokugawa Japan was divided into roughly 300 domains, each governed from a castle town that served as its capital and housed the officials who received petitions. (a) Locations of the 264 capitals active in 1868, plotted over terrain; the archipelago runs about 1,900 km southwest to northeast. (b) The number of active capitals each year, 1600–1868, which changed as the shogunate created, merged, transferred, and abolished domains. (c) Villages recorded as resisting, colored by whether they exited (left the domain or petitioned an outside authority) or confronted their lord directly; gray points are villages present through the end of the period with no recorded resistance. (d) Recorded episodes per decade.

The three forms of direct confrontation differed in intensity (Aoki 1975). Coercive appeals were mass marches on domain offices or castles to present demands backed by an implicit threat of violence. Crowds of thousands wore straw raincoats and carried bamboo spears as symbols of collective resolve (Walthall 1997, 131–132). Destruction targeted the property of officials, merchants, or wealthy villagers associated with oppressive policies. Organized groups “smashed” (*uchikowashi*) houses and storerooms (Walthall 1997, 89–130). Rebellion was armed uprising against domain authority, the rarest form at roughly

1% of recorded incidents (White 1995, 142). All three sometimes won concessions but often met severe repression.

Tokugawa collective departure and coercive marches drew on village organization and required physical participation, creating the competition for participants described in §2.3. Appeals by a few representatives could leave most villagers available for action at home; the participation argument therefore does not apply equally to every recorded exit. Both tactics could nevertheless occur within one dispute. The 1853 Morioka records show end-run appeals and confrontation at different points in a broader cycle of resistance (Aoki 1975). Following the action-level distinction in §2.3, a dispute containing both tactics contributes separate event records. A neighboring lord’s shelter or mediation does not turn villagers’ exit into direct confrontation. The classification follows villagers’ actions: an armed clash between rulers would be distinct from villagers’ coercive action against home authorities.

Land-tax bargaining primarily concerned landholding peasants who bore fiscal obligations on registered farmland. Disputes by landless cultivators against landlords were usually treated as *murakata sōdō*. These landlords were local landholders who leased land to cultivators, distinct from the domain lords who levied land tax. Aoki prints peasant uprisings separately from urban disturbances, *murakata sōdō*, and province-wide petitions. The project comparison ends with the uprising chronology; all 294 unambiguous event-page matches, including 115 in the analytic sample, fall there, so excluding the separately printed categories removes no observation (Table A.6).

Household tenure within included entries is less certain. Removing all nineteen incidents mentioning tenants, landlords, or *komae* smallholders leaves the preferred village-choice estimate positive and statistically significant (Table A.7). I therefore interpret the estimates as behavior by recorded village groups, not a landholder-only effect.

3.2.1. Why Non-Compliance Was Consequential

The shogunate heard petitions, arbitrated disputes among lords, and could convert domains to direct rule or strip lords of their territories. Its oversight also extended to officials administering shogunal lands. Exit publicized governance failure more readily than direct confrontation. The Buke Shohatto (Laws of Military Houses) required

domains to be “well-governed,” and failure gave the shogunate grounds to confiscate, reduce, or transfer a lord’s territory (Steele, Paik, and Tanaka 2017, 3–5). Direct confrontation usually remained internal, allowing the daimyō to contain information or control its reporting. Exit brought villagers or their claims before an authority beyond the home ruler’s exclusive control, exposing the dispute to outside scrutiny. A fleeing village thus exposed its lord to rivals and the shogunate, making attainer risk harder to suppress (Steele, Paik, and Tanaka 2017). This publicity gave daimyō strong incentives to moderate extraction where villages could reach outside authority. Mass flight was accordingly “what the warrior class feared most” (Kuromasa 1927, 69) and was prohibited by strict decree.

The shogunate examined petitions from its own lands as well as from daimyō domains. Kurushima documents villagers on shogunal land whose direct petition against a taxation scheme was examined and accepted (Kurushima 2002, 185–198). In the 1642 Aizu uprising, several thousand villagers instead crossed into neighboring domains, publicly petitioned the bakufu, and threatened permanent resettlement. Their lord canceled arrears partly to forestall intervention, and the unrest contributed to his subsequent attainer without constituting its formal charge (Vlastos 1986, 35–38). Both cases invoked shogunal oversight; Aizu also shows how neighboring refuge and the threat of lost cultivators could reinforce an appeal to the superior.

The 1633 rules permitted higher appeal after district-level remedies failed, helping authorities monitor crop failures and official misconduct, but threatened punishment for unauthorized fiscal petitions (Vlastos 1986, 42–43). Direct appeals nevertheless became an accepted, if formally unlawful, means of resolving tax disputes (Vlastos 1986, 55–59). The shogunate’s 1720 petition-box system sent complaints about its officials directly to the shogun for investigation and remained in use through the period’s end (Sng and Moriguchi 2014, 447). Oversight did not ensure acceptance: in the Minamiyama appeal of 1721, the bakufu initially refused to hear tax-policy complaints while investigating charges of embezzlement and malfeasance (Vlastos 1986, 53). Superior authority could thus support a claim against local misconduct while defending the fiscal demands that prompted it.

3.2.2. Landholding, Tax Liability, and the Cost of Exit

The fiscal mechanism applies most directly to registered landholding households. Seventeenth-century cadastral surveys in shogunal territory matched parcels to cultivating stem families, assessed output and tax, and protected household claims conditional on payment (Nakabayashi et al. 2020, 158, 175). Peasants could use, buy, sell, rent, and inherit land; partible inheritance was rare, and the main family generally inherited all or most land (Kumon 2025; 2026, 158, 160; Nakabayashi 2025). A durable cultivation claim and recurring fiscal duty were thus two sides of the same institution.

Tenure nevertheless varied regionally. Less productive areas retained joint village ownership, covering about 30% of assessed national output, under which villages assigned and sometimes reallocated parcels (Nakabayashi et al. 2020, 175–176). Kaga combined corporate repartition with inheritable and transferable household use rights (Brown 1993, 94–95, 107–108). Land access was therefore place-specific—attached either to a registered household or to membership in a landholding village. Prolonged departure could weaken practical access, and permanent resettlement abandoned a valuable asset, but the sources do not show that border crossing automatically extinguished title.

Kaga officials responded to departure by waiving arrears, offering rice, promising to correct abuses, and calling cultivators home (Brown 1993, 138). Nobeoka likewise granted the departing villagers’ demands before they returned (Steele, Paik, and Tanaka 2017, 356). These episodes illustrate the concession-and-return sequence described in §2.2.

3.2.3. Exit as Deliberate Protest

Desertion and end-run appeals both depended on outside authority, and villagers sometimes combined them by petitioning while others sheltered abroad (Hosaka 2002; Miyazaki 1995). Receiving authorities supplied shelter, guards, mediation, or recognition. Continued shelter deprived the home lord of cultivators, publicized the grievance, and preserved possible absorption; return after concessions was therefore a bargaining outcome, not evidence that departure lacked force (Miyazaki 1995; Vlastos 1986). Seven documented exits or attempts ended in return, although the sources do not permit a general estimate of episode duration, a systematic reconstruction of what villagers paid while away, or an estimate of how common return was (Table A.8).

3.2.4. How Exit Worked: Illustrative Cases

Three episodes of peasant resistance show how neighboring rulers made exit effective through shelter, mediation, and possible escalation to the shogunate.

In Nobeoka (1690), repeated floods and heavy taxation prompted villagers to seek relief from their lord. After he rejected their appeal, about 1,418 peasants crossed into neighboring Takanabe. Negotiations then continued through the Takanabe lord for ten months (Steele, Paik, and Tanaka 2017, 356). Takanabe officials sheltered the villagers, guarded their representatives, and helped take the dispute to Edo when negotiations stalled. Nobeoka eventually accepted all demands, including tax relief, and the villagers returned without punishment. The following year, the shogunate transferred the Nobeoka lord to the much smaller Itoigawa Domain (Ishikawa 1981). Outside protection thus sustained an extended negotiation while exposing the dispute to superior authority.

In Yamashirodani (1841), 631 villagers left Tokushima territory for neighboring Imabari with sixteen demands concerning monopolies and restrictions on trade. They refused to return without concessions and a guarantee against punishment. Imabari's lord backed their demands and warned that he would retain them as his own subjects if Tokushima refused. Local officials housed the villagers in a temple and nearby homes and supplied food while Tokushima sent officials to negotiate. After ten days, the villagers withdrew with a written undertaking accepting their demands and promising no punishment (Aoki 1999; Okamura 2009). The receiving lord made continued absence feasible and gave Tokushima a concrete prospect of losing cultivators and revenue (Miyazaki 1995).

In Matsuyama (1741), villagers in the mountainous Kuma district faced rising rice prices, falling tea prices, and burdensome paper-production obligations. An initial appeal brought no effective relief; a second mobilization spread across twenty-six villages and crossed into Ōzu. The villagers rejected assurances that their complaints would be heard if they returned. Matsuyama then enlisted the abbot of Daihōji to mediate. The settlement canceled proposed increases in tea and other levies, eased paper-related debts, and protected the protest leaders from investigation, although it did not grant every demand. The villagers returned in August, and senior domain officials were subsequently

exiled (Ehime Prefectural History Compilation Committee 1986; Nakagawa 1980). Refuge outside the domain enabled villagers to hold out for specified concessions before returning.

Together, these cases show how outside authority turned departure into bargaining through shelter, mediation, and possible absorption. Customary legitimacy and ritual shaped the available repertoire, while relative costs influenced which tactic villages used (Vlastos 1986; Walthall 1997; White 1995).

3.3. Exposure and Exit Occurrence

Two designs examine whether recorded exit occurs more often under greater geographic exposure: a conversion-based difference-in-differences design and an observational grid-year panel.

3.3.1. Conversion-Based Test: Tenryō Conversion

This design associates conversion with 4.3 percentage points more recorded exit among nearby villages governed by surrounding lords and no detectable average change in their direct confrontation. I use the Tokugawa shogunate’s abolition of domains and conversion of their territories into direct shogunal land (*tenryō*), an institutional change that introduced a powerful nearby authority.

Conversion followed the shogunate’s disposition of a ruling house’s tenure, commonly through succession failure, political sanction, or administrative transfer. Supporting Information §§C.1–C.2 reports the case histories, cause typology, and source references (Kimura, Fujino, and Murakami 2015; Kodama and Kitajima 1989).

A *tenryō* conversion brought shogunal authority closer to surrounding villages, creating a nearby potential route for appeals. Access to shogunal officials could reduce the cost of petitioning and expose a neighboring lord’s conduct to superior scrutiny. Conversion also supplied a precedent for territorial intervention, while contiguity with shogunal land could lower its administrative cost. These changes could strengthen villagers’ claims even when their home domain remained under its own lord.

The conversion design examines the prediction that improved access to outside authority encourages exit. A nearby shogunal appeals route lowers the effective cost of exit and should raise exit in villages of neighboring domains. The conditional benchmark

predicts no increase in direct confrontation when grievances, resources, and coordination incentives remain unchanged. Because internal disorder sometimes triggered conversion, the null on nearby confrontation also tests whether unrest simply spilled across domain borders.

The identifying assumption concerns resistance in surrounding villages, which remain governed by other lords and supply all measured outcomes. Internal succession or tenure disputes can change access to shogunal authority without originating in those villages. Regional unrest, information changes, or administrative expectations could nevertheless affect their subsequent resistance. Domain and year fixed effects absorb persistent local differences and common shocks; the event study, timing test, and historical exclusions assess the remaining concern (SI C.13, SI C.3).

Aoki (1975) catalogs 7,664 incidents of peasant resistance from local histories, official documents, and diaries. Each observation records the date, home domain, nearest neighboring domain,^[10] and resistance mode, which I classify as exit or direct confrontation under Section 3.2. I exclude 574 peaceful petitions and inter-village resource or boundary disputes.

The catalog identifies 2,790 sites. I georeference 2,571 of them using several sources and exclude those whose locations remain imprecise.^[11] The resulting data contain 910 resistance episodes involving 2,571 villages and 2,741 tactic choices (Table A.5, SI B). Because large mass exits were more likely than small flights to enter the historical record, unrecorded minor cases are unlikely to account for the main results. Capital locations and the creation and abolition dates of all 294 domains come from Kimura, Fujino, and Murakami (2015) and Kodama and Kitajima (1989) (Table A.10, SI B); I assign branches (shi-han) to their main domains (hon-pan).

I construct annual domain-year observations around conversion events during 1615–1868, including up to five years before and after each event. The 49 retained events occur

^[10]I used Kodama and Kitajima (1989) to identify the neighboring domains existing at the time of each resistance event. The nearest foreign domain is identified using time-varying domain existence data: for each event, only domains active at that date are potential alternative authorities.

^[11]These sources include Honda, Natsume, and Nemoto (2022), the geocoding API provided by the Ministry of Land, Infrastructure, Transport and Tourism of Japan, and the Historical Location Name Comprehensive Database provided by the Center for Open Data in the Humanities.

between 1617 and 1807, so the matched estimation frame spans 1612–1812. For each domain-year, I count exit and confrontation events in villages of other domains within $r \in \{30\text{km}, 40\text{km}, 50\text{km}\}$ of the capital, the zone exposed to the new exit opportunity. Figure 2 illustrates the design.

Consider a domain abolished in year t and converted to tenryo. The outcomes do not come from villages inside that converted territory. They come from villages that remain governed by surrounding lords but lie within radius r of the converted capital. For the treated domain, the design compares resistance in those foreign villages within five years before conversion with resistance within five years afterward. It then compares that before-and-after change with the contemporaneous change in foreign villages surrounding domains that never converted. The converted capital supplies the new outside authority; villages in neighboring domains supply the measured exit and confrontation outcomes. Domain fixed effects absorb persistent differences around focal capitals, and year fixed effects absorb shocks common to all domains. The coefficient estimates whether resistance around a converted capital changed more at conversion than resistance around never-converted capitals observed in those same years.

The treatment sample contains 49 conversion events, listed by year and stated reason in SI C.1 (Kimura, Fujino, and Murakami 2015; Kodama and Kitajima 1989). Source-identified domain and date corrections are documented in SI C.2. I retain available observations within five years before and after first conversion; never-converted controls enter only for the same calendar years. The resulting panel contains 49 treated and 220 control domains. Exposure ends when the chronology records re-enfeoffment, although those observations remain within the event window. I estimate a domain-year linear probability difference-in-differences model with domain and year fixed effects:

$$\text{NonCompliance}_{j,t} = \beta_1 (\text{Converted}_j \times \text{Post}_{j,t}) + \varphi_j + \omega_t + \varepsilon_{j,t}$$

Here, $\text{NonCompliance}_{j,t} \in \{\text{Exit}_{j,t}, \text{Confront}_{j,t}\}$ indicates any nearby exit or confrontation within r kilometers of domain j 's capital in year t . Converted_j identifies ever-converted domains, and $\text{Post}_{j,t}$ identifies years after first conversion under direct shogunal rule.

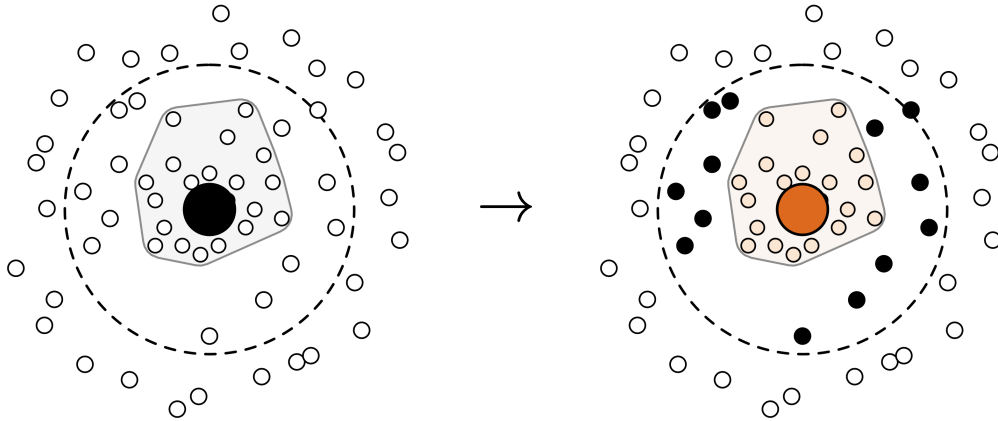


Figure 2: DiD Design: Tenryō Conversion

Note: Stylized illustration. The large central circle is the focal capital, the gray polygon is its territory, small circles are villages, and the dashed circle is the analysis radius r . Left: before conversion, villages of surrounding lords within r lack a nearby shogunal authority. Right: abolition converts the focal capital and territory to direct shogunal rule (*tenryō*, orange); black circles identify surrounding foreign villages inside r , while pale circles inside the gray territory are excluded from the outcomes and open circles fall outside the treated foreign-village set. No village-to-territory arrows are shown because episode destinations are unobserved. The DiD compares changes in exit and confrontation in surrounding foreign villages with changes around never-converted domains.

Destination misclassification motivates the first checks. The design cannot observe the destination of every exit. Sources usually record where villagers came from but rarely where or how they traveled, so proximity identifies converted territory as a plausible outside option rather than a confirmed destination. Figure 2 accordingly depicts exposure within the analysis radius without village-to-territory arrows. At the tighter 40km and 30km radii, the exit coefficients remain positive at 0.0292 ($p = 0.027$) and 0.0183 ($p = 0.039$), respectively, while nearby confrontation remains near zero at every radius. The exit estimate also remains positive when the outcome is restricted to desertion (SI C.5). A general problem of geographically mismeasured unrest would not readily produce all three patterns.

Excluding three conversions associated with documented local unrest or petitioning leaves the exit estimate positive at 0.0472 (cluster $p = 0.0076$); broader five- and nine-case screens yield 0.0448 ($p = 0.0148$) and 0.0493 ($p = 0.0138$). Confrontation remains statistically indistinguishable from zero in each sample. Supporting Information §C.3

Outcome:	Any Nearby Exit			Any Nearby Confrontation		
Sample Radius:	30km	40km	50km	30km	40km	50km
<i>Variables</i>						
Converted $\times$ Post	0.0183	0.0292	0.0427	0.0114	0.0062	0.0126
Cluster SE	(0.0088)**	(0.0131)**	(0.0167)**	(0.0105)	(0.0102)	(0.0125)
Conley SE (70km)	(0.0090)**	(0.0120)**	(0.0144)***	(0.0118)	(0.0122)	(0.0150)
<i>Controls</i>						
Domain FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit Statistics</i>						
Num. Obs.	31,989	31,989	31,989	31,989	31,989	31,989
Converted/Observed Domains	49/269	49/269	49/269	49/269	49/269	49/269
R2	0.034	0.046	0.057	0.070	0.088	0.114
Mean Dep. Var.	0.010	0.017	0.025	0.017	0.025	0.035

Table 1: Domain-to-Tenryo Conversion and Nearby Exit

Note: Linear probability model (LPM) estimates from an annual domain-year panel. “Nearby” denotes resistance in villages of *other* domains within the specified radius. Converted domains contribute available observations within five years before and after conversion and are matched to never-converted controls in the same calendar years. SEs in parentheses are (i) two-way clustered by domain and year and (ii) Conley (70km) on capital coordinates; stars use the corresponding estimator’s p -value. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

documents the cases and exclusion rules; these checks cannot establish the absence of unrecorded unrest.

Table 1 reports the main result. At 50km, tenryo conversion is associated with a 0.0427 increase in nearby exit probability (cluster $p = 0.011$; Conley $p = 0.003$), or 4.3 percentage points. The mean exit probability among matched control observations is 0.025. The confrontation estimate is 0.0126 (cluster $p = 0.315$; Conley $p = 0.401$). The corrected sample therefore shows more exit without a detectable parallel change in direct confrontation.

The conversion-timing checks begin with the assumption that converted and never-converted domains would have followed parallel resistance trends absent conversion. I assess this assumption with an event study:

$$\text{NonCompliance}_{j,t} = \sum_{s \neq -1} \beta_s (\text{Converted}_j \times \mathbf{1}[t - t_j^* = s]) + \varphi_j + \omega_t + \varepsilon_{j,t}$$

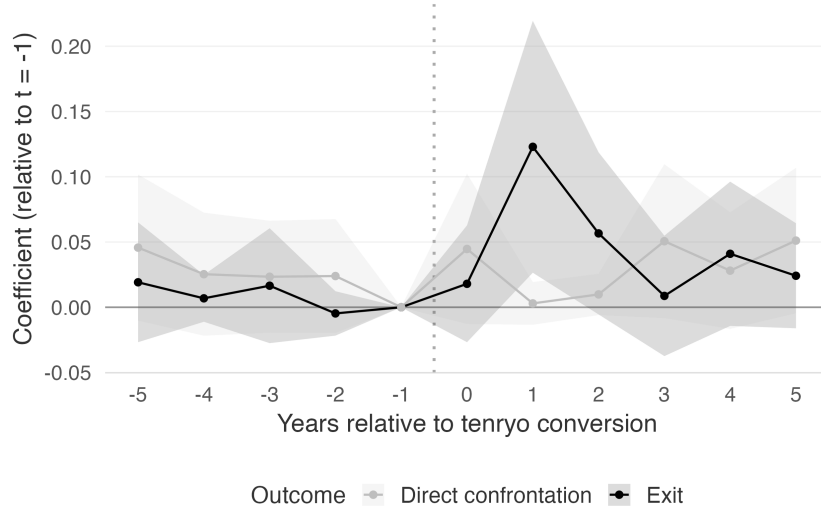


Figure 3: Tenryo-Conversion Event-Study Coefficients

Note: Yearly event-study coefficients for nearby exit (black) and direct confrontation (light gray), with $s = -1$ as the reference. Shaded bands show 95% confidence intervals; the horizontal line marks zero and the vertical dotted line marks conversion between $s = -1$ and $s = 0$. “Nearby” denotes resistance in villages of *other* domains within 50km of the focal capital. Standard errors are two-way clustered by domain and year.

where Converted_j denotes tenryo-converted domains, t_j^* is the first conversion year, and $s = -1$ is the reference period. The event study follows time since initial conversion, whereas the pooled exposure indicator switches off after re-enfeoffment. In Figure 3, the pre-treatment coefficients are jointly insignificant (exit: $F = 0.95$, $p = 0.432$; confrontation: $F = 1.16$, $p = 0.324$). Exit rises at $t = 1$ ($\hat{\beta}_1 = 0.123$, $p = 0.013$); later estimates remain positive but are less precise. No individual post-conversion confrontation coefficient is significant at the 5% level.

Three diagnostics assess the design’s identifying assumptions. Under the Rambachan and Roth (2023) relative-magnitudes restriction, the exit confidence interval excludes zero at tested bounds through $\overline{M} = 0.20$ and includes zero at 0.25 and above (SI C.9, SI C.10). Lagged nearby exit and confrontation do not jointly predict conversion timing in the complete-case hazard sample ($p = 0.436$; SI C.13). The Sun–Abraham first-conversion specification yields positive point estimates for both outcomes, but its clustered covariance does not support valid inference (SI C.6).

The exit coefficient remains positive under alternative Conley bandwidths, count outcomes, and exclusions for potentially exposed controls, although precision varies across

specifications (SI C.14, SI C.15, SI C.4, SI C.11). Placebo reassignment of conversion events also places the observed exit estimate in the upper tail (SI C.12). The pre-1750 cohort contains 45 of the 49 conversions; the later cohorts are too small to support comparable inference (SI C.8).

3.3.2. General Test: Grid-Year Panel

The conversion design estimates the response to a specific improvement in access to shogunal authority. The grid-year panel broadens the comparison: greater geographic exposure is associated with more recorded exit and no detectable increase in confrontation as nearby capitals are established, abolished, or relocated.

I measure location-level exposure with a *distance ratio*:

$$\rho_{i,t}^{\text{grid}} := \frac{d_{i,t}^1}{d_{i,t}^2}$$

where $d_{i,t}^1$ and $d_{i,t}^2$ are straight-line distances from location i to the nearest and second-nearest active capitals in year t . The ratio approaches one when a new capital appears closer than the prior second-nearest or when the nearest capital disappears or moves farther away. Values near zero indicate one dominant capital; values near one indicate two equally accessible rulers. Section 3.4.1 defines the village-level analog.

A village-year panel would be preferable, but domain affiliation is observed only at the end of the Tokugawa period. The governing domain of most non-resisting villages is therefore unknown in earlier years. I address this limitation by assigning villages to fixed 5km hexagons and computing the grid exposure ratio $\rho_{i,t}^{\text{grid}}$ from the two nearest active capitals without using domain affiliation. The 46,086-village late-Tokugawa *Kyūdaka–Kyūryō Torishirabe Chō* catalog defines populated cells (Section 3.4). The resulting panel contains 2,225,802 cell-years across 8,763 cells, with annual counts of exit and confrontation in each hexagon.

The estimating equation is

$$\text{NonCompliance}_{i,t} = \beta_3 \rho_{i,t}^{\text{grid}} + \lambda_i + \eta_t + \varepsilon_{i,t}$$

with hexagon and year fixed effects, where $\text{NonCompliance}_{i,t}$ indicates whether any exit or confrontation occurred in hexagon i in year t . The main estimates are linear probability

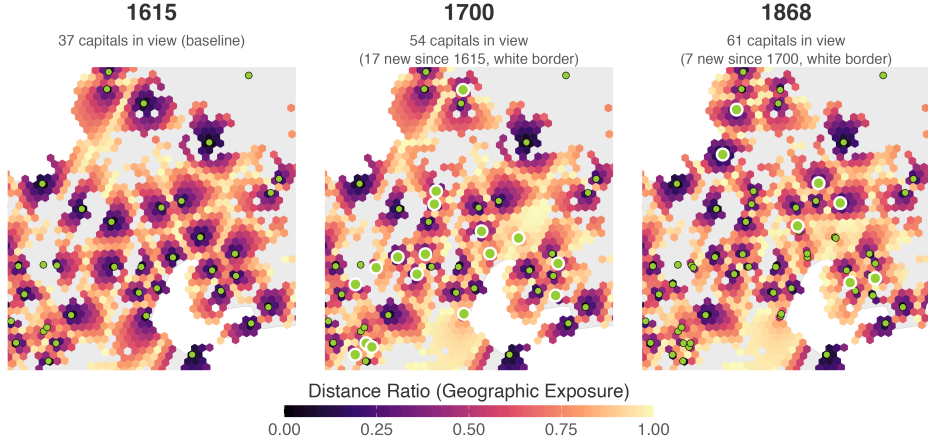


Figure 4: Time-Varying Distance Ratio across 5km Hexagonal Grid: Kinki Region

Note: Each panel maps $\rho_{i,t}^{\text{grid}}$ (nearest-capital distance / second-nearest-capital distance) in a $\approx 200\text{km} \times 200\text{km}$ window of central Japan (lat 34.4–36.2°N, lon 135.5–137.6° E), covering Kyoto, Osaka, Nagoya, and Lake Biwa. This region has the most cells whose ratios changed across the three years. Yellowgreen circles mark active capitals; white borders mark capitals added since the previous snapshot. Yellow indicates greater geographic exposure and purple a dominant nearest capital. Changes reflect domain creation and dissolution.

models; separate count specifications appear in [SI D.2.10](#). Identification comes from capital dissolutions, transfers, and reestablishments that reorder the nearest authorities. [Figure 4](#) maps $\rho_{i,t}^{\text{grid}}$ in three years: broad spatial patterns persist, while domain creation and dissolution change exposure locally. Hexagon and year fixed effects absorb time-invariant geography and common shocks.

[Table 2](#) shows that the distance ratio is associated with exit but not confrontation, whereas remoteness from any authority predicts neither. In columns (1)–(3), a one-unit increase in the ratio is associated with a 0.035-percentage-point increase in cell-year exit probability. The estimate is significant at 5% with both cell-and-year clustering ($p = 0.048$) and Conley inference ($p = 0.037$). The smaller positive estimate for confrontation is not statistically significant. Because the baseline exit probability is only 0.022%, a one-standard-deviation increase in the ratio ($\text{SD} = 0.26$) corresponds to roughly 40% of that mean.

Columns (4)–(6) distinguish geographic exposure from simple remoteness by using distance to the closest active capital $d_{i,t}^1$ and discarding the second-nearest. The exit coefficient is small and marginal with clustered standard errors ($p = 0.092$) but null with

Outcome:	Grid exposure ratio ($\rho_{i,t}^{\text{grid}}$)			Closest-capital dist. ($d_{i,t}^1$, km)		
	(1)	(2)	(3)	(4)	(5)	(6)
	Any exit	Any confr.	Any confr. (strict)	Any exit	Any confr.	Any confr. (strict)
Coefficient	0.00035	0.00026	-0.00012	0.00000	0.00000	0.00000
Cluster SE	(0.00017)**	(0.00019)	(0.00011)	(0.00000)*	(0.00000)	(0.00000)
Conley SE (10km)	(0.00017)**	(0.00023)	(0.00012)	(0.00000)	(0.00000)	(0.00000)
<i>Controls</i>						
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit Statistics</i>						
Num.Obs.	2,225,802	2,225,802	2,225,802	2,225,802	2,225,802	2,225,802
R ²	0.006	0.007	0.006	0.006	0.007	0.006
Mean Dep. Var.	0.00022	0.00064	0.00026	0.00022	0.00064	0.00026

Table 2: Geographic Exposure Predicts Recorded Exit

Note: LPMs on any-event indicators at the 5km hexagon-year level for all 8,763 cells. Columns (1)–(3) use the annual grid exposure ratio $\rho_{i,t}^{\text{grid}} = \frac{d_{i,t}^1}{d_{i,t}^2}$ from the two closest active capitals. Columns (4)–(6) use distance to the closest active capital $d_{i,t}^1$ (km), which measures remoteness without the second-nearest capital. “Confrontation (strict)” includes only destruction and uprising. All models include grid and year FE. SEs are (i) two-way clustered by cell and year and (ii) Conley (10km). See [SI D.2.11](#) for ever-active cells. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

Conley standard errors ($p = 0.162$); confrontation is also null. The contrast between the two sets of columns points to the relative geometry of competing rulers, rather than distance from the nearest ruler, as the relevant condition.

Across alternative samples and measures, the exit coefficient remains positive, although inference is weaker in some specifications. This holds among the 1,466 ever-active cells, in Poisson and Negative Binomial fixed-effects models, for desertion alone, with terrain-weighted least-cost-path distances, and when *daikansho* enter the choice set (SI §§D.2.9–D.2.12 and D.2.14).

3.4. Exposure Shifts the Form of Non-Compliance Toward Exit

The preceding tests ask whether exit occurrence is higher under greater exposure. This section conditions on recorded non-compliance and asks which tactic a resisting village chose: exit or direct confrontation. It estimates composition within the resisting sample,

not whether exposure makes a village resist. Separating these margins matters because more exit events could reflect more disputes rather than substitution between tactics.

The village and tax analyses use 46,086 villages compiled primarily from the *Kyū-daka-Kyūryō Torishirabe Chō* (Ledgers of Former Assessments and Territories, hereafter LFAT), a Meiji land census that records each village’s home domain at the end of the Tokugawa period. Village locations come from [Honda, Natsume, and Nemoto \(2022\)](#). When several domains taxed one village, I assign the village to the domain holding the largest share of its assessed productivity (*kokudaka*, a rice-denominated index of aggregate agricultural output rather than literal rice harvest alone). The data cover virtually all arable land in Japan except high-altitude, non-cultivable regions ([Figure 1\(a\)–\(b\)](#); [Figure 6](#)).

3.4.1. Measuring Exposure to Alternative Authority

I measure each village’s exposure with ρ_i , the distance to its own lord’s capital divided by the distance to the nearest other lord’s capital. Unlike the grid exposure ratio $\rho_{i,t}^{\text{grid}}$ in [Section 3.3.2](#), which uses the two nearest active capitals, ρ_i anchors on the village’s known home domain. Proximity to an outside authority lowered exit costs by facilitating the intervention on which mass exit depended ([Section 3.2](#)): guards or mediators could protect fleeing peasants, villagers could more easily reach neighboring capitals or shogunal offices, and authorities could more readily transfer allegiance. Geographic exposure proxies for access to authority rather than observed bargaining or willingness to receive migrants. The measure is:

$$\rho_i := \frac{d_i^{\text{HC}}}{d_i^{\text{FC}}}$$

Read the ratio as relative access. A village 10km from its home capital and 20km from the nearest foreign capital has $\rho_i = 0.5$; reversing those distances gives $\rho_i = 2$. Higher values place outside authority closer relative to home authority. Because the ratio rises when the home capital is farther away or the foreign capital is closer, Models 6 and 7 later separate the two distances. This decomposition helps distinguish outside access from weak home control. Here, d_i^{HC} is the straight-line distance from village i to its home-domain capital and d_i^{FC} is the distance to the nearest active non-home capital

at the time of resistance. The data do not reconstruct village-specific historical roads, passes, checkpoints, guards, or realized travel routes, so the ratio measures geographic access to administrative centers rather than each village’s exact travel cost. I calculate both distances from the georeferenced resistance data and the time-varying capital panel in [Section 3.3.1](#). Active capitals include existing daimyō seats and the former seats of domains converted to tenryo, where a shogunal intendant (*daikan*) replaced the daimyō ([Vlastos 1986, 49](#)). The latter remain in the choice set throughout the tenryo period.^[12]

3.4.2. Empirical Strategy: Home-Neighbor Dyad Fixed Effects

The home-neighbor dyad design compares otherwise similar resisting villages facing the same pair of rulers. Each observation is a village involved in recorded resistance in [Aoki \(1975\)](#), and the outcome indicates whether it used exit or direct confrontation. [Figure 5](#) illustrates the comparison.

Home-neighbor dyad fixed effects isolate within-pair variation in the distance ratio. Each dyad is an ordered pair of home and nearest-rival domains (for example, A–B and A–C in [Figure 5](#)). Dyad fixed effects absorb time-invariant home- and neighbor-domain characteristics, including persistent differences in coercive capacity, ruler ideology, agricultural endowments, and administrative quality, as well as stable pair-specific cross-border ties, transit corridors, and border controls. They remove persistent features shared by villages with the same home lord and nearest rival; they do not make those villages identical or absorb village geography. Within an A–B dyad, the estimate compares A villages whose nearest rival is B but whose relative position between the two capitals differs, as shown within each dashed circle in [Figure 5](#). The controls account for the measured local geography listed below, and decade fixed effects absorb period-wide shocks.

The main specification is:

$$\text{Exit}_i = \beta_2 \ln(\rho_i) + X_i' \gamma_2 + \underbrace{\varphi_{j,k}}_{\text{home-neighbor dyad FE}} + \underbrace{\omega_t}_{\text{decade FE}} + \varepsilon_i$$

^[12]In some cases, the shogunate may have administered converted territory from a nearby *daikansho* (a shogunal intendant’s district office) rather than from the former castle town. To address this, [SI D.2.13](#), [SI D.2.14](#), and [SI D.3.1](#) replicate all main analyses with the nearest capitals redefined to include *daikansho* locations.

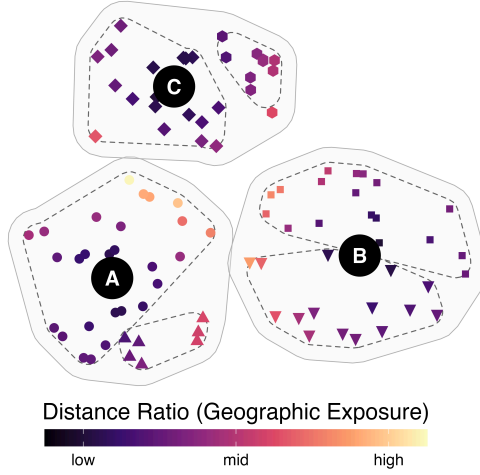


Figure 5: How Home-Neighbor Dyad Fixed Effects Isolate Within-Pair Variation

Note: A schematic, not data. Three domains (A, B, C) are drawn with their capitals as black circles and their villages as shapes shaded by the distance ratio ρ_i , the village's distance to its own lord's capital divided by its distance to the nearest other lord's capital. Dark shading marks villages sitting close to their own capital and far from any rival, light shading the reverse. A village's home domain and its nearest rival together define a dyad, shown by the dashed circles. Including a fixed effect for each dyad discards comparisons across dyads and retains only comparisons among villages facing the same pair of lords, so the estimate cannot be driven by differences between regions or between particular lords.

Here, i indexes a village-event, while j and k index its home and nearest-rival domains and t its decade. Exit_i equals one for exit and zero for direct confrontation. The logged ρ_i uses the capitals active at event i .^[13] X_i contains latitude, longitude, their interaction, logged elevation and ruggedness, logged distance to the nearer of Edo or Kyoto, and logged wet-rice suitability.^[14] The dyad fixed effects $\varphi_{j,k}$ absorb time-invariant features of each ordered home-rival pair, while ω_t absorbs decade-specific shocks.

3.4.3. Geographic Exposure Predicts Exit Share

Table 3 shows that exposure predicts exit rather than confrontation conditional on resistance. The coefficients are positive throughout and statistically significant in the preferred controlled specifications. In Model 5, with home-neighbor dyad fixed effects and full controls, a one-standard-deviation increase in the log distance ratio (≈ 1.58)

^[13]The logged ratio constrains the coefficients on the two logged distances to equal magnitude with opposite signs. Linear hypothesis tests find no evidence against this restriction (SI D.3.5).

^[14]Wet rice suitability is measured as potential output under irrigation and medium input, based on the Caloric Suitability Index developed by Galor and Özak (2016).

Outcome:	Exit _i						
	Distance Ratio					Abs. Distances	
Models:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Variables</i>							
Distance ratio ($\ln(\rho_i)$)	0.0458	0.0519	0.0201	0.0212	0.0244		
Cluster SE	(0.0154)***	(0.0141)***	(0.0071)***	(0.0072)***	(0.0068)***		
Conley SE (10km)	(0.0117)***	(0.0093)***	(0.0078)***	(0.0078)***	(0.0085)***		
$\ln(d_i^{\text{HC}})$						0.0160	0.0203
Cluster SE						(0.0104)	(0.0083)**
Conley SE (10km)						(0.0106)	(0.0113)*
$\ln(d_i^{\text{FC}})$						−0.0306	−0.0309
Cluster SE						(0.0129)**	(0.0135)**
Conley SE (10km)						(0.0138)**	(0.0148)**
<i>Controls</i>							
Decade FE	No	Yes	Yes	Yes	Yes	Yes	Yes
Home Domain FE	No	No	Yes	Yes	No	Yes	No
Neighbor Domain FE	No	No	Yes	Yes	No	Yes	No
Home-Nbr Dyad FE	No	No	No	No	Yes	No	Yes
Geography	No	No	No	Yes	Yes	Yes	Yes
<i>Fit Statistics</i>							
Num.Obs.	2,739	2,739	2,739	2,739	2,739	2,739	2,739
R2	0.028	0.225	0.621	0.626	0.678	0.627	0.678
Mean Dep. Var.	0.246	0.246	0.246	0.246	0.246	0.246	0.246

Table 3: Exposure to Alternative Authority Predicts Exit (Exit versus Confrontation)

Note: OLS estimates of each village’s exit-versus-confrontation choice conditional on resisting. Models (1)–(5) use the log distance ratio; Models (6)–(7) replace it with the two absolute distances (d_i^{HC} : home capital; d_i^{FC} : nearest foreign capital). Full results in Table A.30. SEs in parentheses: (i) clustered on home, nearest-neighbor, decade; (ii) Conley (10 km) on village coordinates; stars reflect that estimator’s p -value. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

raises exit probability by 3.9 percentage points, or 16% of the 24.6% baseline rate. The less restrictive home- and neighbor-FE specification (Model 4) yields a statistically indistinguishable 3.4-point estimate.

Models 6 and 7 separate the two distances because the ratio combines two distinct accounts. Greater distance from the home capital could increase exit through weaker administration at the periphery; shorter distance to the foreign capital instead increases access to an outside authority. In the preferred dyadic specification (Model 7), the home-capital coefficient is 0.0203 (cluster $p = 0.022$; Conley $p = 0.073$), while the foreign-capital coefficient is −0.0309 (cluster $p = 0.031$; Conley $p = 0.037$). The two coefficients are statistically consistent with the ratio’s equal-and-opposite restriction ($p = 0.531$; SI D.3.5). A one-standard-deviation decrease in $\ln(d_i^{\text{FC}})$ (≈ 0.92) corresponds to 2.8 percentage points more exit, or 11% of the baseline rate. Foreign-capital proximity is

therefore the larger and more robust component, although the home-capital estimate does not permit ruling out weaker peripheral administration.

The border-distance result further clarifies the mechanism. Distance to the nearest domain border does not significantly predict exit choice, while controlling for it leaves the capital-based exposure estimate little changed (SI D.3.2). This pattern supports the institutional rationale in §3.1: villagers gained leverage by reaching an authority able to provide protection, mediation, or recognition. The location of that authority mattered beyond the ease of crossing into another domain.

Measurement and sample composition motivate one set of robustness checks. The preferred exit coefficient remains positive with alternative productivity and political-distance controls (SI D.2.1), after excluding repeat-resistance villages (SI D.2.2), and when dropping cross-island dyads whose straight-line routes cross sea straits (SI D.2.4). It also remains positive when adding shogunal intendant offices (*daikansho*) to the measured choice set (SI D.2.13).

Restricting exit to desertion retains 58–70% of the pooled point estimates across all three behavioral designs. The 50km conversion estimate is 0.0295 (cluster $p = 0.035$; Conley $p = 0.031$), the preferred village-choice coefficient is 0.0141 (cluster $p = 0.051$; Conley $p = 0.041$), and the grid-year coefficient is 0.000241 (cluster $p = 0.082$; Conley $p = 0.048$). These estimates isolate horizontal exit by observed action without inferring unrecorded destinations (SI §C.5, §D.2.8, and §D.2.9).

A sequential choice decomposition merges the resisting sample with the comprehensive village ledger. It finds no detectable association between exposure and resistance rather than compliance in the main specification and yields an exit-versus-confrontation estimate similar to Model 5. A symmetric time-averaged exposure measure produces a small negative first-step estimate while preserving the positive composition estimate (SI D.3.3). The evidence therefore distinguishes resistance incidence from tactic composition rather than treating compliance as absent.

3.5. Exposure and Constrained Taxation

If viable exit disciplines rulers, exposed domains should tax less to retain their population. The economic sequence begins with competition over scarce labor. A neighboring ruler able to employ additional cultivators has reason to receive them, making the village's exit option credible. The home ruler then anticipates losing both cultivators and the tax base they support and moderates extraction. When neighbors lack the capacity to absorb labor, geographic access alone creates a weaker threat. Systematic tax data survive only as a late-period cross-section, so this section seeks a suggestive fiscal trace rather than a causal estimate: a pooled relationship between exposure and taxation cannot distinguish exit from state-capacity decay. Neighbor characteristics provide a sharper test. The exit mechanism predicts a negative tax gradient only where neighboring domains are attractive enough to absorb migrants. Generic peripheral state-capacity decay instead predicts lower taxation where home administration is weak, regardless of neighboring labor demand, and therefore predicts no such asymmetry. I measure domain exposure by averaging the village distance ratio within each domain. Figure 6 maps the log ratio (panel a) and reported effective tax rates (panel b).

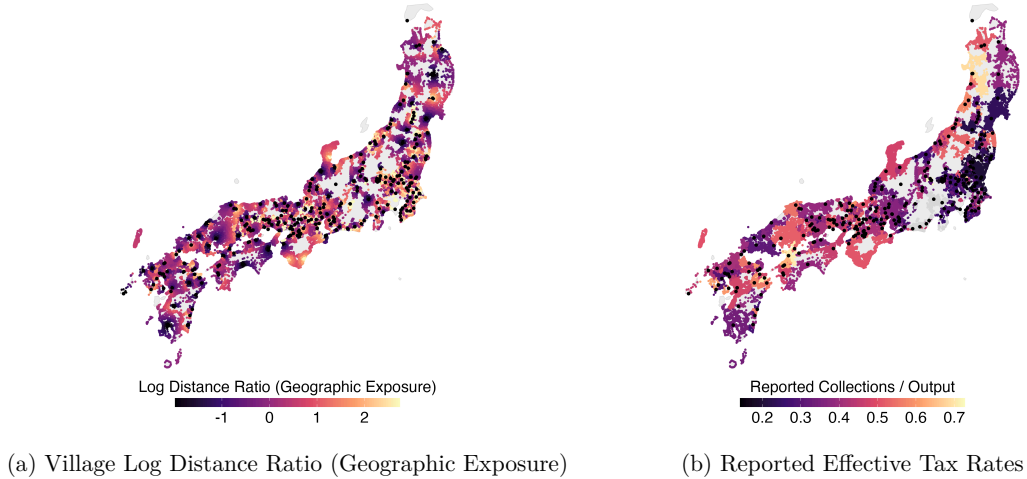


Figure 6: Geographic Exposure and Tax Rates Across Villages

Note: Panel (a) plots village locations shaded by the log distance ratio $\ln(\rho_i) = \ln(d_i^{\text{HC}}/d_i^{\text{FC}})$, where HC is the home-domain capital and FC is the nearest foreign-domain capital; higher values indicate greater geographic exposure. Panel (b) shades those villages by their domain's reported collections divided by reported output, the effective tax-rate measure used in the analysis. Gray points denote villages in domains without tax data, and black dots mark domain capitals.

3.5.1. Empirical Strategy

Capital locations could be endogenous if rulers placed capitals or exchanged border villages to deter exit. Most capitals, however, were inherited from the Sengoku period (Hall 1955) and fixed by the 1615 one-castle-per-domain edict (*ikkoku ichijō rei*); later relocations required shogunal approval and were rare. The shogunate also placed fudai vassals strategically and kept tozama lords on the periphery (Ravina 1999). I therefore control for each daimyō’s alliance status.^[15] These controls account for observed political ties that could affect both territorial decisions and extractive capacity. Unobserved factors influencing capital placement, village transfers, and taxation can still bias the association.

The home-capital distance could also proxy for domain scale, peripheral state capacity (Sng and Moriguchi 2014), or village development. Controls therefore include domain population,^[16] agricultural productivity, territory size, the *gōshi system* (lower-ranking samurai serving as village officials, which raised monitoring costs), and the number of villages named “shinden” (new rice field). Tax-delivery costs are unlikely to drive the result because villages used nearby storage facilities rather than domain capitals.^[17] Tokugawa institutions also limited local agents’ authority (Section 3.1). These controls cannot eliminate unobserved differences in scale or coercive capacity, and the direction of any remaining bias is not identified.

The outcome is the domain effective tax rate from Kodama and Kitajima (1989) (pp. 425–431): *shunodaka* (reported collections in rice equivalents) divided by *uchidaka* (the domain’s internal record of actual production, distinct from its public *kokudaka* assessment). This ratio measures the share of reported output collected.^[18] The Meiji government compiled these data in 1869–1870 from each domain’s average production

^[15]Daimyō are classified according to their relationship with the shogunate: *shinpan* (relatives), *fudai* (hereditary vassals who fought for the shogunate at Sekigahara in 1600), and *tozama* (outside lords who became allies only after the battle).

^[16]Population is measured at the end of the Tokugawa period, so variables that use it should be read as late-period proxies rather than time-invariant characteristics.

^[17]Fukui Prefecture (1994: 335–346) documents this practice in Fukui Domain.

^[18]The measure is reported collections relative to reported output, not the ruler’s chosen extraction rate. It absorbs accounting conventions, regional assessment traditions, incentives when reporting to the new Meiji government, and late-period fiscal disturbances. I therefore treat the tax results as associational and emphasize neighbor heterogeneity rather than the pooled coefficient.

and collections over the preceding five years. Averaging reduces harvest noise, while widespread *jōmen-hō* (fixed-rate taxation) kept many domains' rates administratively stable in the final Tokugawa decades (Iinuma 1974, 20–24; Smith 1958).

Because *shunodaka* and *uchidaka* cover 1865–69, they measure a late-period average rather than annual change. This window remains informative: *jōmen-hō* fixed nominal rates across decades (Smith 1958), limiting year-to-year variation, while the fiscal argument concerns long-run extraction rather than fiscal shocks. Mid-Tokugawa tax scatters in Smith (1958) are qualitatively consistent with the late-period cross-section, but comprehensive mid-period domain data are unavailable. Tax rates also varied within domains, yet systematic village-level data do not survive. The analysis therefore estimates domain-average relationships and cannot test discriminatory village-level tax schedules. [19]

I exclude non-daimyō territories, including shogunal *goryō* and shrine- or temple-administered *jisharyō*, because they lack comprehensive fiscal data and had limited tax autonomy (Murakami 1996). ρ_j averages the village distance ratio from Section 3.4.1 within each domain using end-period capitals.

3.5.2. Taxation and Neighboring Outside Options

The pooled benchmark regresses domain tax rates on exposure and controls:

$$\tau_j = \alpha_4 + \beta_4 \ln(\rho_j) + X_j' \gamma_4 + \varepsilon_j$$

Here τ_j is domain j 's tax rate, ρ_j is its average village distance ratio, and X_j contains three control groups. Core controls are logged territory size, population, and wet-rice suitability,^[20] the number of new-farmland villages, shogunal alliance, and the gōshi system. Geographic controls are capital latitude and longitude and their interaction, logged elevation, ruggedness, and distance to the Edo–Kyoto axis, and the monthly

^[19]Public goods could in principle substitute for tax relief, making tax rates an incomplete measure of oppression. Yet public works required corvée labor (*bueki*) from villages and thus added to their burdens, while petitions and resistance records center on excessive taxes and unfair assessments rather than missing infrastructure (Aoki 1975).

^[20]The wet-rice measure is an agro-climatic potential-yield surface (Galor and Özak 2016) rather than an observed annual output series; I use it only as a proxy for labor-absorption capacity.

shares of rainy and snowy days.^[21] Neighbor controls are logged weighted means of adjacent domains' territory, population, and rice productivity, the weighted number of new-farmland villages, and the share of neighbors with family ties to the focal domain; weights count the focal villages for which each domain is the nearest neighbor.

Table 4 reports a negative cross-sectional association between exposure and reported tax rates. Across Models 1–4, a one-standard-deviation increase in the log distance ratio ($SD = 1.34$) corresponds to a 1.5–3.2 percentage-point decrease. The association remains negative with the full set of controls and in a pure spatial-lag model, although it is only marginal in the fully controlled spatial-lag model (Anselin and Rey 2014) (Models 5–6). Its direction matches Steele, Paik, and Tanaka (2017), who find that collective desertions, but not the broader category of insurrection, predict lower taxes at the domain-period level.

Several checks assess whether model choice or omitted variables account for the pooled association. The sign remains negative with alternative covariates, including kokudaka, samurai share, and population relative to wet-rice suitability (SI E.2.1). Applying the sensitivity analysis of Diegert, Masten, and Poirier (2022) shows that, when exposure- and outcome-side confounding are bounded equally, both SD ratios must reach about 0.554 before the identified set admits a nonnegative coefficient (SI E.2.2). Double/debiased machine learning (DML), following Chernozhukov et al. (2018), tempers the result: ridge nuisance models retain negative estimates, but flexible stacked and random-forest specifications produce estimates near zero with wide confidence intervals (SI E.2.3). The sign also remains negative when daikansho enter the choice set (SI D.3.1). Thus the pooled OLS association is stable across conventional specifications but is not uniformly corroborated by flexible DML; the neighbor-heterogeneity test below asks whether the OLS pattern reflects exit rather than state-capacity decay.

To test the exit channel, I interact exposure with $\hat{y}_{-j}$, a discrete measure of neighboring domains' outside-option quality:

^[21] Weather variables represent proportion of rainy/snowy days per month, imputed as weighted averages using daily weather data from 40 locations, with distance to domain capital as weight. Data from [Historical Weather Database](#) by Professor Minoru Yoshimura.

Outcome:	τ_j					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
Distance ratio ($\ln(\rho_j)$)	−0.0241	−0.0203	−0.0110	−0.0112	−0.0140	−0.0094
HC SE	(0.0053)***	(0.0058)***	(0.0055)**	(0.0055)**	(0.0047)***	(0.0051)*
Conley SE (70km)	(0.0066)***	(0.0067)***	(0.0056)**	(0.0055)**	(0.0058)**	(0.0050)*
<i>Controls</i>						
Core	No	Yes	Yes	Yes	No	Yes
Geography	No	No	Yes	Yes	No	Yes
Neighbors	No	No	No	Yes	No	Yes
τ_j Spatial Lag	No	No	No	No	Yes	Yes
<i>Fit Statistics</i>						
Num.Obs.	232	230	230	230	230	230
R2	0.074	0.141	0.370	0.389	0.312 ^[a]	0.439 ^[a]
Mean Dep. Var.	0.392	0.394	0.394	0.394	0.394	0.394

Table 4: Pooled Benchmark: Domain-Level Geographic Exposure and Tax Rates

Note: OLS estimates (ML in Models 5–6) of reported effective tax rates (collections divided by reported output) on the log domain-average distance ratio. Full results in Table A.50. SEs in parentheses: (i) HC1 (ML SEs in 5–6); (ii) Conley (70 km) on capital coordinates, computed via two-step filtered-residual approximation (Anselin and Rey 2014, pp. 34–36); stars reflect that estimator’s p -value. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1. ^[a] Nagelkerke pseudo-R2

$$\tau_j = \alpha_5 + \beta_5 \ln(\rho_j) + \delta_5 \ln(\rho_j) \mathbb{1}_{\{\hat{y}_{-j} = \text{High}\}} + X_j' \gamma_5 + \varepsilon_j$$

I proxy outside-option quality with neighboring domains’ demand for labor. Labor-scarce rulers attracted workers through land redistribution, tools and rations, irrigation, and other public goods (Hyugano 1957; Miyazaki 1995), with stronger incentives under greater scarcity. Persistent shortages in Kantō led domains to adopt peasant recruitment policies (*iri-byakushō seisaku*) offering tax exemptions, food and seed grants, and other settlement inducements (Gorai 1950).

The moderator operationalizes this labor-absorption capacity. $\hat{y}_{-j}$ equals “Low” when neighbors’ weighted average labor-to-productivity ratio is above its median and “High” otherwise.^[22] I weight each neighbor by the number of domain j villages for which its

^[22] Hainmueller, Mummolo, and Xu (2019) show that continuous multiplicative-interaction models rest on a linear-interaction assumption that frequently fails in practice, and recommend replacing the continuous moderator with a discrete one precisely for settings like this, in which the moderating relationship is inherently nonlinear.

capital is nearest and divide its population by potential wet-rice yield from [Galor and Özak \(2016\)](#).^[23] High $\hat{y}_{-j}$ thus denotes neighbors with greater capacity and incentives to absorb migrants. The full interaction specification retains neighboring rice yield and territory size among the controls in X_j .

The interaction probes the retention mechanism. Exposure should induce the home ruler to moderate extraction most strongly when a labor-scarce neighbor can credibly absorb departing cultivators. A countervailing possibility is that departure reduces the number of villagers available for simultaneous confrontation, weakening pressure on the home ruler when the neighboring jurisdiction is unattractive. The interaction does not identify this channel or imply that exit and confrontation are substitutes across a longer sequence. Its observable prediction is that the negative tax gradient should be strongest where absorption is credible.

Exposure is negatively associated with tax rates in the High-outside-option group across all four specifications. With full controls (Model 4), the High-group slope remains statistically significant at -0.0158 ($p = 0.023$), compared with -0.0085 ($p = 0.182$) in the Low group. The consistently more negative High-group point estimates accord with the retention mechanism, although the direct slope contrast is imprecise ($p = 0.323$). These cross-sectional associations provide suggestive fiscal evidence, subject to domain- and region-level confounding. The Appendix repeats the interaction analysis using neighboring domains' population density as an alternative proxy for labor-absorption capacity ([SI E.3.3](#)).

3.5.3. Additional Analyses

The border-distance null also appears in the tax analysis: distance to the nearest border does not significantly predict tax rates, while capital-based exposure remains negatively associated with them ([SI E.3.2](#)).^[24] Together with the village result in §3.4, this pattern supports the importance of access to administrative authority for protection and mediation. Adding the log of average home-capital distance, the predictor implied

^[23]I use the potential yield measure under irrigated conditions with medium inputs.

^[24]The boundary data were constructed following the procedure described in ([Honda, Natsume, and Nemoto 2022](#)).

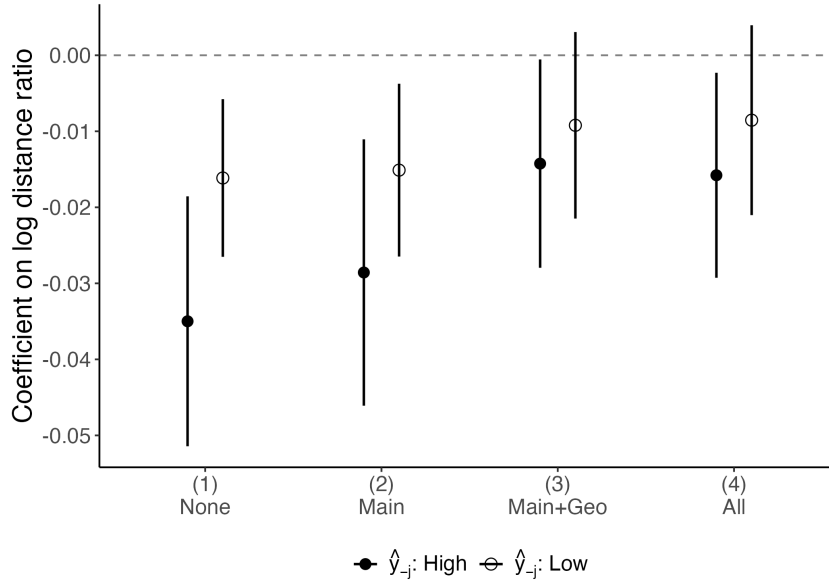


Figure 7: Tax-Exposure Slopes by Outside-Option Quality

Note: Group-specific OLS slopes on $\ln(\rho_j)$. Filled points marked High denote below-median neighbor labor-to-potential-yield ratios, hence greater capacity and incentive to absorb migrants; open points marked Low denote above-median ratios. Lines are 95% heteroskedasticity-consistent confidence intervals. Controls are cumulative: (1) none; (2) domain; (3) +geography; (4) +neighbor. Full table in [Table A.51](#).

by state-capacity decay ([Sng and Moriguchi 2014](#)), also leaves the exposure coefficient negative ([SI E.2.4](#)); replacing the ratio with remoteness produces no significant negative association once geographic controls enter ([SI E.3.1](#)).

4. CONCLUSION

The four designs recover successive links between outside access, exit, tactic choice, and extraction. Tenryō conversion and the grid-year panel associate improved outside access with more recorded exit. Among resisting villages, proximity to a rival capital supplied the more robust component of the shift toward exit. Exposure is associated with lower tax rates, and the estimated gradient is more negative where neighbors could absorb migrants, although the difference between groups is imprecise. The behavioral evidence supports the role of outside access in communities' use of exit; the fiscal results remain suggestive rather than causal evidence of ruler moderation.

The paper explains how communities can use alternative authorities to challenge extraction within a common hierarchy. Prohibited migration and attachment to land

did not preclude temporary departure: shelter could sustain absence while villagers sought concessions and a return home. Hierarchical oversight added pressure when the superior accepted the grievance, but could also defend fiscal authority. The quantitative evidence supports the behavioral role of outside access; the cases illustrate the distinct contributions of shelter and oversight. Neither the interaction between these routes nor their long-run institutional consequences is separately identified.

The mechanism of discipline through a credible outside option suggests two applications beyond the case. In civil wars with rebel governance, civilians may shift support and resources between governments and insurgent authorities; rival protection can constrain predation only where another armed actor can govern and protect those who move (Kalyvas 2006). In competition for foreign investment, firms rather than villages are mobile, and governments face discipline when another jurisdiction offers a credible productive location (Acharya and Lee 2023; Kleven et al. 2020). The Tokugawa designs identify neither relationship. They show why both analogies require an alternative authority that mobile actors can reach and reasonably expect to receive them.

DATA AVAILABILITY STATEMENT

Replication data and code will be deposited in the International Organization Dataverse (<https://dataverse.harvard.edu/dataverse/IOJ>) upon conditional acceptance. The release will include the documented intermediate geographic inputs, scripts, and codebook needed to reproduce the reported tables and figures.

ETHICAL STATEMENT

This research uses publicly available historical records and archival data on Tokugawa-era Japan and does not involve human subjects. No IRB review was required.

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A. FORMAL ANALYSIS

A.1. Setup

This model isolates the horizontal outside-option mechanism: access to a neighboring domain changes villages' resistance choices and rulers' tax incentives. Receiving rulers' decisions to shelter or admit villagers, superior intervention, post-exit bargaining, and the terms of return are outside the game. The model therefore supports the behavioral and fiscal implications of outside access; the institutional argument about temporary protection and hierarchical enforcement is developed in the main text.

Consider two neighboring domains $j \in \{1, 2\}$. All primitives are common knowledge. Each ruler governs a continuum of villages of measure one and simultaneously chooses a common domain-level tax component $\tau_j \in [0, 1]$. A single rate matches the domain-average empirical estimand but excludes schedules keyed to observed village exit costs. Rulers cannot make side payments. The ruler equilibrium is Nash, and an enforceable tax cartel lies outside the model.

The timing is: (i) rulers simultaneously set taxes; and (ii) villages observe both rates and choose $a_{i_j} \in \{\text{comply}, \text{fight}, \text{exit}\}$. Ordinary petitioning is outside the game; the village choice represents the response after petitioning fails to change the tax demand. Here, *fight* is the model's compact label for direct confrontation. Let μ_j^a denote the share choosing action a , and let each village in domain j produce yield $y_j > 0$.

A compliant village receives $y_j(1 - \tau_j) + \varepsilon_{i_j}$, where $\varepsilon_{i_j} \sim \text{Uniform}[-\delta, \delta]$ within each exit-cost group and $\delta > 0$. A fight succeeds with probability μ_j^{fight} , yields y_j if successful and zero otherwise, and costs $k > 0$. Exit yields the neighboring domain's net-of-tax income less village-specific exit cost: $y_{-j}(1 - \tau_{-j}) - c_{i_j}$, where $c_{i_j} \in \{c_L, c_H\}$ and $c_H > c_L > 0$. The share $\theta_j \in [0, 1]$ with cost c_L measures geographic exposure to a rival capital; the remaining $1 - \theta_j$ face cost c_H .

Exit removes labor and payments from the home tax base over the period represented by the payoffs. Neighboring net income values the material outside option during that period. Whether villagers subsequently return or settle elsewhere is not determined by the model.

Village utility is

$$u_{i_j}(a_{i_j}; \tau_j, \tau_{-j}) = \begin{cases} y_j(1 - \tau_j) + \varepsilon_{i_j} & \text{if } a_{i_j} = \text{comply} \\ y_j \mu_j^{\text{fight}} - k & \text{if } a_{i_j} = \text{fight} \\ y_{-j}(1 - \tau_{-j}) - c_{i_j} & \text{if } a_{i_j} = \text{exit} \end{cases}.$$

Ruler j collects taxes only from compliant villages and maximizes

$$u_j(\tau_j, \tau_{-j}) = [1 - \mu_j^{\text{fight}} - \mu_j^{\text{exit}}] y_j \tau_j = \mu_j^{\text{comply}} y_j \tau_j.$$

The analysis follows the village continuation in which high-exit-cost villages choose between compliance and fight, whereas low-exit-cost villages choose between compliance and exit. The following assumption makes this continuation an equilibrium after every tax pair. It does not rule out other continuations of the full coordination game; the comparative statics hold with this selection fixed.

ASSUMPTION 1: Let $0 < c_L < k < \delta$ and $0 < \theta_j < 1$. For each $j \in \{1, 2\}$,

$$\frac{k - c_L}{\delta - c_L} > \frac{y_j(1 - \theta_j)}{2\delta} \quad \text{and} \quad c_H > y_{-j} + \delta.$$

For the equilibrium comparative statics, the tax candidate derived below must satisfy $0 < \tau_j^* < 1$, $0 < \mu_j^{*, \text{fight}} < 1 - \theta_j$, and $0 < \mu_j^{*, \text{exit}} < \theta_j$ in both domains. These strict inequalities ensure interior cutoffs and persist under small parameter changes. The proof establishes global ruler best responses at such a candidate, including deviations that move a village cutoff to a boundary. Uniqueness is asserted only among interior candidates under the selected village continuation.

A.2. Equilibrium Results

PROPOSITION 1 (Resistance Composition under Neighboring-Option Changes):

Under [Assumption 1](#) and interior village cutoffs, for fixed tax rates $(\tau_j, \tau_{-j}) \in (0, 1)^2$:

- (i) $\frac{\partial \mu_j^{\text{exit}}}{\partial y_{-j}} > 0$ and $\frac{\partial \mu_j^{\text{fight}}}{\partial y_{-j}} = 0$; (ii) $\frac{\partial \mu_j^{\text{exit}}}{\partial \tau_{-j}} < 0$ and $\frac{\partial \mu_j^{\text{fight}}}{\partial \tau_{-j}} = 0$.

Only the exit payoff contains neighboring-domain quantities. The proposition is therefore a fixed-tax outside-option comparative static that motivates, but does not structurally reproduce, the conversion-based test: conversion is not a primitive in the model, and the empirical design does not identify a change in y_{-j} or τ_{-j} . Its zero fight derivatives are not a general

claim: comparison-based grievances or outside resources, shelter, and political pressure could instead raise confrontation. The empirical null concerns average incidence and does not rule out those channels in particular episodes.

PROPOSITION 2 (Geographic Exposure and Exit): Under [Assumption 1](#) and the stated interiority conditions: (a) whenever greater exposure lowers the home ruler's equilibrium tax, the equilibrium exit share among non-compliant villages is strictly increasing in θ_j ; and (b) the equilibrium exit rate is strictly increasing in θ_j regardless of the tax response.

Greater exposure expands the low-cost exit pool and contracts the high-cost fight pool. When the ruler cuts taxes, that response offsets only part of the resulting increase in exit because further tax reductions eventually sacrifice more revenue per compliant village than they recover through retention.

PROPOSITION 3 (Outside-Option Quality and Equilibrium Taxes): Greater exposure lowers the equilibrium tax exactly when the neighboring domain's equilibrium net income exceeds a threshold determined by home-domain primitives. Under [Assumption 1](#) and the stated interiority conditions, let $z_{-j}^* := y_{-j}(1 - \tau_{-j}^*)$. Then

$$\kappa_j := c_L - k + \frac{y_j(1 - \theta_j)[4\delta - y_j(1 - \theta_j)]}{2\delta - y_j(1 - \theta_j)^2},$$

$$\frac{\partial \tau_j^*}{\partial \theta_j} < 0 \quad \text{if and only if} \quad z_{-j}^* > \kappa_j.$$

The tax response is zero at equality and positive below the threshold. If $\kappa_j \leq 0$, it is negative throughout the interior regime.

Within the model's single-action benchmark, exposure tightens the exit constraint but also shrinks the contemporaneous pool available for direct confrontation. A sufficiently attractive net outside option makes the retention force dominate, so the ruler lowers taxes; below the threshold, confrontation relief permits higher taxes. The criterion accounts for both rulers' tax adjustments and is not a claim that gross neighboring productivity has a unique threshold. This event-level abstraction does not rule out the complementary use of exit and confrontation across a multi-stage protest sequence.

A.3. Equilibrium and Proofs

Write $[x]_0^1 := \min\{1, \max\{0, x\}\}$ and $D_j := 2\delta - y_j(1 - \theta_j) > 0$. The selected continuation has clipped resistance shares

$$\begin{aligned}\mu_j^{\text{fight}} &= (1 - \theta_j) \left[\frac{\delta - k - y_j(1 - \tau_j)}{D_j} \right]_0^1, \\ \mu_j^{\text{exit}} &= \theta_j \left[\frac{y_{-j}(1 - \tau_{-j}) - c_L - y_j(1 - \tau_j) + \delta}{2\delta} \right]_0^1.\end{aligned}$$

To verify this continuation, a high-cost village's exit payoff is below $-\delta$, whereas compliance pays at least $-\delta$. Given that low-cost villages do not fight, the high-cost fight share is the fixed point of $(1 - \theta_j) \left[\frac{\delta - k - y_j(1 - \tau_j) + y_j \mu_j^{\text{fight}}}{2\delta} \right]_0^1$. Assumption 1 implies $y_j(1 - \theta_j) < 2\delta$, so this map is a contraction. Its unique fixed point is the clipped expression above. Moreover, its maximal value, attained at $\tau_j = 1$, is bounded by $(1 - \theta_j) \frac{\delta - k}{D_j}$. Consequently,

$$y_j \mu_j^{\text{fight}} - k \leq \frac{y_j(1 - \theta_j)(\delta - k)}{D_j} - k \leftarrow c_L.$$

The strict inequality is equivalent to the first inequality in [Assumption 1](#). Low-cost exit pays at least $-c_L$, so low-cost villages strictly prefer exit to fight on this continuation. This verifies the selection without claiming that other village equilibria are absent.

When both cutoffs are interior, these shares become

$$\begin{aligned}\mu_j^{\text{fight}}(\tau_j) &= \frac{(1 - \theta_j)[\delta - k - y_j(1 - \tau_j)]}{2\delta - y_j(1 - \theta_j)}, \\ \mu_j^{\text{exit}}(\tau_j, \tau_{-j}) &= \frac{\theta_j}{2\delta} [y_{-j}(1 - \tau_{-j}) - c_L - y_j(1 - \tau_j) + \delta].\end{aligned}$$

A.3.1. Proof of Proposition 1

PROOF: Fix j and hold (τ_j, τ_{-j}) fixed. Differentiating the resistance-share expressions above gives

$$\frac{\partial \mu_j^{\text{exit}}}{\partial y_{-j}} = \theta_j \frac{1 - \tau_{-j}}{2\delta} > 0, \quad \frac{\partial \mu_j^{\text{exit}}}{\partial \tau_{-j}} = -\theta_j \frac{y_{-j}}{2\delta} < 0.$$

The fight expression contains neither y_{-j} nor τ_{-j} , so both corresponding derivatives are zero.

A.3.2. Equilibrium Tax Rates

Define

$$\begin{aligned} A_j &:= \frac{(1 - \theta_j)y_j}{2\delta - y_j(1 - \theta_j)} + \frac{\theta_j y_j}{2\delta}, \\ B_j &:= \frac{(1 - \theta_j)(\delta - y_j - k)}{2\delta - y_j(1 - \theta_j)} + \frac{\theta_j}{2\delta} [y_{-j} - c_L - y_j + \delta], \\ C_j &:= \frac{\theta_j y_{-j}}{2\delta}. \end{aligned}$$

Then total resistance is $B_j + A_j\tau_j - C_j\tau_{-j}$, and ruler utility is

$$u_j(\tau_j, \tau_{-j}) = y_j\tau_j [1 - B_j + C_j\tau_{-j} - A_j\tau_j].$$

On this interior segment, $\frac{\partial^2 u_j}{\partial \tau_j^2} = -2y_j A_j < 0$, and the first-order condition is

$$\tau_j = p_j + q_j\tau_{-j}, \quad p_j = \frac{1 - B_j}{2A_j}, \quad q_j = \frac{C_j}{2A_j}.$$

Moreover, $A_j \geq \theta_j \frac{y_j}{2\delta}$ implies $q_j \leq \frac{y_{-j}}{2y_j}$ and hence $q_1 q_2 \leq \frac{1}{4} < 1$. The two interior first-order conditions therefore have the unique candidate

$$\tau_j^* = \frac{p_j + q_j p_{-j}}{1 - q_j q_{-j}}.$$

This interior candidate is a global best response for each ruler under the selected continuation.

To see this, hold the other ruler at its candidate tax. Positive fight in the neighboring domain implies $z_{-j}^* < \delta - k$. Hence the low-cost exit cutoff is below one for every home tax in $[0, 1]$.

Assumption 1 also implies

$$y_j(1 - \theta_j) < \frac{2\delta(k - c_L)}{\delta - c_L} < \delta + k,$$

where the last inequality follows from $(\delta - k)(\delta + c_L) > 0$. The high-cost fight cutoff is therefore below one at every home tax as well. Each resistance share is consequently a positive multiple of the maximum of zero and an affine, increasing function of τ_j . On every smooth segment, ruler revenue has second derivative $-2y_j$ times the sum of the active resistance slopes, which is nonpositive. At each cutoff's activation, its first derivative jumps downward by $y_j\tau_j$ times the newly active slope. Revenue is thus globally concave on $[0, 1]$ and strictly concave around the interior candidate. The first-order condition identifies the unique global best response to the other candidate tax. Applying this argument to both rulers establishes

the candidate as a Nash equilibrium under the selected continuation; it does not exclude equilibria involving other continuations or boundary cutoffs.

A.3.3. Proof of Proposition 2

PROOF: At fixed tax rates, $\frac{\partial \mu_j^{\text{exit}}}{\partial \theta_j} = \frac{\mu_j^{\text{exit}}}{\theta_j} > 0$. Writing $D_j = 2\delta - y_j(1 - \theta_j) > 0$ and $N_j = \delta - k - y_j(1 - \tau_j) > 0$, differentiation gives $\frac{\partial \mu_j^{\text{fight}}}{\partial \theta_j} = -2\delta \frac{N_j}{D_j^2} < 0$.

First suppose $\frac{d\tau_j^*}{d\theta_j} \geq 0$. The other ruler's interior first-order condition gives $\frac{d\tau_{-j}^*}{d\theta_j} = q_{-j} \frac{d\tau_j^*}{d\theta_j} \geq 0$. Let $W_j = y_j - y_{-j}q_{-j}$. Since $A_{-j} \geq \theta_{-j} \frac{y_{-j}}{2\delta}$, $W_j \geq \frac{y_j}{2} > 0$. For $E_j = y_{-j}(1 - \tau_{-j}^*) - c_L - y_j(1 - \tau_j^*) + \delta$, therefore, $\frac{dE_j}{d\theta_j} = W_j \frac{d\tau_j^*}{d\theta_j} \geq 0$, and

$$\frac{d\mu_j^{*,\text{exit}}}{d\theta_j} = \frac{E_j}{2\delta} + \theta_j \frac{W_j}{2\delta} \cdot \frac{d\tau_j^*}{d\theta_j} > 0.$$

Now suppose $\frac{d\tau_j^*}{d\theta_j} < 0$. At the optimum, compliance equals $A_j \tau_j^*$. Because $A_j' < 0$,

$$\frac{d(\mu_j^{\text{fight}} + \mu_j^{\text{exit}})}{d\theta_j} = -A_j' \tau_j^* - A_j \frac{d\tau_j^*}{d\theta_j} > 0.$$

Meanwhile,

$$\frac{d\mu_j^{\text{fight}}}{d\theta_j} = \frac{[-2\delta N_j + (1 - \theta_j)y_j D_j \frac{d\tau_j^*}{d\theta_j}]}{D_j^2} < 0.$$

Thus exit increases because total resistance increases while fight decreases, proving part (b). For part (a), the quotient-rule numerator for the exit share is $\left(\frac{d\mu_j^{*,\text{exit}}}{d\theta_j}\right) \mu_j^{*,\text{fight}} - \mu_j^{*,\text{exit}} \left(\frac{d\mu_j^{*,\text{fight}}}{d\theta_j}\right) > 0$.

A.3.4. Proof of Proposition 3

PROOF: Separate the neighboring net income from the home ruler's coefficients.

Define

$$\tilde{B}_j := \frac{(1 - \theta_j)(\delta - y_j - k)}{2\delta - y_j(1 - \theta_j)} + \frac{\theta_j}{2\delta}(\delta - c_L - y_j), \quad \tilde{p}_j := \frac{1 - \tilde{B}_j}{2A_j}.$$

Then $B_j = \tilde{B}_j + \theta_j \frac{y_{-j}}{2\delta}$, so $p_j = \tilde{p}_j - q_j$. The two equilibrium first-order conditions can be written exactly as

$$\tau_j^* = \tilde{p}_j - q_j(1 - \tau_{-j}^*), \quad \tau_{-j}^* = p_{-j} + q_{-j}\tau_j^*.$$

When θ_j changes, p_{-j} and q_{-j} are fixed because they depend on the other ruler's exposure and the two yields, not on θ_j . They are not fixed when y_{-j} changes.

Let a prime denote differentiation with respect to θ_j . We have

$$A'_j = \frac{y_j}{2\delta} - \frac{2\delta y_j}{D_j^2} < 0, \quad K_j := \frac{A_j - \theta_j A'_j}{4\delta A_j^2} > 0, \quad q'_j = y_{-j} K_j.$$

Differentiating both equilibrium conditions, substituting $(\tau_{-j}^*)' = q_{-j}(\tau_j^*)'$, and collecting terms gives

$$(1 - q_j q_{-j})(\tau_j^*)' = \tilde{p}'_j - q'_j(1 - \tau_{-j}^*) = \tilde{p}'_j - K_j z_{-j}^*.$$

This retains the other ruler's endogenous tax response. Since $1 - q_j q_{-j} > 0$, the sign is determined by $\frac{\tilde{p}'_j}{K_j} - z_{-j}^*$.

To calculate the threshold, write $a_j := 1 - \theta_j$ and $H_j := 2\delta - y_j \theta_j a_j > 0$. Combining terms in A_j gives $A_j = y_j \frac{H_j}{2\delta D_j}$, and direct differentiation yields

$$K_j = \frac{\delta(2\delta - y_j a_j^2)}{y_j H_j^2},$$

$$\tilde{p}'_j = \frac{\delta[(c_L - k)(2\delta - y_j a_j^2) + y_j a_j(4\delta - y_j a_j)]}{y_j H_j^2}.$$

Thus $\frac{\tilde{p}'_j}{K_j} = \kappa_j$, with κ_j as stated in the proposition, and

$$\frac{\partial \tau_j^*}{\partial \theta_j} = \frac{K_j}{1 - q_j q_{-j}}(\kappa_j - z_{-j}^*).$$

The multiplier is strictly positive, proving all three sign cases. If $\kappa_j \leq 0$, then $z_{-j}^* > 0 \geq \kappa_j$, so the derivative is strictly negative. The comparison is local in exposure along the interior equilibrium: no claim of existence or uniqueness of a cutoff in gross neighboring productivity is required.

A.4. Non-Vacuousness

Both signs of the tax response occur on non-empty open sets. For tax restraint, take $\delta = 10$, $y_1 = y_2 = 18$, $k = \frac{7}{2}$, $c_L = \frac{1}{2}$, $c_H = 29$, and $\theta_1 = \theta_2 = \frac{5}{6}$. The interior coefficients are $A_j = \frac{63}{68}$, $B_j = \frac{77}{272}$, $C_j = \frac{3}{4}$, $p_j = \frac{65}{168}$, and $q_j = \frac{17}{42}$. The equilibrium has $\tau_j^* = \frac{13}{20}$, $\mu_j^{*,\text{fight}} = \frac{1}{510}$, and $\mu_j^{*,\text{exit}} = \frac{19}{48}$. Neighboring net income is $z_{-j}^* = \frac{63}{10}$, exceeding $\kappa_j = \frac{35}{13}$, and $\frac{\partial \tau_j^*}{\partial \theta_j} = -\frac{5628}{36875} < 0$.

For a tax increase, take $\delta = 10$, $y_1 = y_2 = 12$, $k = 7$, $c_L = \frac{1}{2}$, $c_H = 23$, and $\theta_1 = \theta_2 = \frac{1}{3}$. Then $\tau_j^* = \frac{7}{8}$, $\mu_j^{*,\text{fight}} = \frac{1}{12}$, and $\mu_j^{*,\text{exit}} = \frac{19}{120}$. Here $z_{-j}^* = \frac{3}{2} < \kappa_j = \frac{241}{22}$, so greater exposure raises the tax. In both examples, all primitive and interiority inequalities are strict, and the global-best-response argument above applies. Continuity therefore preserves the respective signs and equilibrium conditions on open neighborhoods of these parameter vectors.

B. DESCRIPTIVE STATISTICS

B.1. Village Resistance Data

	Unique	Missing %	Mean	SD	Min	Median	Max	Histogram	Source
Basic									
Home Ruler	168	0	—	—	—	—	—	—	Aoki (1975)
Nbr Ruler	224	0	—	—	—	—	—	—	Aoki (1975)
Year	218	0	1,777.3	59.0	1,615.0	1,782.0	1,867.0		Aoki (1975)
Decade	26	0	1,772.7	59.2	1,610.0	1,780.0	1,860.0		Aoki (1975)
Outcome									
Exit	2	0	0.2	0.4	0.0	0.0	1.0		Aoki (1975)
Direct Confrontation	2	0	0.8	0.4	0.0	1.0	1.0		Aoki (1975)
Predictor									
ρ_i	2,529	0	7.1	61.3	0.0	1.0	2,792.0		—
$\ln(\rho_i)$	2,529	0	0.2	1.5	-5.6	0.0	7.9		—
Dist. to Home Cap	2,517	0	45.8	85.5	0.0	21.2	1,096.0		—
Dist. to Nbr Cap	2,508	0	24.0	29.7	0.0	16.4	494.5		—
ρ_i^{LCP}	168	0	7.2	28.1	0.2	1.7	601.7		—
Controls									
Latitude	2,458	0	35.7	2.0	27.7	35.4	42.0		Honda, Natsume, and Nemoto (2022) and other sources
Longitude	2,445	0	136.5	3.3	128.9	136.6	142.0		Honda, Natsume, and Nemoto (2022) and other sources
Dist. to Edo	2,484	0	413.6	235.6	29.4	413.2	1,356.8		—
Dist. to Edo or Kyoto	2,484	0	234.0	152.4	0.0	191.1	1,041.7		—
Elevation	103	0	19.9	20.9	0.0	12.0	108.0		Geospatial Information Authority of Japan
Ruggedness	146	0	3.7	3.8	0.0	2.5	22.6		—
Wet-Rice Suitability	716	0	18,549.9	3,417.3	0.0	19,003.2	22,570.0		Galor and Özak (2016)
Avg. Caloric Suitability	1,133	0	2,221.3	242.7	941.1	2,297.5	2,531.2		Galor and Özak (2016)

Table A.5: Summary Statistics of the Village Resistance Data ($N = 2,741$). The distance ratios and capital distances summarize the stored source-data measures. The preferred regressions reconstruct the nearest active foreign capital at each event date using augmented domain spells, so their exposure measure and estimation sample differ from these descriptive statistics.

B.2. Source-Category and Actor Audit

Source section	Printed pages	PDF pages	OCR start	OCR end	Used in audit
Peasant uprisings	7–364	9–366	51	23323	Yes
Urban disturbances	365–410	367–412	23324	25841	No
Village conflicts (murakata sodo)	411–620	413–622	25842	36731	No
Province-wide petitions	621–636	623–638	36732	37490	No

Table A.6: Source-Section Boundary in Aoki’s Chronology

Note: Aoki prints separate chronologies. The project comparison script reads OCR lines 51–23,324 (lines[50:23324] in Python’s zero-based indexing); the last included line is the heading of the next section, so substantive comparison content ends at line 23,323. The separately printed *murakata sōdō* chronology begins at PDF p. 413 (printed p. 411; OCR line 25,842) and is outside that window. Page boundaries were checked against the PDF’s form-feed markers and title pages. This audit establishes exclusion of the separate source category, not the household tenure status of every actor named within included peasant-uprising entries. Source: (Aoki 1975).

Sample	Estimate	Cluster p	Conley p	Events	Rows
Baseline	0.0244	0.0013	0.0041	909	2739
Exclude tenure-keyword events	0.0218	0.0008	0.0113	890	2672

Table A.7: Village-Choice Sensitivity to Tenure-Related Narratives

Note: The restricted sample removes entire resistance events whose demand narrative contains *kosaku* (tenant), *jinushi* (landlord), or *komae* (smallholder). This excludes nineteen events and sixty-seven village rows, only one of which is coded as exit. Both rows use the preferred home-neighbor dyad and decade fixed effects with the main geographic controls. Cluster inference is three-way by home ruler, neighboring ruler, and decade; Conley inference uses a 10 km cutoff. The screen is a conservative content proxy, not a household-level tenure classification.

B.3. Recorded Return after Exit

ID	Year	Home	Village	Recorded end	Aoki p.
809	1759	松山	久妙寺	Returned after reconciliation	123
2822	1788	宇和島	惣川村西	Required to return; later punishment	166
801	1800	松山	波万	Persuaded and returned that day	181
2823	1813	宇和島	岩木	Domain persuasion; return and imprisonment	196
2922	1818	吉田	俵津蒲	Returned on the twenty-fifth	204
2924	1820	吉田	奥野川	Stopped or persuaded; return and repression	207
2824	1827	宇和島	富野川	Persuaded to return; later exile or relocation	218

Table A.8: Exit Events Explicitly Recording Return

Note: These seven analytic-sample narratives explicitly record return after flight or attempted flight. They establish that some recorded exits were temporary, but they do not identify a representative return rate because Aoki did not code duration or final disposition systematically. Only ID 809 states that return followed reconciliation; the other entries record persuasion, an order to return, administrative transfer, or subsequent punishment. Source: (Aoki 1975).

B.4. Audit of Destination Codes

The destination field provides incomplete information about where villagers went or whom they petitioned. This section reports its full distribution, coverage, and the treatment of repeated village records. The underlying incident descriptions come from (Aoki 1975).

Sample and counting units. The source file contains 776 records coded as exit. Excluding 11 records dated before 1615 leaves 765 village records dated 1615–1868; no exit-coded record has a missing year or a year after 1868. These records represent 328 unique events. Panel A of Table A.9 collapses records sharing the same incident identifier to one event, while Panel B retains the village records, so events involving more recorded villages receive greater weight. The denominators include every exit-coded record or event in this period, including attempted actions and those without a destination label; they do not depend on completeness of regression covariates.

Tactic	Foreign	Edo shogunate	Kyoto	Other	Uncoded	Total
<i>Panel A: Unique events</i>						
Desertion	55	1	0	0	126	182
End-run appeal	7	112	1	0	26	146
Total	62	113	1	0	152	328
<i>Panel B: Village records</i>						
Desertion	86	1	0	0	222	309
End-run appeal	18	365	1	0	72	456
Total	104	366	1	0	294	765

Table A.9: Destination Codes Are More Complete for End-Run Appeals Than for Desertions

Note: Counts cover exit-coded records dated 1615–1868. Panel A counts each `revolt_id` once; Panel B retains every village record. Column headings report the recorded destinations; Edo denotes the Edo shogunate. Other includes nonmissing labels outside Foreign, Edo, and Kyoto; Uncoded denotes a missing label. Each tactic includes attempted actions, consistent with the pooled exit outcome in the main analyses. Every category, including zero counts, is displayed. Source: (Aoki 1975).

Coverage and missingness. A nonmissing label is available for 176 of 328 events (53.7%) and 471 of 765 village records (61.6%). The remaining 152 events and 294 records are uncoded. Coverage differs substantially across tactics: 56 of 182 desertion events (30.8%) and 120 of 146 end-run appeals (82.2%) have a label. Among coded events, 55 of 56 desertions carry Foreign, while 112 of 120 end-run appeals are coded as directed to the Edo shogunate. These proportions describe the coded subset; the unequal coverage prevents treating them as destination shares for all exits. Missing labels indicate unavailable coding, not evidence that no destination existed or that movement remained within the home domain.

Classification and event consistency. The report retains the existing exit indicator and groups actions by their source codes. Desertion includes 逃散 and attempted desertion (逃散未遂); end-run appeal includes 越訴 and attempted appeal (越訴未遂). This follows the main analyses, which count both attempted forms as exit. All events have a recorded action type. The two attempted desertions and six attempted appeals remain identifiable in the event register. Destination labels are standardized for letter case and grouped into Foreign, Edo, Kyoto, Other, and Uncoded. Edo denotes the Edo shogunate. There are no Other labels in the retained sample. Before collapsing records, I compare all nonmissing tactic, destination, and year values within each incident identifier. None of the 328 events has conflicting values or a mixture of missing and nonmissing values for these fields. Consequently, the collapse does not select one destination or tactic over another or fill missing village-level codes from another village. The accompanying event register retains all 328 identifiers, record counts, original labels, domain names, narratives, and source-row references, including uncoded events.

Attempted movement. The narratives also distinguish intentions from completed journeys. The 1853 Morioka entry describes an intended appeal to Sendai that was stopped at Miyako (920); the 1853 Sendai entry describes villagers stopped while heading toward Sendai (2274). A nonmissing label therefore cannot establish that the participants reached the named destination, entered another ruler's territory, or received protection there.

Implications for the analysis. The table reports destinations among events with recorded information. Missing destinations limit its coverage, and the codes alone do not establish that villagers entered territory converted to direct shogunal rule. The separate desertion-only robustness analyses use the narrower derived subclass, which excludes attempted desertions; the descriptive counts here include attempts to match the main pooled exit outcome.

B.5. Domain Data

	Unique	NA %	Mean	SD	Min	Median	Max	Hist.	Source
Outcome									
Exits	23	0	2.2	5.5	0.0	0.0	37.0		Aoki (1975)
Direct Confrontations	39	0	7.0	14.4	0.0	1.0	164.0		Aoki (1975)
Resistance Events	45	0	9.2	17.7	0.0	2.0	201.0		Aoki (1975)
Exit Rate	56	40	0.3	0.4	0.0	0.1	1.0		Aoki (1975)
Tax Rate (τ_j)	256	7	0.4	0.1	0.1	0.4	0.7		Kodama and Kijima (1989)
Predictor									
ρ_j	241	12	7.3	15.3	0.1	2.2	138.5		—
$\ln(\rho_j)$	241	12	1.0	1.3	-2.4	0.8	4.9		—
Dist. to Home Cap	241	12	49.0	66.7	2.5	22.4	442.5		—
Dist. to Nbr Cap	242	12	16.1	10.8	3.9	12.5	80.2		—
ρ_j^{LCP}	242	12	239.0	1,988.8	0.1	2.7	24,956.3		—
Controls: Main									
Wet-Rice Suitability	242	12	3,011,700.3	4,937,787.1	48,040.8	1,232,259.1	41,226,133.2		Galor and Özak (2016)
Territory Size	258	6	811.0	1,785.9	9.5	257.1	16,262.8		Honda, Natsume, and Nemoto (2022) and other sources
Population	255	7	87,781.0	159,087.3	2,466.0	30,833.0	1,065,910.0		LFAT
Pop. Density	249	9	166.5	125.1	11.9	150.2	1,285.0		—
# New Land	32	12	7.8	27.9	0.0	1.0	232.0		—
Shogunate Kin (0/1)	2	0	0.6	0.5	0.0	1.0	1.0		Kimura, Fujino, and Murakami (2015)
Goushi System (0/1)	2	0	0.1	0.3	0.0	0.0	1.0		Kimura, Fujino, and Murakami (2015)

Table A.10: Summary Statistics of the Domain Data ($N = 275$)

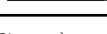
	Unique	Missing %	Mean	SD	Min	Median	Max	Histogram	Source
Controls: Geography									
Latitude	265	1	35.5	1.7	31.6	35.3	41.4		Kimura, Fujino, and Murakami (2015)
Longitude	265	1	136.5	3.3	128.8	136.6	141.5		Kimura, Fujino, and Murakami (2015)
Mean Elevation	242	12	16.1	13.6	2.6	12.0	88.4		—
Mean TRI	240	12	3.2	1.8	0.2	2.9	8.9		—
Dist. to Edo or Kyoto	242	12	188.5	148.9	25.2	135.0	671.3		—
% Rainy Days	251	3	0.1	0.0	0.1	0.1	0.2		Yoshimura HWDB ^[25]
% Snowy Days	251	3	0.0	0.0	0.0	0.0	0.1		Yoshimura HWDB
Controls: Neighbor									
Prop. Same Fam.	65	12	0.1	0.2	0.0	0.0	1.0		Kimura, Fujino, and Murakami (2015)
Avg. Count of New-Farmland Villages	78	12	32.9	69.2	0.0	7.5	421.0		—
Avg. Wet-Rice Suitability	242	12	2,135,287.7	3,139,273.3	288,682.9	1,350,670.1	41,057,071.8		Galor and Özak (2016)
Avg. Territory Size	242	12	433.8	602.5	12.4	278.0	6,926.3		Honda, Natsume, and Nemoto (2022) and other sources
Avg. Population	239	13	56,178.5	84,147.0	6,264.6	31,723.2	1,061,486.4		LFAT
Avg. Pop. Density	239	13	175.5	107.1	39.9	161.0	1,028.7		—
Avg. Pop. Density (Prod.)	239	13	0.0	0.0	0.0	0.0	0.2		—
Controls: Alternative									
Samurai Pop.	230	16	70,945.3	134,463.8	614.0	21,519.5	896,808.0		Kimura, Fujino, and Murakami (2015)
# Villages	170	6	196.5	304.5	1.0	82.5	2,243.0		LFAT
Dist. to Edo	242	12	370.8	244.7	42.2	309.4	1,039.5		—

Table A.11: Summary Statistics of the Domain Data (Cont.)

^[25]Historical Weather Database made available online by Minoru Yoshimura: <http://tk2-202-10627.vs.sakura.ne.jp/>

C. EVENT STUDY ROBUSTNESS

C.1. Tenryo-Conversion Universe

Conversion was a general instrument of shogunal lordship, not a policy directed at neighboring villagers. Daimyō governed fiefs as invested vassals: the shogunate could revoke, reduce, or transfer their holdings, but could not tax the inhabitants while the territory remained daimyō land. When a ruling house failed to produce an heir or a lord lost his tenure through misconduct or political sanction, conversion put territory under shogunal authority and preserved its discretion to re-enfeoff the territory later. This explains why the same disposition followed heterogeneous internal triggers. Attainder and transfer also fell disproportionately on smaller *fudai* houses whose tenure the shogunate could alter most readily, rather than on domains selected because neighboring peasants had become unusually mobilized (Kimura, Fujino, and Murakami 2015; Kodama and Kitajima 1989; Ravina 1999, 17–21).

The treatment sample contains 49 events drawn from domain histories (Kimura, Fujino, and Murakami 2015; Kodama and Kitajima 1989). Table A.12 lists each event, its date, stated cause, and source locator. The source check corrects three event dates and removes six records assigned to the wrong domain, to a year without the coded transition, or to a different governing authority (SI C.2). All conversion estimates use this corrected sample.

The cause typology distinguishes succession and household matters, misconduct and political sanction, and administrative or strategic reorganization. These formal reasons describe the disposition of a ruling house; they do not establish that the surrounding region was free of unrest. Potential links to unrest are examined separately in SI C.3.

Year	Domain	Province	Formal reason	Local source
1617	Itabashi 板橋藩	下野	Administrative transfer	HD v.2 p.194 (PDF 210); HS p.102 (PDF 104)
1618	Nagamine 長峰藩	越後	Administrative relocation	HD v.3 p.105 (PDF 119)
1619	Musashi Komuro 武蔵小室藩	武蔵	No heir	HD v.2 p.632 (PDF 648)
1632	Omori 大森藩	近江	Succession restriction	HD v.5 p.30 (PDF 46); HS pp.258-259 (PDF 260-261)
1639	Fukozu 深溝藩	三河	Family partition and relocation	HD v.4 p.314 (PDF 330)
1642	Nariwa 成羽藩	備中	Administrative transfer	HD v.6 p.255 (PDF 271)
1645	Fuchu 府中藩	常陸	No heir	HD v.2 p.39 (PDF 55)
1650	Kaibara 柏原藩	丹波	No heir	HD v.5 p.429 (PDF 445); HS p.275 (PDF 277)
1653	Toyooka 豊岡藩	但馬	No heir	HD v.5 p.463 (PDF 479)
1656	Funai 府内藩	豊後	No heir	HD v.7 p.402 (PDF 418)
1657	Marugame 丸亀藩	讃岐	No heir	HD v.6 p.447 (PDF 463); chronology p.451 (PDF 467)
1658	Kakegawa 掛川藩	遠江	No heir	HD v.4 p.177 (PDF 193)
1658	Yashima 矢島藩	出羽	Testamentary family partition	HD v.1 p.389 (PDF 405)
1660	Sakura 佐倉藩	下総	Official misconduct	HD v.2 p.453 (PDF 469); Aoki printed p.33 (PDF 35)
1665	Fukumoto 福本藩	播磨	Family partition	HD v.5 p.530 (PDF 546); HS p.295 (PDF 297)
1665	Saijo 西条藩	伊予	Official dereliction	HD v.6 p.469 (PDF 485); Aoki printed p.38 (PDF 40)
1666	Miyazu 宮津藩	丹後	Household conflict and misgovernment	HD v.5 p.443 (PDF 459); Aoki printed p.39 (PDF 41)

Table A.12: 49 Tenryō Conversions Used in Table 1 (Part 1 of 3)

Note: The table lists the first qualifying conversion of each treated domain, with corrected dates and source-stated reasons. HD denotes *Hanshi Daijiten* (Kimura, Fujino, and Murakami 2015); HS denotes *Hanshi Sōran* (Kodama and Kitajima 1989). Locators give volume and printed page, with PDF pages in parentheses. Table A.16 lists the coding corrections.

Year	Domain	Province	Formal reason	Local source
1668	Shimabara 島原藩	肥前	Misgovernment	HD v.7 p.212 (PDF 228); Aoki printed p.40 (PDF 42)
1670	Harima Shingu 播磨新宮藩	播磨	Succession restriction	HD v.5 p.583 (PDF 599)
1676	Aso 麻生藩	常陸	No heir	HD v.2 p.110 (PDF 126)
1678	Yamazaki 山崎藩	播磨	No heir	HD v.5 p.562 (PDF 578); narrative p.557 (PDF 573)
1679	Kururi 久留里藩	上総	Mental incapacity	HD v.2 p.530 (PDF 546); chronology p.533 (PDF 549)
1679	Niwase 庭瀬藩	備中	No heir	HD v.6 p.199 (PDF 215)
1681	Numata 沼田藩	上野	Misgovernment and official failure	HD v.2 p.257 (PDF 273); Aoki printed pp.47,49 (PDF 49,51)
1681	Takada 高田藩	越後	Household conflict	HD v.3 p.113 (PDF 129)
1683	Tatebayashi 館林藩	上野	Succession termination	HD v.2 p.280 (PDF 296); chronology p.288 (PDF 304); Aoki printed p.45 (PDF 47)
1684	Sanuki 佐貫藩	上総	Official misconduct	HD v.2 p.546 (PDF 562)
1687	Karasuyama 烏山藩	下野	Succession-rule violation	HD v.2 p.165 (PDF 181)
1687	Soumi 沢海藩	越後	Misconduct	HD v.3 p.54 (PDF 68); HS p.154 (PDF 156)
1689	Fukami 深見藩	相模	Official dereliction	HD v.2 p.681 (PDF 697)
1689	Kitami 喜多見藩	武蔵	Collective punishment after violence	HD v.2 p.675 (PDF 691)
1692	Hachiman 八幡藩	美濃	No heir	HD v.4 p.33 (PDF 49)
1692	Takayama 高山藩	飛騨	Administrative and political transfer	HD v.4 p.3 (PDF 19)
1693	Takahashi 備中松山藩	備中	No heir	HD v.6 p.243 (PDF 259)

Table A.13: 49 Tenryō Conversions Used in Table 1 (Part 2 of 3)

Note: Continuation of Table A.12. Source abbreviations and page conventions are unchanged.

Year	Domain	Province	Formal reason	Local source
1698	Fukuyama 福山藩	備後	No heir	HD v.6 p.267 (PDF 283); Aoki printed p.57 (PDF 59)
1698	Sano 佐野藩	下野	Administrative relocation	HD v.2 p.236 (PDF 252)
1698	Yoshii 吉井藩	上野	Administrative relocation	HD v.2 p.336 (PDF 352)
1701	Ako 赤穂藩	播磨	Violence	HD v.5 p.608 (PDF 624)
1702	Nagashima 長島藩	伊勢	Violence and mental derangement	HD v.4 p.385 (PDF 401)
1703	Tamanawa 玉縄藩	相模	Administrative relocation	HD v.2 p.683 (PDF 699); HS p.146 (PDF 148)
1704	Yamura 谷村藩	甲斐	Administrative relocation	HD v.3 p.322 (PDF 338); HS p.181 (PDF 183)
1712	Hojo 北条藩	安房	Misgovernment and extraction	HD v.2 p.569 (PDF 585); chronology p.572 (PDF 588); Aoki printed p.66 (PDF 68)
1724	Kofu 甲府藩	甲斐	Administrative transfer	HD v.3 pp.331–332 (PDF 345–346)
1725	Matsumoto 松本藩	信濃	Violence and mental derangement	HD v.3 p.451 (PDF 467); Aoki printed p.70 (PDF 72)
1739	Takayanagi 高柳藩	越後	Appointment-related land exchange	HD v.3 p.122 (PDF 136); HS p.156 (PDF 158)
1767	Obata 小幡藩	上野	Political punishment	HD v.2 p.361 (PDF 377); chronology p.367 (PDF 383)
1786	Sagara 相良藩	遠江	Political fall	HD v.4 p.161 (PDF 177); chronology p.165 (PDF 181)
1788	Omi Komuro 近江小室藩	近江	Official misconduct and corruption	HD v.5 p.5 (PDF 21); HS pp.259-260 (PDF 261-262); Aoki printed p.146 (PDF 148)
1807	Matsumae 松前藩	蝦夷	Strategic and military transfer	HD v.1 p.5 (PDF 21); chronology p.16 (PDF 32)

Table A.14: 49 Tenryō Conversions Used in Table 1 (Part 3 of 3)

Note: Continuation of Table A.12. Source abbreviations and page conventions are unchanged.

C.2. Source Corrections and Conversion Causes

The source check corrects Fukumoto from 1687 to 1665, Kōfu from 1704 to 1724, and Takayanagi from 1746 to 1739. The Mibu and Omigawa records refer to events in other domains and therefore attach the event to the wrong capital; Numazu records no transition in 1724. Ichinoseki returned to Sendai, Hamada came under Chōshū occupation, and the Kaifuchi/Jōzai confiscation was undertaken by the new government after shogunal rule. Those six records are excluded from the conversion sample. [Table A.16](#) gives the corrections and source locators.

[Table A.15](#) summarizes the source-stated causes for the 49 events used in the analysis. Formal causes and evidence of unrest are coded separately: a succession or political-sanction entry does not rule out a contributing dispute, and a nearby resistance episode does not by itself establish why the shogunate acted.

Source-stated cause	Conversions
Administrative or strategic reorganization	11
Misconduct, misgovernment, or political sanction	17
Succession, partition, or incapacity	21
Total	49

Table A.15: Source-Stated Causes of the 49 Tenryō Conversions

Note: Categories group the formal reasons recorded in the local domain histories. They describe the corrected treatment sample in [Table A.12](#). Possible connections to unrest or petitioning are assessed separately in [Table A.17](#).

Domain	Original year	Corrected year	Correction	Reason	Local source
Mibu	1665	—	Remove	Recorded abolition concerns Minagawa; the panel uses Mibu capital and outcomes.	HD v.2 p.226 (PDF 242); HS p.114 (PDF 116); Mibu entry v.2 p.210 (PDF 226)
Omigawa	1677	—	Remove	Recorded abolition concerns Owa; the panel uses Omigawa capital and outcomes.	HS p.75 (PDF 77); Omigawa entry HD v.2 p.427 (PDF 443)
Numazu	1724	—	Remove	No conversion occurred in 1724; the earlier direct-administration episode is not an abolition of an independent Numazu domain.	HD v.4 p.115 (PDF 131); p.116 (PDF 132)
Ichinoseki	1671	—	Remove	Holdings explicitly returned to Sendai domain.	HD v.1 p.135 (PDF 151)
Hamada	1866	—	Remove	Choshu occupation, not conversion to shogunal administration.	HD v.6 p.119 (PDF 135); chronology p.128 (PDF 144); Aoki printed pp.291,303,305 (PDF 293,305,307)
Kaifuchi	1868	—	Remove	New-government confiscation after the shogunate, not shogunal conversion.	HD v.2 p.526 (PDF 542); chronology p.527 (PDF 543)
Fukumoto	1687	1665	Date	Domain abolition occurred in 1665; 1687 was a later hatamoto division.	HD v.5 p.530 (PDF 546); HS p.295 (PDF 297)
Kofu	1704	1724	Date	Direct shogunal administration began in 1724; 1704 changed the ruling house.	HD v.3 pp.331–332 (PDF 345–346)
Takayanagi	1746	1739	Date	Former holding reverted in 1739; 1746 was a later move to Mikusa.	HD v.3 p.122 (PDF 136); HS p.156 (PDF 158)

Table A.16: Corrections to Conversion Coding

Note: The table records changes to event dates and the removal of events that were not conversions to direct shogunal rule. A dash indicates removal without a replacement date. HD denotes *Hanshi Daijiten* (Kimura, Fujino, and Murakami 2015); HS denotes *Hanshi Sōran* (Kodama and Kitajima 1989). Locators give volume and printed page, with PDF pages in parentheses.

C.3. Potentially Unrest-Related Conversions

Conversion endogeneity motivates a separate exclusion exercise. A page-level local-source and Aoki chronology audit identifies three cases with documented unrest or petitioning near conversion—Saijō (1665), Shimabara (1668), and Hōjō (1712)—and two further cases, Numata (1681) and Ōmi Komuro (1788). Removing the first three increases the exit estimate from 0.0427 to 0.0472 (two-way cluster $p = 0.0076$); removing all five yields 0.0448 ($p = 0.0148$). A maximal screen also removes four conversions preceded within ten years by an Aoki event in the same named domain or province, even though the histories do not identify those events as causes; the exit estimate is 0.0493 ($p = 0.0138$). Confrontation remains statistically indistinguishable from zero in all three samples (Table A.17). These screens show that the exit result does not depend on cases with documented or potential links to unrest, although domain histories cannot establish the absence of unrecorded unrest.

Specification	Outcome	Estimate	Cluster SE	Cluster p	Conley p	Treated
Baseline	Exit	0.0427	0.0167	0.0113	0.0031	49
Baseline	Confrontation	0.0126	0.0125	0.3147	0.4013	49
Exclude 3 documented cases	Exit	0.0472	0.0174	0.0076	0.0011	46
Exclude 3 documented cases	Confrontation	0.0129	0.0133	0.3338	0.4339	46
Exclude 5 documented/mixed cases	Exit	0.0448	0.0182	0.0148	0.0013	44
Exclude 5 documented/mixed cases	Confrontation	0.0112	0.0142	0.4292	0.4995	44
Maximal 10-year history screen	Exit	0.0493	0.0198	0.0138	0.0012	40
Maximal 10-year history screen	Confrontation	0.0140	0.0153	0.3632	0.4333	40

Table A.17: Sensitivity to Excluding Potentially Unrest-Related Conversions

Note: The narrow set excludes Saijō (1665), Shimabara (1668), and Hōjō (1712), with documented unrest or petitioning near conversion. This timing does not establish that unrest caused each conversion. The broader set also excludes Numata (1681) and Ōmi Komuro (1788) as mixed cases. The maximal screen additionally removes Sakura, Miyazu, Tatebayashi, and Matsumoto because Aoki records an event in the same named domain or province during the preceding ten years, even though the local histories do not identify those events as causes of conversion. Cluster standard errors are two-way by domain and year; Conley p -values use a 70 km cutoff. Silence in a short domain history is not treated as proof that no unrest occurred. Sources: (Aoki 1975; Kimura, Fujino, and Murakami 2015; Kodama and Kitajima 1989).

C.4. Tenryo-Conversion Count Robustness

Outcome:	Nearby Exit Count			Nearby Confrontation Count		
Radius:	30km	40km	50km	30km	40km	50km
Converted $\times$ Post	0.0387	0.0635	0.0835	0.0179	0.0275	0.0346
Cluster SE	(0.0264)	(0.0359)*	(0.0440)*	(0.0155)	(0.0207)	(0.0263)
Conley SE (70km)	(0.0292)	(0.0395)	(0.0460)*	(0.0122)	(0.0193)	(0.0292)
Num. Obs.	31,989	31,989	31,989	31,989	31,989	31,989
Domain FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Converted/Observed Domains	49/269	49/269	49/269	49/269	49/269	49/269

Table A.18: Count Robustness for Domain-to-Tenryo Conversions

Note: OLS fixed-effects estimates with event counts (rather than binary indicators) as outcomes, using the same annual domain-year panels as Table 1. Tenryo-converted domains contribute available observations within five years before and after conversion and are matched to never-converted controls observed in the same calendar years represented by the conversion window. Two SE rows: (i) two-way clustered on domain and year; (ii) Conley spatial standard errors with a 70 km cutoff on domain capital coordinates. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

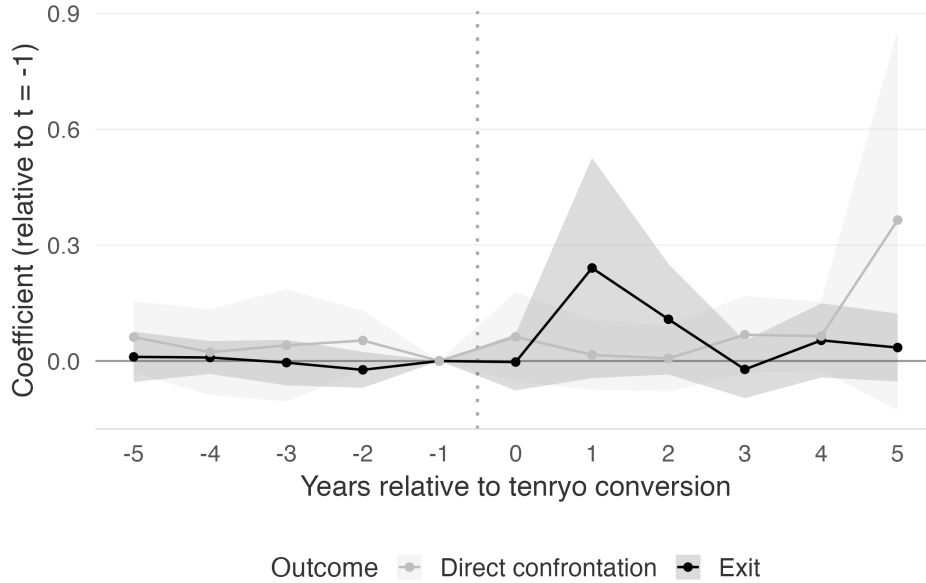


Figure A.8: OLS Count Event-Study Coefficients (50km)

Note: Year-by-year OLS coefficients from the specification $\text{count}_{j,t} = \sum_{s \neq -1} \beta_s (\mathbf{1}[t - t_j^* = s]) + \varphi_j + \omega_t + \varepsilon_{j,t}$, where the outcome is the count of nearby exit (black) or direct-confrontation (light gray) events within 50km of domain j 's capital and $s = -1$ is the reference. Shaded bands show 95% confidence intervals; the horizontal line marks zero and the vertical dotted line marks conversion between $s = -1$ and $s = 0$. Standard errors are two-way clustered by domain and year.

C.5. Disaggregating Exit into Desertion and End-Run Appeal

Table A.19 replicates Table 1 with the exit outcome restricted to desertion only (*chōsan*), excluding end-run appeals.

Outcome:	Any Nearby Desertion		
Sample Radius:	30km	40km	50km
<i>Variables</i>			
Converted × Post	0.0127	0.0196	0.0295
Cluster SE	(0.0072)*	(0.0114)*	(0.0138)**
<i>Controls</i>			
Domain & Year FE	Yes	Yes	Yes
<i>Fit Statistics</i>			
Num. Obs.	31,989	31,989	31,989
R2	0.031	0.043	0.054
Mean Dep. Var.	0.004	0.007	0.010

Table A.19: Desertion-Only DiD

Note: Replicates Table 1 with the exit outcome restricted to desertion (*chōsan*), excluding end-run appeals. Specification matches Table 1. Two-way clustered (domain & year) standard errors. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

C.6. Alternative First-Conversion Estimates

Table A.20 compares the pooled estimate with the interaction-weighted estimator of Sun and Abraham (2021). The latter follows first-conversion timing, while the main exposure indicator can switch off after re-enfeoffment. Its exit point estimate is 0.0454, compared with the main 0.0427, and its confrontation point estimate is 0.0314. The unadjusted two-way clustered covariance is not positive semidefinite; the aggregate confrontation variance is negative. I therefore report these point estimates without cluster standard errors or p -values. With 49 treated domains distributed over 40 conversion cohorts, this sparse specification does not provide an independent inferential check on the main contrast between tactics.

Estimator	Nearby Exit			Nearby Confrontation		
	ATT	SE	p	ATT	SE	p
TWFE	0.0427	(0.0167)**	0.0113	0.0126	(0.0125)	0.3147
Sun & Abraham (2021)	0.0454	—	—	0.0314	—	—

Table A.20: Pooled Exposure and First-Conversion Point Estimates

Note: The TWFE row reproduces the main 50km specification from Table 1. The Sun and Abraham row uses first-conversion cohorts and never-converted controls; exposure timing therefore differs when domains were re-enfeoffed. Two-way clustered (domain & year) inference is reported for TWFE. Dashes mark unavailable Sun–Abraham inference because the unadjusted clustered covariance is invalid; no inference from the package’s covariance repair is substituted. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

C.7. Goodman-Bacon Decomposition of the TWFE Estimator

Table A.21 reports the Goodman-Bacon (2021) decomposition for a balanced, full-history panel with 30 converted and 157 never-converted domains (50,303 observations). This diagnostic codes treatment as beginning at first conversion and remaining on thereafter. It therefore examines a different time window and exposure definition from the main specification, which permits re-enfeoffment. Treated-versus-never-treated comparisons receive 94.9% of the weight; already-treated and not-yet-treated comparisons receive 3.5% and 1.6%, respectively. These weights describe this balanced diagnostic and do not decompose the main windowed coefficient or establish homogeneous effects.

Comparison type	Weight	Avg. 2×2 estimate
Treated vs. never-treated	94.9%	0.0102
Treated vs. already-treated	3.5%	0.0013
Treated vs. yet-to-be-treated	1.6%	0.0182

Table A.21: Goodman-Bacon Decomposition of the 50 km Tenryo DiD

Note: The decomposition uses a balanced 50 km domain-year panel. Weights describe the contribution of each type of two-by-two comparison to the two-way fixed-effects estimate. A large weight on comparisons with never-treated domains limits the role of comparisons between differently timed treated cohorts; it does not establish homogeneous effects across cohorts.

C.8. Cohort-Split Sun and Abraham ATT

To inspect heterogeneity, [Table A.22](#) estimates the Sun and Abraham (2021) first-conversion specification separately for pre-1750, 1750–1799, and post-1800 conversions. The groups contain 45, 3, and 1 treated domains, respectively. Their exit point estimates are 0.0482, -0.0064 , and 0.0708. Most identifying cases belong to the first group; the small later groups, especially the single post-1800 domain, do not support a general comparison across historical periods. The covariance and timing qualifications in [SI C.6](#) also apply here.

Cohort bin	Cohorts	Treated units	ATT	SE	<i>p</i>
All cohorts	40	49	0.0454	—	—
pre-1750	36	45	0.0482	—	—
1750–1799	3	3	-0.0064	—	—
post-1800	1	1	0.0708	—	—

Table A.22: Cohort-Split Sun and Abraham (2021) ATT on Nearby Exit (50 km)

Note: Each row re-runs the interaction-weighted estimator on the sub-sample of treated domains whose tenryo-conversion year falls in the indicated bin (plus all never-treated controls). An em dash marks inference omitted because the original covariance matrix clustered by domain and year is not valid for that fit. Point estimates are retained; no covariance repair is used for reported inference.

C.9. Rambachan-Roth Relative-Magnitudes Breakdown

Table A.23 reports the relative-magnitudes sensitivity curve of Rambachan and Roth (2023) for the average post-conversion exit effect at 50 km. $\overline{M}$ bounds the largest post-period deviation from parallel trends relative to the largest pre-period deviation. The 95% robust interval excludes zero at tested values through $\overline{M} = 0.20$, where it is $[0.0047, 0.1040]$, and includes zero at 0.25. Thus $[0.20, 0.25]$ brackets the breakdown on the tested grid; 0.20 is not an exact threshold. The pre-period joint test gives $F = 0.95$, $p = 0.432$. Nonrejection of that test does not establish parallel trends, which is why the sensitivity bounds are reported separately.

Restriction	Bound	95% robust CI	Zero in CI?
Δ^{RM}	0.00	$[0.0133, 0.0773]$	No
Δ^{RM}	0.20	$[0.0047, 0.1040]$	No
Δ^{RM}	0.50	$[-0.0407, 0.1567]$	Yes
Δ^{RM}	1.00	$[-0.1313, 0.2493]$	Yes
Δ^{RM}	2.00	$[-0.3153, 0.4333]$	Yes
Δ^{SD}	0.0000	$[0.0242, 0.0793]$	No
Δ^{SD}	0.0010	$[0.0129, 0.0882]$	No
Δ^{SD}	0.0018	$[0.0004, 0.0988]$	No
Δ^{SD}	0.0020	$[-0.0027, 0.1014]$	Yes
Δ^{SD}	0.0030	$[-0.0182, 0.1143]$	Yes
Δ^{SD}	0.0050	$[-0.0488, 0.1391]$	Yes
Δ^{SD}	0.0100	$[-0.1318, 0.1822]$	Yes

Table A.23: Sensitivity of the Average Post-Conversion Exit Effect

Note: Rambachan–Roth (2023) intervals for the average post-conversion exit effect at 50 km. Under Δ^{RM} , $\overline{M}$ bounds post-period deviations relative to the largest pre-period deviation. Under Δ^{SD} , M bounds the second difference of the trend. The largest tested bounds retaining a positive lower confidence limit are $\overline{M} = 0.20$ and $M = 0.0018$. These are grid-based thresholds. The exit pre-trend joint Wald test gives $F = 0.95$ and $p = 0.432$.

C.10. Rambachan-Roth Sensitivity at Canonical Benchmarks

Table A.24 reports the robust 95% interval at relative-magnitudes bounds $\overline{M} \in \{0.5, 1, 2\}$: $[-0.0407, 0.1567]$, $[-0.1313, 0.2493]$, and $[-0.3153, 0.4333]$, respectively. All include zero. Under the smoothness restriction Δ^{SD} , the last tested second-difference bound whose interval excludes zero is $M = 0.0018$; the interval includes zero at $M = 0.0020$. At the empirical pre-period coefficient dispersion, $\sigma_{\text{pre}} = 0.0108$, the interval is $[-0.1453, 0.1849]$. These bounds describe how much deviation the inference tolerates under each restriction; they do not establish the magnitude of the actual post-period departure from parallel trends.

Restriction	Bound or benchmark	95% robust CI	Zero in CI?
Δ^{RM}	0.00	[0.0133, 0.0773]	No
Δ^{RM}	0.20	[0.0047, 0.1040]	No
Δ^{RM}	0.50	$[-0.0407, 0.1567]$	Yes
Δ^{RM}	1.00	$[-0.1313, 0.2493]$	Yes
Δ^{RM}	2.00	$[-0.3153, 0.4333]$	Yes
Δ^{SD}	0	[0.0242, 0.0793]	No
Δ^{SD}	0.0018 (grid threshold)	[0.0004, 0.0988]	No
Δ^{SD}	$\sigma_{\text{pre}} = 0.010825$	$[-0.1453, 0.1849]$	Yes

Table A.24: Wider Trend-Deviation Bounds Include Zero

Note: Extension of Table A.23. Relative-magnitude bounds are shown at the stated benchmarks. Smoothness bounds include zero, the largest tested value retaining a positive lower confidence limit, and the empirical standard deviation of the pre-period event-study coefficients, $\sigma_{\text{pre}} = 0.010825$. The smoothness interval at that empirical benchmark is evaluated using its value rounded to six decimal places. All intervals concern the average post-conversion exit effect at 50 km.

C.11. SUTVA and Spatial Interference

Because conversions cluster in time and space, I examine possible interference by removing converted domains whose capitals lie within 100 km and five years of another conversion, removing never-converted controls within 50 km of a converted capital, and imposing both restrictions. Table A.25 reports the results. Exit coefficients remain positive, ranging from 0.0285 to 0.0427. Excluding potentially exposed controls alone leaves a statistically significant estimate; removing overlapping treated domains produces wider intervals that include zero. Confrontation estimates remain statistically indistinguishable from zero in all four specifications.

Sample	Nearby Exit		Nearby Confrontation	
	$\hat{\beta}$	SE	$\hat{\beta}$	SE
Full sample ($N = 31,989$)	0.0427	(0.0167)**	0.0126	(0.0125)
	(0.0144)***			(0.0150)
Drop overlap ($N = 31,673$)	0.0337	(0.0256)	0.0333	(0.0215)
	(0.0260)			(0.0230)
Clean controls ($N = 13,110$)	0.0421	(0.0178)**	0.0085	(0.0123)
	(0.0148)***			(0.0156)
Strictest ($N = 12,794$)	0.0285	(0.0264)	0.0266	(0.0207)
	(0.0269)			(0.0250)
Domain FE: Yes Year FE: Yes				

Table A.25: SUTVA / Spatial Interference Robustness

Note: Each row re-estimates the main 50 km DiD specification on progressively restricted samples. “Drop overlap” removes converted domains within 100 km and 5 years of another conversion. “Clean controls” removes never-converted domains within 50 km of any converted capital. “Strictest” applies both exclusions. Two SE rows are reported in parentheses: (i) two-way clustered on domain and year and (ii) Conley spatial standard errors with a 70 km cutoff. Significance stars on each SE row reflect that estimator’s p -value. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

C.12. Permutation Test (Full-Pool Randomization)

I reassign 49 placebo conversion events across the 288 eligible domains, drawing years from the corrected conversion-date distribution and reapplying the five-year window and matched-control-year restriction. Although each draw assigns 49 treatment labels, only 33–48 treated domains remain in its estimation frame (mean 40.6), compared with 49 in the observed design. The placebo samples therefore do not hold the effective number of treatments fixed. Across 500 assignments, no absolute placebo coefficient reaches the observed exit estimate of 0.0427; the finite-simulation correction gives $\frac{0+1}{500+1} \approx 0.002$. The placebo distribution has mean -0.0004 , standard deviation 0.0150, and 2.5th–97.5th percentiles of $[-0.0302, 0.0296]$. This is a placebo-assignment diagnostic: conversions were not randomly assigned, and placebo exposure remains on after assignment whereas observed exposure allows re-enfeoffment. It does not establish the design’s causal identifying assumptions.

Full-pool permutation test (50 km, nearby exit)	Result
Actual coefficient (Converted $\times$ Post)	0.0427
Actual clustered p -value	0.0113
Valid / requested iterations	500 / 500
Eligible domains	288
Placebo events per iteration	49
Permutation mean	-0.0004
Permutation SD	0.0150
Permutation 2.5% quantile	-0.0302
Permutation 97.5% quantile	0.0296
Two-sided extreme draws / valid draws	0 / 500
Two-sided Monte Carlo p -value	0.0020
Right-tail Monte Carlo p -value	0.0020

Table A.26: The Observed Exit Estimate Lies Beyond the Permutation Distribution

Note: Each iteration draws 49 placebo conversion events without replacement from the 288 eligible domains, including the actually treated domains. Dates are drawn from the empirical conversion-year distribution. The same five-year event window and matched calendar years used in Table 1 are applied before estimating the 50 km two-way fixed-effects exit coefficient. The two-sided tail counts draws whose absolute estimate is at least as large as the actual absolute estimate. Monte Carlo p -values use $\frac{k+1}{B+1}$, where k is the number of extreme draws and B is the number of valid draws.

C.13. Hazard Model for Conversion Timing

The F-test on pre-period event-time coefficients reported in the main text (Table 1) bounds one threat to identification: a secular differential trend between treated and control domains over the five-year window leading up to conversion. A distinct concern is whether attainer decisions responded to **recent** local unrest, making conversion timing endogenous to nearby exit or confrontation. I test this with a hazard model using the corrected conversion dates. After requiring covariates and five years of lagged history, the estimation sample contains 29 conversions across 233 domains (51,313 domain-years). The outcome indicates conversion to tenryo in year t ; domains are censored at conversion or 1868. The covariates are lagged 5-year moving averages of nearby exit and confrontation within 50 km, plus territory size, distance to the Edo–Kyoto axis, and alliance status. Neither lagged exit intensity ($p = 0.2066$ Cox, $p = 0.4912$ logit) nor confrontation intensity ($p = 0.8093$ Cox, $p = 0.3432$ logit) predicts conversion timing; the joint Wald test fails to reject at $p = 0.4355$. Fudai/kin status is the one significant predictor, consistent with the historiography of attainer politics (Ravina 1999). The hazard model, pre-trend test, and permutation analysis provide complementary checks on timing.

Model	Variable	$\hat{\beta}$	SE	p	Ratio
Cox	Nearby exit MA(5)	-4.4807	3.5476	0.2066	0.0113
Cox	Nearby confrontation MA(5)	0.6075	2.5178	0.8093	1.8359
Cox	Log area	-0.2176	0.1728	0.2078	0.8044
Cox	Log distance to Edo–Kyoto axis	0.2430	0.3158	0.4416	1.2750
Cox	Fudai/kin (vs. tozama)	1.2120	0.5163	0.0189	3.3602
Logit	Nearby exit MA(5)	-2.4002	3.4863	0.4912	0.0907
Logit	Nearby confrontation MA(5)	-2.2383	2.3614	0.3432	0.1066
Logit	Log area	-0.1556	0.1685	0.3558	0.8559
Logit	Log distance to Edo–Kyoto axis	0.2100	0.3110	0.4997	1.2336
Logit	Fudai/kin (vs. tozama)	1.1734	0.5198	0.0240	3.2331
Cox	Joint Wald: exit and confrontation	$\chi^2 = 1.663$		0.4355	—

Table A.27: Lagged Nearby Resistance Does Not Predict Conversion Timing

Note: Cox proportional-hazards and discrete-time logit estimates. The complete-case timing sample contains 51313 domain-years across 233 domains, including 29 observed conversions from the 49 treated domains in the primary analysis. Resistance variables are lagged five-year moving averages within 50 km of the focal capital. SEs are from the fitted Cox and logit models; ratios are hazard ratios for Cox and odds ratios for logit. The Cox joint Wald test has two degrees of freedom.

C.14. Conley Bandwidth Sweep (DiD)

The 70 km Conley cutoff used for the main DiD standard errors is a researcher choice. [Table A.28](#) reports cutoffs of 30, 50, 70, 100, and 150 km. The point estimate remains 0.0427; Conley standard errors range from 0.0117 to 0.0144, with all p -values below 0.005.

Inference	Cutoff	$\hat{\beta}$	SE	p	N
Cluster (domain+year)	–	0.0427	0.0167	0.0113	31989
Conley	30 km	0.0427	0.0138	0.0020	31989
Conley	50 km	0.0427	0.0140	0.0023	31989
Conley (paper)	70 km	0.0427	0.0144	0.0031	31989
Conley	100 km	0.0427	0.0121	0.0004	31989
Conley	150 km	0.0427	0.0117	0.0003	31989

Table A.28: Conley Bandwidth Sweep for the Tenryo DiD (50 km)

Note: Tenryo DiD exit coefficient with standard errors computed under the cluster estimator (the main-text default) and Conley spatial HAC standard errors at five cutoffs. The point estimate does not depend on the inference method; only the SE and p -value vary. The hand-picked 70 km cutoff used in [Table 1](#) sits near the middle of the sweep.

C.15. Kelly (2020) Data-Driven Conley Bandwidth

As an alternative to the hand-picked cutoffs used in the main text (10 km at the village and grid levels, 70 km at the domain and DiD levels), I apply the Kelly (2019) data-driven rule, which selects the bandwidth from the empirical residual correlogram: the smallest distance at which the mean pairwise residual correlation falls below $\exp(-1) \approx 0.37$. Table A.29 reports this bandwidth alongside the hand-picked choice for three analyses: the village cross-section (Table 3 Model 5), the domain-level tax regression (Table 4 Model 4), and the tenryo DiD (Table 1 50 km). The data-driven choice is tighter than the hand-picked one in all three cases (6 km for villages, 10 km for tax, 10 km for DiD). Conley SEs change by less than 0.001 between the two choices in every regression. The village and DiD estimates remain significant at $p < 0.01$, while the tax estimate remains significant at $p < 0.05$.

Analysis	Bandwidth rule	Cutoff	$\hat{\beta}$	Conley SE	Conley p	N
<i>Village-level exit share (Table 3 Model 5 dyadic-FE)</i>						
	hand-picked	10km	0.0244	0.0085	0.0041	2739
	Kelly (2020) data-driven	6km	0.0244	0.0089	0.0062	2739
<i>Domain tax regression (Table 4 Model 4)</i>						
	hand-picked	70km	-0.0112	0.0055	0.0432	230
	Kelly (2020) data-driven	10km	-0.0112	0.0055	0.0449	230
<i>Tenryo DiD (Table 1 50 km)</i>						
	hand-picked	70km	0.0427	0.0144	0.0031	31989
	Kelly (2020) data-driven	10km	0.0427	0.0143	0.0029	31989

Table A.29: Kelly (2020) Data-Driven vs. Hand-Picked Conley Bandwidth

Note: The Kelly bandwidth is the smallest distance at which the empirical residual correlogram falls below $\exp(-1) \approx 0.37$. Pairwise residual products are aggregated into 40 distance bins for the village and DiD regressions and 25 bins for the tax regression. The data-driven bandwidth is tighter than the hand-picked cutoff in all three analyses, especially for the tax regression (10 km vs. 70 km). Point estimates are identical across bandwidth choices; only SEs vary. The data-driven choice moves SEs by less than 0.001 in absolute terms across all three regressions. The village and DiD estimates remain significant at $p < 0.01$; the tax estimate remains significant at $p < 0.05$.

D. VILLAGE EXIT CHOICE ANALYSIS

D.1. Full Table

Outcome:	exit _i					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
ln(ρ_i)	0.0458 (0.0154)***	0.0235 (0.0117)*	0.0143 (0.0089)	0.0201 (0.0071)***	0.0212 (0.0072)***	0.0244 (0.0068)***
lat					-1.4681 (0.6101)**	-2.0087 (1.4745)
lon					-0.2692 (0.1515)*	-0.4046 (0.3957)
ln(Dist. Edo/Kyoto)					0.2237 (0.0419)***	0.0784 (0.0793)
ln(Rice)					-0.0055 (0.0045)	-0.0018 (0.0046)
ln(Elev.)					-0.0228 (0.0198)	-0.0335 (0.0193)*
ln(Rugg.)					0.0289 (0.0152)*	0.0330 (0.0146)**
Lat × Lon					0.0100 (0.0047)**	0.0141 (0.0114)
<i>Controls</i>						
Home (j) Domain FE	No	Yes	Yes	Yes	Yes	No
Neighbor (k) Domain FE	No	No	Yes	Yes	Yes	No
Decade (t) FE	No	No	No	Yes	Yes	Yes
Home-Neighbor Dyad (j, k) FE	No	No	No	No	No	Yes
<i>Fit Statistics</i>						
Num.Obs.	2,739	2,739	2,739	2,739	2,739	2,739
R2	0.028	0.390	0.543	0.621	0.626	0.678

Table A.30: Villages Exposed to Alternative Authority Opt for Exit Over Direct Confrontation

(Full Table)

Note: The coefficients represent OLS estimates of the relationship between each village's geographic exposure and its choice of exit conditional on recorded resistance. Clustered (home ruler, nearest neighboring ruler, and decade) standard errors are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.2. Robustness Checks

D.2.1. Alternative Covariates

Outcome:	exit _i							
	Cal. Suit.					Dist. to Edo		
Models:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Variables</i>								
ln(ρ_i)	0.0434 (0.0150)***	0.0162 (0.0090)*	0.0209 (0.0072)***	0.0234 (0.0069)***	0.0439 (0.0149)***	0.0135 (0.0092)	0.0199 (0.0075)**	0.0240 (0.0068)***
lat	−1.3123 (0.3162)***	−0.1980 (0.7180)	−1.6336 (0.5745)***	−2.0788 (1.3142)	−1.7772 (0.6161)***	−2.7623 (1.0481)**	−2.9088 (0.7516)***	−2.9748 (1.3385)**
lon	−0.2880 (0.0781)***	0.1762 (0.2098)	−0.2967 (0.1291)**	−0.4006 (0.3399)	−0.4525 (0.1775)**	−0.6267 (0.3681)	−0.7051 (0.2625)**	−0.6908 (0.3908)*
ln(Dist. Edo/Kyoto)	−0.0187 (0.0451)	0.4531 (0.1668)**	0.2295 (0.0576)***	0.1037 (0.0744)				
ln(Cal.)	−0.0614 (0.1949)	0.0953 (0.1834)	0.1490 (0.1591)	0.2386 (0.1564)				
ln(Elev.)	0.0121 (0.0260)	−0.0146 (0.0193)	−0.0261 (0.0184)	−0.0410 (0.0175)**	−0.0005 (0.0203)	−0.0198 (0.0200)	−0.0257 (0.0194)	−0.0350 (0.0191)*
ln(Rugg.)	0.0328 (0.0241)	0.0222 (0.0161)	0.0288 (0.0155)*	0.0324 (0.0146)**	0.0441 (0.0209)**	0.0302 (0.0153)*	0.0329 (0.0149)**	0.0349 (0.0141)**
Lat × Lon	0.0091 (0.0022)***	0.0001 (0.0056)	0.0112 (0.0044)**	0.0146 (0.0101)	0.0127 (0.0045)***	0.0197 (0.0084)**	0.0210 (0.0061)***	0.0215 (0.0104)**
ln(Dist. Edo)					−0.1374 (0.0755)*	−0.3334 (0.3733)	−0.1806 (0.2859)	−0.2381 (0.4100)
ln(Rice)					−0.0085 (0.0067)	−0.0077 (0.0062)	−0.0050 (0.0044)	−0.0018 (0.0047)
Const.	42.3502 (11.2821)***				64.0829 (24.2964)**			
<i>Controls</i>								
Home (<i>j</i>) Domain FE	No	Yes	Yes	No	No	Yes	Yes	No
Neighbor (<i>k</i>) Domain FE	No	Yes	Yes	No	No	Yes	Yes	No
Decade (<i>t</i>) FE	No	No	Yes	Yes	No	No	Yes	Yes
Home-Neighbor Dyad (<i>j, k</i>) FE	No	No	No	Yes	No	No	No	Yes
<i>Fit Statistics</i>								
Num.Obs.	2,739	2,739	2,739	2,739	2,739	2,739	2,739	2,739
R2	0.061	0.551	0.627	0.679	0.069	0.549	0.626	0.678

Table A.31: Villages Exposed to Alternative Authority

Opt for Exit Over Direct Confrontation (Alternative Covariates)

Note: Clustered (Home & Neighbor Domains & Decade) standard errors in parentheses.
Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.2.2. Excluding Villages with More Than One Recorded Resistance Episode

Outcome:	exit _{<i>i</i>}					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
ln(ρ_i)	0.0488 (0.0153)***	0.0270 (0.0119)**	0.0228 (0.0104)**	0.0229 (0.0094)**	0.0234 (0.0094)**	0.0231 (0.0080)***
lat					-0.8936 (0.7595)	-0.4750 (1.5455)
lon					-0.0921 (0.2281)	0.0445 (0.4405)
ln(Dist. Edo/Kyoto)					0.1885 (0.0942)*	0.1062 (0.1155)
ln(Rice)					-0.0031 (0.0039)	-0.0003 (0.0047)
ln(Elev.)					-0.0077 (0.0221)	-0.0227 (0.0203)
ln(Rugg.)					0.0211 (0.0161)	0.0309 (0.0159)*
Lat × Lon					0.0061 (0.0057)	0.0029 (0.0118)
Const.	0.2255 (0.0373)***					
<i>Controls</i>						
Home (<i>j</i>) Domain FE	No	Yes	Yes	Yes	Yes	No
Neighbor (<i>k</i>) Domain FE	No	No	Yes	Yes	Yes	No
Home-Neighbor Dyad (<i>j, k</i>) FE	No	No	No	No	No	Yes
Decade (<i>t</i>) FE	No	No	No	Yes	Yes	Yes
Geography	No	No	No	No	Yes	Yes
<i>Fit Statistics</i>						
Num.Obs.	2,295	2,295	2,295	2,295	2,295	2,295
R2	0.032	0.431	0.590	0.672	0.675	0.732

Table A.32: Villages Exposed to Alternative Authority Opt for Exit Over Direct Confrontation
(Excluding Repeat Villages)

Note: Clustered (Home & Neighbor Domains & Decade) standard errors in parentheses. Signif. Codes:
***: 0.01, **: 0.05, *: 0.1.

D.2.3. Absolute Distances

Outcome:	exit _i					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
ln(Dist. to Home Capital _i)	0.0399 (0.0223)*	0.0248 (0.0161)	0.0127 (0.0109)	0.0231 (0.0096)**	0.0160 (0.0104)	0.0203 (0.0083)**
ln(Dist. to Foreign Capital _i)	-0.0596 (0.0140)***	-0.0203 (0.0218)	-0.0172 (0.0199)	-0.0147 (0.0147)	-0.0306 (0.0129)**	-0.0309 (0.0135)**
lat					-1.5650 (0.6458)**	-2.0841 (1.5054)
lon					-0.2878 (0.1658)*	-0.4180 (0.4041)
ln(Dist. Edo/Kyoto)					0.2343 (0.0430)***	0.0869 (0.0839)
ln(Rice)					-0.0055 (0.0046)	-0.0017 (0.0047)
ln(Elev.)					-0.0211 (0.0206)	-0.0319 (0.0199)
ln(Rugg.)					0.0305 (0.0163)*	0.0340 (0.0153)**
Lat × Lon					0.0107 (0.0050)**	0.0147 (0.0116)
Const.	0.2984 (0.0788)***					
<i>Controls</i>						
Home (<i>j</i>) Domain FE	No	Yes	Yes	Yes	Yes	No
Neighbor (<i>k</i>) Domain FE	No	No	Yes	Yes	Yes	No
Home-Neighbor Dyad FE	No	No	No	No	No	Yes
Decade (<i>t</i>) FE	No	No	No	Yes	Yes	Yes
Geography	No	No	No	No	Yes	Yes
<i>Fit Statistics</i>						
Num.Obs.	2,739	2,739	2,739	2,739	2,739	2,739
R2	0.030	0.390	0.543	0.621	0.627	0.678

Table A.33: Villages Exposed to Alternative Authority Opt for Exit Over Direct Confrontation
(Abs. Dist. Measures)

Note: Clustered (Home & Neighbor Domains & Decade) standard errors in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.2.4. Water-Crossing Subsample

The main village distance-ratio regression uses straight-line great-circle distances from each village to its home-domain capital and nearest foreign-domain capital, and these lines can traverse sea straits such as Kanmon, Bungo, Naruto, or Sado. I therefore drop every resistance-event row whose village, home capital, or nearest foreign capital lies on a different main island (Honshu, Kyushu, Shikoku, Sado, or Awaji), classified from latitude and longitude. This removes 150 of 2,739 rows (5.5%). On this water-safe subsample the coefficient on $\log(\rho_i)$ remains positive across all six specifications, including the dyadic home-neighbor fixed-effects column that mirrors Table 3 Model 5 ($\hat{\beta} = 0.0253$, clustered $p < 0.001$; baseline 0.0244, $p = 0.001$).

Outcome:	exit _i					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
Distance ratio ($\ln(\rho_i)$)	0.0460	0.0232	0.0160	0.0209	0.0207	0.0253
Cluster SE	(0.0156)***	(0.0127)*	(0.0105)	(0.0082)**	(0.0080)**	(0.0063)***
Conley SE (10km)	(0.0121)***	(0.0113)**	(0.0094)*	(0.0079)***	(0.0079)***	(0.0086)***
<i>Controls</i>						
Home Domain FE	No	Yes	Yes	Yes	Yes	No
Neighbor Domain FE	No	No	Yes	Yes	Yes	No
Home-Nbr Dyad FE	No	No	No	No	No	Yes
Decade FE	No	No	No	Yes	Yes	Yes
Geography	No	No	No	No	Yes	Yes
<i>Fit Statistics</i>						
Num. Obs.	2,589	2,589	2,589	2,589	2,589	2,589
R ²	0.029	0.391	0.552	0.635	0.639	0.683
Mean Dep. Var.	0.251	0.251	0.251	0.251	0.251	0.251

Table A.34: Village Distance-Ratio Regression: Water-Crossing Subsample

Note: Re-runs the six village distance-ratio specifications on the subsample that drops resistance-event rows whose straight-line village-to-home-capital or village-to-foreign-capital line crosses a sea strait (i.e., the village, the home capital, and the nearest foreign capital are not all on the same main island). Island membership (Honshu, Kyushu, Shikoku, Sado, Awaji) is assigned from lat/lon bounding boxes. 150 (5.5%) of 2,739 rows are classified as sea-crossing and dropped. The coefficient on $\log(\rho_i)$ remains positive across all six specifications and statistical significance is preserved in Model 6 (dyadic home-neighbor FE plus geography controls), which corresponds to Table 3 Model 5. SE rows: (i) three-way clustered on (home, neighbor, decade); (ii) Conley spatial SE with a 10 km cutoff on village coordinates. Stars reflect the corresponding estimator's p -value. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.2.5. Conley Standard Error Bandwidth Sensitivity

Table A.35 reports the bandwidth sensitivity of Conley spatial standard errors for the main village distance-ratio regression (Table 3 Model 5, dyadic home-neighbor FE). The coefficient on $\log(\rho_i)$ is stable across all bandwidths from 25 to 150 km and remains significant at the 5% level in every specification.

Standard error	Estimate	SE	p
Cluster (home + nbr + decade, paper)	0.0244	(0.0068)	0.001
Conley 25 km	0.0244	(0.0089)	0.006
Conley 50 km	0.0244	(0.0097)	0.012
Conley 75 km	0.0244	(0.0102)	0.017
Conley 100 km	0.0244	(0.0099)	0.014
Conley 150 km	0.0244	(0.0078)	0.002

Table A.35: Conley SE Bandwidth Sensitivity: Village Distance-Ratio Regression

Note: The coefficient on $\log(\rho_i)$ from Table 3 Model 5 (dyadic home-neighbor FE) is stable across all bandwidths from 25 to 150 km and remains significant at the 5% level in every specification. The first row reports the three-way cluster standard error used in the main table.

D.2.6. Small- G Cluster Sensitivity of the Decade Dimension

Table A.36 reports a small- G cluster-sensitivity check for the village distance-ratio regression (Table 3 Model 5, dyadic home-neighbor FE). The decade dimension has $G = 26$ clusters, below the conventional safe threshold for multi-way cluster-robust inference, motivating a check of whether this dimension is load-bearing for the reported significance. Dropping the decade dimension entirely and clustering only on the two higher- G dimensions (home ruler and neighbor ruler) yields $p = 0.0024$, close to the three-way number ($p = 0.0013$); clustering on decade alone yields $p = 0.0014$. The main-text significance is therefore preserved under every cluster specification considered.

Cluster specification	Estimate	SE	p
Three-way (home + nbr + decade, paper)	0.0244	(0.0068)	0.001
Two-way (home + nbr, dropping decade)	0.0244	(0.0079)	0.002
One-way on decade only ($G = 26$)	0.0244	(0.0068)	0.001

Table A.36: Small- G Cluster Sensitivity of the Decade Dimension

Note: Re-runs Table 3 Model 5 (dyadic home-neighbor FE) under three cluster specifications. The decade dimension has $G = 26$ clusters, below the conventional safe threshold for multi-way clustering. The key finding is that the paper’s three-way cluster SE is robust to dropping the decade dimension entirely — two-way clustering on home and neighbor gives nearly identical inference — indicating the decade cluster is not load-bearing for the reported p -value.

D.2.7. Hatamoto Kantō Exclusion

Morris (1990) documents that hatamoto (direct shogunal retainers holding small fiefs) were particularly dense in four Kantō provinces: Musashi, Sagami, Kazusa, and Shimōsa. Because the baseline distance-ratio measure in [Section 3.4.2](#) uses active daimyō and former-domain capitals but omits individual hatamoto fiefs, a village in these provinces may have had a hatamoto landlord closer than its nearest measured outside authority. Excluding the 40 resistance-event rows (1.5% of the sample) leaves the coefficient positive and statistically significant ([Table A.37](#)).

Sample	Num. Obs.	Estimate	SE	<i>p</i>
Full sample (paper)	2739	0.0244	(0.0068)	0.001
Excluding Musashi, Sagami, Kazusa, Shimōsa	2699	0.0245	(0.0069)	0.002

Table A.37: Village Distance-Ratio Regression: Excluding Dense-Hatamoto Kantō Provinces

Note: Re-runs [Table 3](#) Model 5 (dyadic home-neighbor FE) on the subsample that excludes resistance-event rows in the four Kantō provinces with the highest documented hatamoto density – Musashi, Sagami, Kazusa, and Shimōsa (Morris 1990) – where the distance-ratio measure may under-capture the nearest outside authority because hatamoto fiefs are not in the baseline daimyō/former-domain-capital geography. The coefficient on $\log(\rho_i)$ remains positive and statistically significant.

D.2.8. Desertion-Only Village Cross-Section

Table A.38 re-estimates the preferred specifications with the exit outcome restricted to desertion only (*chōsan*), excluding end-run appeals. The preferred coefficient is positive, marginal under three-way clustering ($p = 0.051$), and significant with Conley inference ($p = 0.041$), so the narrower definition preserves the direction and most of the pooled magnitude with some inference sensitivity.

Outcome:	Desertion _{<i>i</i>}	
Models:	(4)	(5)
<i>Variables</i>		
Distance ratio ($\ln(\rho_i)$)	0.0127	0.0141
Cluster SE	(0.0060)**	(0.0068)*
Conley SE (10km)	(0.0062)**	(0.0069)**
<i>Controls</i>		
Home + Neighbor + Decade FE + Geography	Yes	No
Dyadic + Decade FE + Geography	No	Yes
<i>Fit Statistics</i>		
Num.Obs.	2,727	2,727
R2	0.598	0.674

Table A.38: Desertion-Only Village Cross-Section

Note: Replicates Table 3 Models 4–5 with exit restricted to desertion (*chōsan*), excluding end-run appeals. Two SE rows: (i) three-way clustered; (ii) Conley spatial SE (10 km). Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.2.9. Desertion-Only Grid-Year Panel

Table A.39 replicates Table 2 with exit restricted to desertion (*chōsan*) only. The desertion-specific coefficients track the pooled exit estimates closely in both sign and significance, across the full panel and the ever-active subsample, confirming that the within-cell exit response is driven by the prototypical desertion cases rather than by end-run appeals.

Outcome:	Full sample		Ever-active cells	
	(1)	(2)	(3)	(4)
	Desertion	Pooled exit	Desertion	Pooled exit
Grid exposure ratio ($\rho_{i,t}^{\text{grid}}$)	0.00024	0.00035	0.00152	0.00198
Cluster SE	(0.00014)*	(0.00017)**	(0.00087)*	(0.00109)*
Conley SE (10km)	(0.00012)**	(0.00017)**	(0.00075)**	(0.00103)*
<i>Controls</i>				
Grid & Year FE	Yes	Yes	Yes	Yes
<i>Fit Statistics</i>				
Num.Obs.	2,225,802	2,225,802	372,364	372,364
R ²	0.005	0.006	0.006	0.008
Mean Dep. Var.	0.00009	0.00022	0.00054	0.00131

Table A.39: Desertion-Only Grid-Year Panel

Note: LPM on binary any-event-of-type indicators. Replicates Table 2 with exit restricted to desertion (*chōsan*), excluding end-run appeals. Pooled exit columns reproduce the main specification for comparison. Columns (1)–(2) use the full panel of 8,763 cells; columns (3)–(4) restrict to the 1,466 ever-active hexagons. Grid and year FE. SEs: (i) two-way clustered (cell, year); (ii) Conley (10 km). Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.2.10. Grid-Panel Count Models: Poisson and Negative Binomial

Table A.40 replicates the main-text Table 2 using two count-data estimators with hexagon and year fixed effects: Poisson (columns 1–3) and the negative binomial (columns 4–6). Because both estimators use a log link ($\log \mathbb{E}[y_{i,t} | X_{i,t}] = \beta X_{i,t} + \alpha_i + \tau_t$), the raw coefficient $\hat{\beta}$ sits on the log-count scale. The *incidence rate ratio*, $\text{IRR} \equiv \exp(\hat{\beta})$, back-transforms this to the count scale: a one-unit increase in the regressor multiplies the expected count by IRR. Both estimators give positive exit coefficients, although the Poisson estimate is imprecise (clustered $p = 0.1472$; Conley $p = 0.1239$). The exit coefficient is positive and corresponds to a multiplicative association of roughly 4–5 $\times$ per unit distance ratio (Poisson $\text{IRR} = 4.020$; NB $\text{IRR} = 5.357$). The confrontation coefficient is small and statistically indistinguishable from zero under both estimators (Poisson $p = 0.5676$; NB $p = 0.4891$), and the strict-confrontation coefficient turns negative (IRRs below one). Both estimators drop cells with no events of the relevant outcome. Their retained samples therefore vary by outcome and contain fewer cells than the full 8,763-cell panel.

Outcome:	Poisson			Neg. Binom.		
	(1)	(2)	(3)	(4)	(5)	(6)
	Exit count	Confronta- tion count	Confronta- tion count (strict)	Exit count	Confronta- tion count	Confronta- tion count (strict)
Grid exposure ratio ($\rho_{i,t}^{\text{grid}}$)	1.3912	0.3240	−0.4231	1.6784	0.4453	−0.5370
IRR	4.020	1.383	0.655	5.357	1.561	0.584
Cluster SE	(0.9599)	(0.5668)	(0.5299)	(0.8860)*	(0.6437)	(0.7985)
Conley SE (10km)	(0.9043)	(0.5257)	(0.6052)	(0.8601)*	(0.6620)	(0.7938)
<i>Controls</i>						
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit Statistics</i>						
Num.Obs.	62,900	202,130	40,810	62,900	202,130	40,810
Pseudo R ²	0.135	0.130	0.127	0.093	0.092	0.087
Mean Dep. Var.	0.0003	0.0009	0.0004	0.0003	0.0009	0.0004

Table A.40: Grid-Panel Count Models: Poisson and Negative Binomial with FE

Note: Columns (1)–(3) estimate a Poisson model and columns (4)–(6) a negative binomial, both with hexagon and year fixed effects on the 5 km grid-year panel (1615–1868). Coefficients are on the log-count scale; $\text{IRR} = \exp(\hat{\beta})$ gives the multiplicative effect on the expected count. Both estimators drop all-zero outcome groups, so Num. Obs. reflects the retained sample for each outcome rather than the full 8,763-cell panel. SEs: (i) two-way clustered on cell and year; (ii) Conley (10 km). Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.2.11. Ever-Active Cells Subsample

Table A.41 replicates Table 2 on the 1,466 “ever-active” hexagons — cells with at least one exit or confrontation event between 1615 and 1868. This restriction filters out the 83% of cells that recorded neither form of resistance and sharpens within-cell variation among active locales while using the same binary-outcome LPM and fixed-effects structure. Columns (1)–(3) use the grid exposure ratio $\rho_{i,t}^{\text{grid}}$; columns (4)–(6) use the closest-capital distance $d_{i,t}^1$ (km).

The distance-ratio coefficient on exit rises to 0.00198 ($p = 0.054$ Conley, $p = 0.070$ clustered), while the confrontation and strict-confrontation coefficients remain small and statistically indistinguishable from zero. The closest-capital regressor remains a null predictor of exit on this subsample ($p = 0.179$ clustered, $p = 0.259$ Conley). It carries a marginal negative signal for strict confrontation (Conley $p = 0.028$), plausibly because uprisings and castle attacks require a nearby state target. Remoteness therefore contains some information about where direct confrontation is mechanically feasible but not where exit is chosen; relative geographic exposure is associated with exit.

Outcome:	Grid exposure ratio ($\rho_{i,t}^{\text{grid}}$)			Closest-capital dist. ($d_{i,t}^1$, km)		
	(1)	(2)	(3)	(4)	(5)	(6)
	Any exit	Any con- frontation	Any con- frontation (strict)	Any exit	Any con- frontation	Any con- frontation (strict)
Coefficient	0.00198	0.00096	−0.00097	0.00004	−0.00004	−0.00004
Cluster SE	(0.00109)*	(0.00105)	(0.00067)	(0.00003)	(0.00003)	(0.00002)*
Conley SE (10km)	(0.00103)*	(0.00105)	(0.00062)	(0.00004)	(0.00002)*	(0.00002)**
<i>Controls</i>						
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit Statistics</i>						
Num.Obs.	372,364	372,364	372,364	372,364	372,364	372,364
R ²	0.008	0.011	0.011	0.008	0.011	0.011
Mean Dep. Var.	0.00131	0.00384	0.00155	0.00131	0.00384	0.00155

Table A.41: Grid-Year Panel on Ever-Active Cells: Distance Ratio vs. Closest-Capital Distance

Note: Replicates Table 2 on the 1,466 “ever-active” hexagons — cells with at least one exit or confrontation event between 1615 and 1868 — which sharpens within-variation among active locales while filtering out cells with no recorded resistance. Columns (1)–(3) use the time-varying grid exposure ratio $\rho_{i,t}^{\text{grid}} = \frac{d_{i,t}^1}{d_{i,t}^2}$. Columns (4)–(6) replace it with the raw closest-capital distance $d_{i,t}^1$ (km). Grid and year FE. SEs: (i) two-way clustered (cell, year); (ii) Conley (10 km). Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.2.12. Least-Cost-Path Distance Ratio

Table A.42 replicates Table 2 with the great-circle grid exposure ratio $\rho_{i,t}^{\text{grid}} = \frac{d_{i,t}^1}{d_{i,t}^2}$ replaced by a terrain-weighted least-cost-path variant $\rho_{i,t}^{\text{grid,LCP}}$, computed from a $(1 + \text{slope})$ cost surface on the SRTM elevation raster. For each of the 8,763 5 km hexagon centroids and each unique capital location, I run a single-source Dijkstra over the cost surface and record the cumulative terrain-weighted travel cost; $\rho_{i,t}^{\text{grid,LCP}}$ is the ratio of the nearest-to-second-nearest such cost among capitals active in year t . The two regressors capture overlapping but distinct information: their Pearson correlation is 0.44, and the LCP ratio has mean 0.82 and standard deviation 0.24 (vs. 0.63 and 0.26 for the great-circle ratio), reflecting the common terrain component that travel costs to multiple capitals share when paths cross the same ridges and valleys.

The exit coefficient preserves its sign and standardized magnitude: a one-standard-deviation increase in $\rho_{i,t}^{\text{grid,LCP}}$ corresponds to $0.00045 \times 0.24 \approx 0.011\%$ -points, close to the 0.009%-points implied by Table 2 column (1). The exit-versus-confrontation asymmetry also survives: confrontation coefficients remain small and statistically indistinguishable from zero across the full panel and ever-active subsample. Statistical significance on exit falls below conventional thresholds ($p = 0.278$ clustered; $p = 0.203$ Conley). Terrain friction adds a shared component to the cost of reaching any capital and compresses the contrast between the nearest two, making the LCP ratio a noisier proxy for relative geographic exposure. The alternative measure therefore preserves sign and standardized magnitude but yields less precise inference.

Outcome:	<i>Full sample</i>			<i>Ever-active cells</i>		
	(1)	(2)	(3)	(4)	(5)	(6)
	Any exit	Any con- frontation	Any con- frontation (strict)	Any exit	Any con- frontation	Any con- frontation (strict)
Grid exposure ratio (LCP, $\rho_{i,t}^{\text{grid,LCP}}$)	0.00045	0.00053	−0.00008	0.00173	0.00177	−0.00051
Cluster SE	(0.00041)	(0.00060)	(0.00031)	(0.00169)	(0.00221)	(0.00122)
Conley SE (10km)	(0.00035)	(0.00068)	(0.00026)	(0.00141)	(0.00192)	(0.00074)
<i>Controls</i>						
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit Statistics</i>						
Num.Obs.	2,225,802	2,225,802	2,225,802	372,364	372,364	372,364
R ²	0.006	0.007	0.006	0.008	0.011	0.011
Mean Dep. Var.	0.00022	0.00064	0.00026	0.00131	0.00384	0.00155

Table A.42: Grid-Year Panel with Least-Cost-Path Distance Ratio

Note: LPM on binary any-event-of-type indicators. Replicates Table 2 with the haversine grid exposure ratio replaced by a terrain-weighted least-cost-path (LCP) variant: $\rho_{i,t}^{\text{grid,LCP}}$ equals the ratio of nearest-to-second-nearest LCP cost from each hexagon centroid to any active domain capital in year t , where cost is cumulative across pixels of a $(1 + \text{slope})$ cost surface derived from the SRTM elevation raster. Columns (1)–(3) use the full panel; (4)–(6) restrict to ever-active hexagons. Grid and year FE. SEs: (i) two-way clustered (cell, year); (ii) Conley (10 km). Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.2.13. Daikansho-Inclusive Distance Ratio (Village)

Table A.43 replicates Table 3 with d_i^{FC} redefined as the distance to the nearest non-home administrative center of any type, including shogunal intendant offices (*daikansho*). The preferred dyadic-FE specification (Model 5) is virtually unchanged ($\hat{\beta} = 0.0268$ vs. 0.0244 in Table 3).

Outcome:	Exit _i				
Models:	(1)	(2)	(3)	(4)	(5)
<i>Variables</i>					
Distance ratio ($\ln(\rho_i)$)	0.0422	0.0496	0.0201	0.0202	0.0268
Cluster SE	(0.0148)***	(0.0137)***	(0.0074)**	(0.0075)**	(0.0067)***
Conley SE (10km)	(0.0117)***	(0.0093)***	(0.0077)***	(0.0079)**	(0.0085)***
<i>Controls</i>					
Decade FE	No	Yes	Yes	Yes	Yes
Home Domain FE	No	No	Yes	Yes	No
Neighbor Domain FE	No	No	Yes	Yes	No
Dyadic FE	No	No	No	No	Yes
Geography	No	No	No	Yes	Yes
<i>Fit Statistics</i>					
Num.Obs.	2,739	2,739	2,739	2,739	2,739
R2	0.024	0.222	0.633	0.638	0.686
Mean Dep. Var.	0.246	0.246	0.246	0.246	0.246

Table A.43: Village Exit Choice with Daikansho-Inclusive Distance Ratio

Note: Replicates Table 3 with d_i^{FC} redefined to include shogunal intendant offices (*daikansho*). Two SE rows: (i) clustered on home, nearest neighboring domain, and decade; (ii) Conley spatial standard errors (10 km). Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.2.14. Daikansho-Inclusive Distance Ratio (Grid)

Table A.44 replicates Table 2 with the daikansho-inclusive distance ratio. Adding daikansho to the capital set compresses the ratio toward unity and reduces within-cell variation; all grid-level coefficients attenuate and lose statistical significance. The village-level and tax results are unaffected by this inclusion (SI D.2.13; SI D.3.1).

		Full sample				Ever-active cells	
Outcome:		(1)	(2)	(3)	(4)	(5)	(6)
		Any exit	Any confrontation	Any confrontation (strict)	Any exit	Any confrontation	Any confrontation (strict)
Grid exposure ratio ($\rho_{i,t}^{\text{grid}}$)		0.00019	0.00023	−0.00006	0.00095	−0.00012	−0.00096
Cluster SE		(0.00019)	(0.00021)	(0.00012)	(0.00111)	(0.00111)	(0.00068)
Conley SE (10km)		(0.00018)	(0.00021)	(0.00012)	(0.00107)	(0.00097)	(0.00060)
<i>Controls</i>							
Grid & Year FE		Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit Statistics</i>							
Num.Obs.		2,225,802	2,225,802	2,225,802	372,364	372,364	372,364
R ²		0.006	0.007	0.006	0.008	0.011	0.011
Mean Dep. Var.		0.00022	0.00064	0.00026	0.00131	0.00384	0.00155

Table A.44: Grid-Year Panel with Daikansho-Inclusive Distance Ratio

Note: LPM on binary any-event-of-type indicators. Replicates Table 2 with the daikansho-inclusive distance ratio. Columns (1)–(3) use the full panel of 8,763 cells; columns (4)–(6) restrict to the 1,466 ever-active hexagons. See Table 2 notes for details. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

D.3. Additional Analyses

D.3.1. Daikansho-Inclusive Distance Ratio (Tax)

Table A.45 replaces the exposure measure in Table 4 with the daikansho-inclusive distance ratio. The controlled spatial-lag specification also uses neighboring population in levels rather than logs, so that comparison changes both exposure and a control transformation. Models 1–2 remain significant at $p < 0.01$; models 3–4 show modest p -value drift (from $p < 0.05$ to $p \approx 0.07$) with stable point estimates, consistent with daikansho offices compressing the distance ratio’s variance. The spatial-lag estimates (Models 5–6) remain negative. Model 5 is statistically significant (likelihood $p = 0.0055$; Conley $p = 0.0244$), whereas Model 6 is imprecise (likelihood $p = 0.107$; Conley $p = 0.114$).

Outcome:	Tax Rate (τ_j)					
	<i>OLS</i>				<i>Spatial Lag</i>	
Models:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
Distance ratio ($\ln(\rho_j)$)	−0.0233	−0.0192	−0.0098	−0.0096	−0.0130	−0.0080
HC SE	(0.0054)***	(0.0058)***	(0.0054)*	(0.0054)*	(0.0047)***	(0.0050)
<i>Controls</i>						
Domain Characteristics	No	Yes	Yes	Yes	No	Yes
Geography	No	No	Yes	Yes	No	Yes
Neighbor Characteristics	No	No	No	Yes	No	Yes
τ_j Spatial Lag	No	No	No	No	Yes	Yes
<i>Fit Statistics</i>						
Num.Obs.	232	230	230	230	230	230
R2 / Pseudo R2	0.071	0.138	0.368	0.386	0.309	0.437
Mean Dep. Var.	0.391	0.391	0.391	0.391	0.394	0.394

Table A.45: Domain Tax Rates with Daikansho-Inclusive Distance Ratio

Note: Tax models using the daikansho-inclusive distance ratio. Models 1–4: OLS with heteroskedasticity-robust standard errors. Models 5–6: spatial lag (SAR) with neighborhood-weighted tax rates as spatial regressor; standard errors from the spatial likelihood. Model 6 also uses neighboring population in levels rather than the logged control in Table 4, so its comparison changes both exposure and that control transformation. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.3.2. Controlling for Distance to the Border

Outcome:	exit _i					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
ln(ρ_i)	0.0487 (0.0161)***	0.0225 (0.0124)*	0.0152 (0.0098)	0.0206 (0.0068)***	0.0206 (0.0065)***	0.0264 (0.0075)***
ln(Shortest Distance to Border) _{t}	0.0060 (0.0103)	0.0073 (0.0085)	0.0010 (0.0106)	0.0057 (0.0089)	0.0058 (0.0086)	0.0037 (0.0086)
<i>Controls</i>						
Home (j) Domain FE	No	Yes	Yes	Yes	Yes	No
Neighbor (k) Domain FE	No	No	Yes	Yes	Yes	No
Decade (t) FE	No	No	No	Yes	Yes	Yes
Home-Neighbor Dyad (j, k) FE	No	No	No	No	No	Yes
<i>Fit Statistics</i>						
Num.Obs.	2,391	2,391	2,391	2,391	2,391	2,391
R2	0.037	0.377	0.535	0.632	0.638	0.676

Table A.46: Villages Exposed to Alternative Authority Opt for Exit Over Direct Confrontation

(Shortest Distance to Border)

Note: The coefficients relate the distance ratio (home-capital distance divided by distance to the nearest active non-home capital in the baseline daimyō/former-domain-capital geography) to exit choice conditional on recorded resistance, controlling for shortest distance to another domain's border. Border data match 2,391 of the 2,739 preferred-panel rows. On this covered sample, the preferred dyadic estimate is 0.0265 without and 0.0264 with the border control; the border coefficient is 0.0037 ($p = 0.668$). Clustered (home ruler, nearest neighboring ruler, and decade) standard errors are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

D.3.3. Sequential Decomposition of the Village Choice

The main analysis in [Section 3.4.3](#) conditions on recorded resistance and so cannot, on its own, rule out an association between geographic exposure and resistance incidence. I therefore combine the resisting-village data with the comprehensive end-period village ledger, adding compliant villages to the analysis. For each resisting village, I identify its counterpart in the full ledger, using the geographically nearest match when names do not resolve the record, and remove matched ledger rows to avoid duplication. A village-event remains a separate observation when the home or neighboring domain differs or when the event records a distinct tactic. The combined data contain 46,387 observations before covariate exclusions; actions by resisting villages are classified as exit or direct confrontation, while unmatched villages are labeled compliance. The multinomial models retain 46,365 observations with complete covariates.

On this combined sample, I decompose the village's three-way choice into two binary steps: Step 1 estimates recorded resistance versus compliance over the full village sample, and Step 2 estimates exit versus direct confrontation conditional on resistance. This is the two-step structure implied by a nested logit with resistance nest = {exit, confrontation}, and it can be estimated as two internally consistent binary models without the IIA restriction imposed by a flat multinomial logit.

Formally, let i index villages and let the choice set be $\mathcal{J} = \{C, E, F\}$ (compliance, exit, direct confrontation) with nest structure $B_0 = \{C\}$ and $B_1 = \{E, F\}$. Random utility is $U_{ij} = V_{ij} + \varepsilon_{ij}$, with the GEV distribution

$$F(\varepsilon_i) = \exp\left\{-e^{-\varepsilon_{iC}} - \left(e^{-\frac{\varepsilon_{iE}}{\lambda}} + e^{-\frac{\varepsilon_{iF}}{\lambda}}\right)^\lambda\right\},$$

where $\lambda \in (0, 1]$ is the dissimilarity parameter for the resistance nest ($\lambda = 1$ collapses to multinomial logit). With covariates X_i governing resistance incidence and Z_i governing tactic composition, the linear indices are $V_{iC} = 0$ (normalization), $V_{iE} = X_i'\gamma + Z_i'\delta_E$, and $V_{iF} = X_i'\gamma + Z_i'\delta_F$. Defining the inclusive value $I_i = \log\left[\exp\left(\frac{V_{iE}}{\lambda}\right) + \exp\left(\frac{V_{iF}}{\lambda}\right)\right]$, the closed-form choice probabilities are

$$\mathbb{P}(R_i = 1 \mid X_i, Z_i) = \frac{\exp(\lambda I_i)}{1 + \exp(\lambda I_i)}, \quad \mathbb{P}(E_i = 1 \mid R_i = 1, Z_i) = \Lambda\left((\delta_E - \delta_F)' \frac{Z_i}{\lambda}\right),$$

where $\Lambda(\cdot)$ is the logistic CDF. Letting $r_i = \mathbb{1}\{y_i \in \{E, F\}\}$ and $e_i = \mathbb{1}\{y_i = E\}$, the nested-logit log-likelihood factors as

$$\begin{aligned} \log \mathcal{L} = & \underbrace{\sum_i [r_i \log \mathbb{P}(R_i) + (1 - r_i) \log(1 - \mathbb{P}(R_i))]}_{\log \mathcal{L}_1(\psi_1)} \\ & + \underbrace{\sum_{i:r_i=1} [e_i \log \mathbb{P}(E_i | R_i) + (1 - e_i) \log(1 - \mathbb{P}(E_i | R_i))]}_{\log \mathcal{L}_2(\psi_2)}. \end{aligned}$$

Because $\log \mathcal{L}_1$ depends only on the top-nest parameters ψ_1 and $\log \mathcal{L}_2$ only on the within-nest contrast $\psi_2 = \frac{\delta_E - \delta_F}{\lambda}$, joint MLE coincides with the concatenation of two independent maximizations. Reparameterizing the singleton-nest top level into a single binary index $X_{i'}\beta_1$, the two pieces actually estimated are

$$\text{Step 1: } \mathbb{P}(R_i = 1 | X_i) = \Lambda(X_{i'}\beta_1), \quad i = 1, \dots, N,$$

$$\text{Step 2: } \mathbb{P}(E_i = 1 | R_i = 1, Z_i) = \Lambda(Z_{i'}\beta_2), \quad i \in \{i : r_i = 1\}.$$

β_1 identifies correlates of resistance incidence; β_2 identifies the within-resistance contrast $\frac{\delta_E - \delta_F}{\lambda}$, so its sign and significance are interpretable directly while its magnitude is identified jointly with the dissimilarity parameter λ . Estimating Step 1 and Step 2 separately writes the nested logit in sequential form, provided X_i and Z_i enter the linear indices as a clean partition and V_{iC} is normalized to zero.

[Table A.47](#) reports three complementary specifications side by side. Columns (1)–(2) report a flat multinomial logit (no fixed effects) as an IIA-restricted anchor. Columns (3)–(4) and (5)–(6) estimate the sequential decomposition as binary logit and linear probability model respectively. Step 1 (Columns (3) and (5)) runs on one row per unique village and uses no fixed effects because compliant villages drawn from the full ledger have no observed neighboring ruler or event date. Step 2 (Columns (4) and (6)) runs on the full resistance-event panel and matches [Table 3](#) Model 5: home $\times$ neighbor dyadic fixed effects, decade fixed effects, the same geographic controls, and three-way clustering on home ruler, neighbor, and decade.

The sequential decomposition addresses selection into recorded resistance directly. Step 1 starts from 46,296 unique villages and retains 46,274 after excluding 22 with missing controls. The binary logit gives $\beta = +0.028$ ($p = 0.58$), while the LPM gives $\beta = -0.00078$ ($p = 0.12$ heteroskedasticity-robust; $p = 0.42$ Conley); both are small and statistically indistinguishable

from zero. Step 2 is positive in both estimators: the binary logit gives $\beta = 0.276$ (cluster $p = 0.009$; Conley $p = 0.017$) and the LPM gives $\beta = 0.0244$ (cluster $p = 0.001$; Conley $p = 0.004$). The Step 1 logit used for Conley inference omits the latitude and longitude main effects while retaining their interaction, so it changes the specification as well as inference ($\beta = 0.0261$, $p = 0.734$). The Step 2 LPM reproduces Table 3 Model 5. Geographic exposure is therefore associated with tactic composition (exit versus direct confrontation) without a detectable positive association with resistance incidence in this specification.

	<i>Multinomial logit</i>		<i>Seq. binary logit</i>		<i>Seq. LPM</i>	
Outcome:	Exit	Direct confrontation	Non- compliance	Exit non-compli- ance	Non- compliance	Exit non-compli- ance
Models:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
Distance ratio ($\ln(\rho_i)$)	0.2823	−0.0055	0.0281	0.2762	−0.00078	0.0244
Primary SE (see note)	(0.0149)***	(0.0167)	(0.0514)	(0.1054)***	(0.00050)	(0.0068)***
Conley SE (see note)	—	—	(0.0771)	(0.1159)**	(0.00096)	(0.0085)***
<i>Controls</i>						
Geo controls	Yes	Yes	Yes	Yes	Yes	Yes
Home × Neighbor FE	No	No	No	Yes	No	Yes
Decade FE	No	No	No	Yes	No	Yes
<i>Fit Statistics</i>						
Num. Obs.	46,365	46,365	46,274	1,310	46,274	2,739
Mean Dep. Var.	0.014	0.042	0.054	0.246	0.054	0.246

Table A.47: Sequential Decomposition of the Village Choice: Logit and LPM

Note: Columns (1)–(2) report a flat multinomial logit (no fixed effects) as an IIA-restricted anchor; both coefficients come from a single regression with compliance as the baseline category. Columns (3)–(4) and (5)–(6) estimate the sequential decomposition as binary logit and linear probability model respectively. Step 1 (Columns (3) and (5)) runs on one row per unique village (non-compliance vs. compliance); it omits dyadic and decade fixed effects because compliant villages drawn from the full ledger have no observed neighboring ruler or event date. Step 2 (Columns (4) and (6)) runs on the full non-compliance-event panel (exit vs. direct confrontation conditional on non-compliance) and uses the specification of Table 3 Model 5: home × neighbor dyadic fixed effects, decade fixed effects, and the same geographic controls. Two SE rows are reported in parentheses. The first is model-based for Columns (1)–(3), heteroskedasticity-robust for Column (5), and three-way clustered on home ruler, neighbor, and decade for Columns (4) and (6). The second is Conley spatial inference with a 10 km cutoff, omitted for Columns (1)–(2) because no standard implementation exists for the multinomial likelihood. Column (3) Conley inference comes from a separate logit that includes the latitude-longitude interaction but omits their main effects; the reported primary coefficient includes both main effects. The alternate coefficient is 0.0261 ($p = 0.734$, $N = 46,274$). Other Conley rows retain their column specifications. Observation counts refer to fitted samples; dependent-variable means describe the input samples before estimation exclusions. Stars reflect the corresponding estimator’s p -value. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

D.3.4. Sequential Decomposition Under Time-Averaged Distance Ratio

The distance ratio used in the main Step-1 regression (Table A.47) differs across outcomes: resisting villages receive the event-time ratio, while compliant villages from the full ledger receive the end-period (1868) ratio. Table A.48 applies to both groups each village’s mean annual home-capital distance divided by nearest-foreign-capital distance over 1615–1868, with active capitals reconstructed each year. This check also retains the earlier neighboring-domain assignments and dyads, which differ from those reconstructed for the preferred specification. It therefore changes both the timing convention and the assignment of neighboring authorities, and does not isolate the effect of time averaging alone. Under this symmetric construction Step 1 is small and negative ($\hat{\beta}_{\text{logit}} = -0.0303$, $p = 0.0402$; LPM $\hat{\beta} = -0.00155$, $p = 0.0387$). Step 2, exit versus direct confrontation conditional on resistance, retains its positive sign ($\hat{\beta} = 0.01436$, cluster $p = 0.0089$, Conley $p = 0.0524$). Under this combination of time averaging and earlier neighbor assignments, the composition association remains positive, while exposure does not positively predict selection into recorded resistance.

Inference	<i>Step 1: non-compliance vs. compliance</i>					<i>Step 2: exit vs. direct confrontation</i>				
	Logit	LPM (robust)	LPM (Conley 10 km)	10		LPM (3-way cluster)	clus-	LPM (Conley 10 km)	10	
$\hat{\beta}$ on $\ln(\hat{\rho}_i^{\text{avg}})$	-0.0303	-0.00155	-0.00155			0.01436		0.01436		
SE	0.0148	0.00075	0.00072			0.00506		0.00740		
p	0.0402	0.0387	0.0311			0.0089		0.0524		
Geo controls	Yes	Yes	Yes			Yes		Yes		
Home $\times$ Neighbor FE	No	No	No			Yes		Yes		
Decade FE	No	No	No			Yes		Yes		
Num. Obs.	46024	46024	46024			2740		2740		
Mean Dep. Var.	0.0543	0.0543	0.0543			0.2464		0.2464		

Table A.48: Sequential Decomposition Under Time-Averaged Distance Ratio

Note: Replicates the sequential decomposition in Table A.47 with a time-averaged distance ratio applied symmetrically to resisting and compliant villages. The time-averaged ratio is the mean of the annual distance-to-home-domain capital / distance-to-nearest-foreign-domain capital across 1615-1868, with the active-domain panel recomputed each year using the augmented tenryo spells. This symmetric construction removes the measurement asymmetry in the existing Step-1 table, where the distance ratio for resisting villages used event-time capitals while that for compliant villages used end-period positions. Step 1 (non-compliance vs. compliance) is estimated on all 46,024 villages (2,497 non-compliant); Step 2 (exit vs. direct confrontation conditional on non-compliance) uses its own 2,740-event time-averaged analytic sample. Under time-averaged capitals Step 1 is a small negative association ($\hat{\beta}_{\text{logit}} = -0.030$, $p = 0.040$; LPM $\hat{\beta} = -0.00155$, $p = 0.039$), and Step 2 retains its positive sign ($\hat{\beta} = 0.014$, $p = 0.009$ cluster, $p = 0.052$ Conley). The symmetric specification therefore provides no evidence that exposure positively selects villages into recorded non-compliance while preserving the positive composition estimate conditional on non-compliance.

D.3.5. Linear Hypothesis Testing

In the village-level analysis for validating the village mobility measure, I use the log-ratio of home-capital distance to nearby-capital distance:

$$\ln\left(\frac{\text{Dist. to Home Cap.}}{\text{Dist. to Foreign Cap.}}\right) = \ln(\text{Dist. to Home Cap.}) - \ln(\text{Dist. to Foreign Cap.})$$

Implicit in this choice is the restriction: $\beta_{\ln(\text{Dist. to Home Cap.})} + \beta_{\ln(\text{Dist. to Foreign Cap.})} = 0$

To check whether the data are consistent with this restriction, I estimate the unrestricted model including both $\ln(\text{Dist. to Home Cap.})$ and $\ln(\text{Dist. to Foreign Cap.})$ separately, then conduct a Wald test of the linear hypothesis $\beta_{\ln(\text{Dist. to Home Cap.})} + \beta_{\ln(\text{Dist. to Foreign Cap.})} = 0$.

Corresponding Model	Outcome	Restriction	Df	F	p-value
Table 3 (7)	exit _i	$\beta_{\ln(\text{DHC})} + \beta_{\ln(\text{DFC})} = 0$	(1, 25)	0.4035	0.5311

Table A.49: Wald Test of the Log-Ratio Restriction

The Wald test uses the preferred dyadic model and standard errors clustered by home domain, neighboring domain, and decade. It does not reject the equal-and-opposite restriction ($p = 0.531$); nonrejection does not establish that the restriction holds.

E. DOMAIN TAXATION ANALYSIS

E.1. Full Tables

E.1.1. Main Model

Outcome:	τ_j											
Models:	(1)		(2)		(3)		(4)		(5: SPL)		(6: SPL)	
<i>Variables</i>												
$\ln(\rho_j)$	-0.0241	(0.0053)***	-0.0203	(0.0058)***	-0.0110	(0.0055)**	-0.0112	(0.0055)**	-0.0140	(0.0047)***	-0.0094	(0.0051)*
$\ln(\text{Rice})$			-0.0010	(0.0158)	0.0076	(0.0157)	0.0116	(0.0163)			0.0004	(0.0135)
$\ln(\text{Area})$			-0.0260	(0.0112)**	-0.0318	(0.0144)**	-0.0340	(0.0151)**			-0.0227	(0.0130)*
$\ln(\text{Pop.})$			0.0392	(0.0167)**	0.0263	(0.0144)*	0.0232	(0.0150)			0.0222	(0.0147)
New FL			0.0002	(0.0003)	0.0005	(0.0004)	0.0003	(0.0004)			0.0004	(0.0002)
Shog. Ally			-0.0316	(0.0177)*	0.0059	(0.0161)	0.0011	(0.0164)			0.0025	(0.0143)
Goshi			0.0217	(0.0256)	0.0196	(0.0261)	0.0274	(0.0271)			0.0242	(0.0215)
lat					-0.5279	(0.2934)*	-0.4747	(0.3077)			-0.3535	(0.2855)
lon					-0.1758	(0.0754)**	-0.1639	(0.0796)**			-0.1252	(0.0743)*
$\ln(\text{Elev.})$					-0.0118	(0.0160)	-0.0137	(0.0165)			-0.0100	(0.0134)
$\ln(\text{Rugg.})$					0.0398	(0.0160)**	0.0373	(0.0158)**			0.0238	(0.0152)
$\ln(\text{Dist. Edo/Kyoto})$					-0.0370	(0.0158)**	-0.0441	(0.0171)**			-0.0335	(0.0167)**
Rainy					-1.6629	(0.8969)*	-1.5887	(0.9078)*			-1.0143	(0.7369)
Snowy					0.7322	(1.8580)	0.6920	(1.8597)			-0.2419	(1.4193)
Lat $\times$ Lon					0.0042	(0.0021)*	0.0038	(0.0023)*			0.0029	(0.0021)
Kin Nbs.							0.0396	(0.0296)			0.0188	(0.0257)
Nb. FL							0.0002	(0.0001)			0.0001	(0.0001)
$\ln(\text{Nb. Rice})$							-0.0244	(0.0189)			-0.0221	(0.0172)
$\ln(\text{Nb. Area})$							0.0180	(0.0176)			0.0116	(0.0159)
$\ln(\text{Nb. Pop.})$							-0.0047	(0.0207)			-0.0015	(0.0196)
Const.	0.4152	(0.0091)***	0.1760	(0.1247)	23.1744	(10.2659)**	21.8425	(10.7990)**	0.1761	(0.0274)***	16.4558	(10.0938)
<i>Fit Statistics</i>												
Num.Obs.	232		230		230		230		230		230	
R2	0.074		0.141		0.370		0.389		0.312 ^[a]		0.439 ^[a]	

Table A.50: Geographic Exposure and Reported Domain Tax Rates

Note: OLS estimates (maximum likelihood in Models 5–6) of reported effective tax rates on the log domain-average distance ratio. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Models 5–6. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1. ^[a] Nagelkerke pseudo-R2

E.1.2. Interaction Model

Outcome:	τ_j			
Models:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Distance ratio ($\ln(\rho_j) \times \hat{y}_{-j}$): High	−0.0350 (0.0084)***	−0.0286 (0.0089)***	−0.0143 (0.0070)**	−0.0158 (0.0069)**
Distance ratio ($\ln(\rho_j) \times \hat{y}_{-j}$): Low	−0.0161 (0.0053)***	−0.0151 (0.0058)***	−0.0092 (0.0063)	−0.0085 (0.0064)
<i>Controls</i>				
Main	No	Yes	Yes	Yes
Geography	No	No	Yes	Yes
Neighbors	No	No	No	Yes
<i>Fit Statistics</i>				
Num.Obs.	232	230	230	230
R2	0.091	0.150	0.371	0.391
Mean Dep. Var.	0.392	0.394	0.394	0.394

Table A.51: Geographic Exposure and Tax Rates, Interacted with Outside-Option Quality

Note: The table reports group-specific OLS slopes for the log distance ratio. High denotes a below-median weighted labor-to-potential-yield ratio; Low denotes an above-median ratio. HC1 standard errors in parentheses.

E.1.3. Interaction Model (Full Table)

Outcome:	τ_j							
Models:	(1)		(2)		(3)		(4)	
<i>Variables</i>								
$\ln(\rho_j) \times \text{High}$	−0.0350	(0.0084)***	−0.0286	(0.0089)***	−0.0143	(0.0070)**	−0.0158	(0.0069)**
$\ln(\rho_j) \times \text{Low}$	−0.0161	(0.0053)***	−0.0151	(0.0058)***	−0.0092	(0.0063)	−0.0085	(0.0064)
$\ln(\text{Rice})$			0.0014	(0.0156)	0.0083	(0.0153)	0.0122	(0.0159)
$\ln(\text{Area})$			−0.0256	(0.0111)**	−0.0314	(0.0144)**	−0.0336	(0.0151)**
$\ln(\text{Pop.})$			0.0360	(0.0171)**	0.0253	(0.0145)*	0.0221	(0.0151)
New FL			0.0002	(0.0003)	0.0005	(0.0004)	0.0003	(0.0004)
Shog. Ally			−0.0300	(0.0180)*	0.0064	(0.0163)	0.0016	(0.0166)
Goshi			0.0221	(0.0257)	0.0197	(0.0262)	0.0278	(0.0271)
lat					−0.5362	(0.2946)*	−0.4835	(0.3095)
lon					−0.1785	(0.0758)**	−0.1668	(0.0802)**
$\ln(\text{Elev.})$					−0.0111	(0.0160)	−0.0125	(0.0165)
$\ln(\text{Rugg.})$					0.0381	(0.0163)**	0.0348	(0.0161)**
$\ln(\text{Dist. Edo/Kyoto})$					−0.0373	(0.0159)**	−0.0445	(0.0172)**
Rainy					−1.6210	(0.9021)*	−1.5333	(0.9156)*
Snowy					0.6224	(1.8655)	0.5548	(1.8808)
Lat $\times$ Lon					0.0042	(0.0022)*	0.0038	(0.0023)*
Kin Nbs.							0.0404	(0.0297)
Nb. FL							0.0002	(0.0001)
$\ln(\text{Nb. Rice})$							−0.0224	(0.0190)
$\ln(\text{Nb. Area})$							0.0191	(0.0174)
$\ln(\text{Nb. Pop.})$							−0.0081	(0.0206)
Const.	0.4181	(0.0092)***	0.1754	(0.1211)	23.4988	(10.3154)**	22.2086	(10.8672)**
<i>Fit Statistics</i>								
Num.Obs.	232		230		230		230	
R2	0.091		0.150		0.371		0.391	

Table A.52: Geographic Exposure and Reported Tax Rates by Outside-Option Quality
(Full Table)

Note: The coefficients represent OLS estimates of the relationship between the average distance ratio of each domain interacted with $h@(\mathbf{y})_{\{-j\}}$ (indicator of whether the weighted average labor-to-potential-yield ratio of neighboring domains is above its median) and the domain tax rate. Heteroskedasticity-robust standard errors in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

E.2. Robustness Checks

E.2.1. Alternative Covariates

Outcome:	τ_j									
Models:	(1)		(2)		(3)		(4)		(5)	
<i>Variables</i>										
$\ln(\rho_j)$	−0.0099	(0.0049)**	−0.0106	(0.0057)*	−0.0112	(0.0058)*	−0.0108	(0.0055)**	−0.0131	(0.0063)**
$\ln(\text{Kokudaka})$	−0.0788	(0.0160)***								
$\ln(\text{Pop.})$	0.0820	(0.0168)***							0.0282	(0.0156)*
$\ln(\text{Area})$	−0.0050	(0.0127)	−0.0328	(0.0128)**	−0.0251	(0.0142)*	−0.0052	(0.0064)	−0.0350	(0.0166)**
New FL	0.0003	(0.0004)	0.0003	(0.0004)	0.0003	(0.0004)	0.0004	(0.0004)	0.0003	(0.0003)
Shog. Ally	0.0068	(0.0154)	−0.0057	(0.0165)	0.0009	(0.0171)	0.0032	(0.0167)	0.0069	(0.0160)
Goshi	0.0245	(0.0260)	0.0261	(0.0272)	0.0309	(0.0269)	0.0267	(0.0273)	0.0321	(0.0248)
lat	−0.3791	(0.2791)	−0.4525	(0.3160)	−0.5680	(0.3505)	−0.3697	(0.3181)	0.1399	(0.2658)
lon	−0.1412	(0.0729)*	−0.1581	(0.0828)*	−0.1879	(0.0909)**	−0.1392	(0.0834)*	0.0166	(0.0673)
$\ln(\text{Elev.})$	−0.0064	(0.0157)	−0.0144	(0.0161)	−0.0155	(0.0165)	−0.0203	(0.0163)	−0.0081	(0.0167)
$\ln(\text{Rugg.})$	0.0143	(0.0156)	0.0352	(0.0159)**	0.0338	(0.0164)**	0.0296	(0.0157)*	0.0250	(0.0152)
$\ln(\text{Dist. Edo/Kyoto})$	−0.0418	(0.0165)**	−0.0423	(0.0173)**	−0.0435	(0.0172)**	−0.0481	(0.0168)***		
Snowy	1.8194	(1.7508)	0.4931	(1.8551)	0.9999	(1.9673)	1.0110	(1.9217)	−0.8507	(2.0177)
Rainy	−2.0637	(0.8820)**	−1.4692	(0.8923)	−1.7103	(0.9270)*	−1.6807	(0.9375)*	−1.0287	(0.9469)
Kin Nbs.	0.0266	(0.0301)	0.0500	(0.0283)*	0.0418	(0.0292)	0.0468	(0.0290)	0.0177	(0.0304)
Nb. FL	0.0001	(0.0001)	0.0002	(0.0001)	0.0002	(0.0001)	0.0002	(0.0001)	0.0002	(0.0001)*
$\ln(\text{Nb. Rice})$	−0.0050	(0.0174)	−0.0296	(0.0189)	−0.0291	(0.0199)	−0.0222	(0.0186)	−0.0190	(0.0187)
$\ln(\text{Nb. Area})$	−0.0019	(0.0170)	0.0176	(0.0167)	0.0177	(0.0176)	0.0100	(0.0163)	0.0073	(0.0182)
$\ln(\text{Nb. Pop.})$	−0.0073	(0.0186)	0.0018	(0.0203)	0.0005	(0.0215)	0.0011	(0.0202)	−0.0039	(0.0206)
Lat × Lon	0.0031	(0.0021)	0.0036	(0.0023)	0.0044	(0.0026)*	0.0030	(0.0023)	−0.0008	(0.0020)
$\ln(\text{Rice})$			−0.0017	(0.0250)	0.0239	(0.0151)			0.0047	(0.0181)
$\ln(\text{Num. Vils.})$			0.0373	(0.0250)						
% Samurai Pop					−0.1259	(0.1834)				
Population/wet-rice suitability							−0.0040	(0.2763)		
$\ln(\text{Dist. Edo})$									0.0677	(0.0261)**
Const.	18.8458	(9.8989)*	21.3229	(11.1659)*	25.2642	(12.3202)**	18.7213	(11.2921)*	−3.1359	(8.9956)
<i>Fit Statistics</i>										
Num.Obs.	230		231		226		230		230	
R2	0.446		0.397		0.383		0.374		0.399	

Table A.53: Geographic Exposure and Reported Domain Tax Rates
(Alternative Covariates)

Note: OLS estimates relating each domain's average distance ratio to its reported tax rate. The distance ratio measures geographic exposure, not observed competition or bargaining between rulers. Heteroskedasticity-robust standard errors are in parentheses. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

E.2.2. Sensitivity Analysis

I assess the robustness of the domain-level results to omitted variable bias using the breakdown frontier framework of Diegert, Masten, and Poirier (2022). This method improves upon the widely used approach of Oster (2019) in a critical respect: it remains valid when the included control variables are correlated with the omitted confounders—a condition Diegert, Masten, and Poirier (2022) term *endogenous controls*. Standard sensitivity methods assume that the controls are exogenous, an assumption that is difficult to defend in observational cross-sections where controls such as population, productivity, and daimyō characteristics may themselves be influenced by unobserved factors. As Diegert, Masten, and Poirier (2022) show, residualization-based methods can produce incorrect robustness conclusions when this assumption is violated.

The key sensitivity parameter is $\bar{r}_X$, the ratio of the standard deviation of unobservables' influence on geographic exposure to that of observables. Figure A.9 displays the admissible set for the exposure coefficient as $\bar{r}_X$ varies under the most conservative assumptions. The negative sign remains identified for $\bar{r}_X < 0.213$: at that threshold, the upper endpoint of the DMP identified set first includes zero. This is an SD ratio, not a share of predictive power. The calculation evaluates the observational coefficient; it does not convert the cross-section into a causal design.

When the impact of unobservables on the outcome is also bounded ($\bar{r}_Y \leq \bar{r}_X$, the “equal-importance” assumption), the identified set tightens considerably. Figure A.10 plots the breakdown frontier in the $(\bar{r}_X, \bar{r}_Y)$ plane. The diagonal sign-identification breakdown is $\bar{r} = 0.554$: both SD ratios must reach about this level before the identified set admits a nonnegative coefficient.

Those breakdowns concern population sign identification, not sampling significance. To answer the latter question transparently, Table A.54 also reports auxiliary thresholds that hold the baseline HC1 standard error fixed and ask when the upper DMP endpoint plus 1.96 standard errors reaches zero. The thresholds are $\bar{r}_X = 0.007$ with no outcome-side bound and $\bar{r}_X = \bar{r}_Y = 0.096$ under equal importance. These are fixed-SE diagnostics rather than fully bias-adjusted DMP confidence sets. They show that the observational association is much easier to render statistically imprecise than to make its negative sign unidentified.

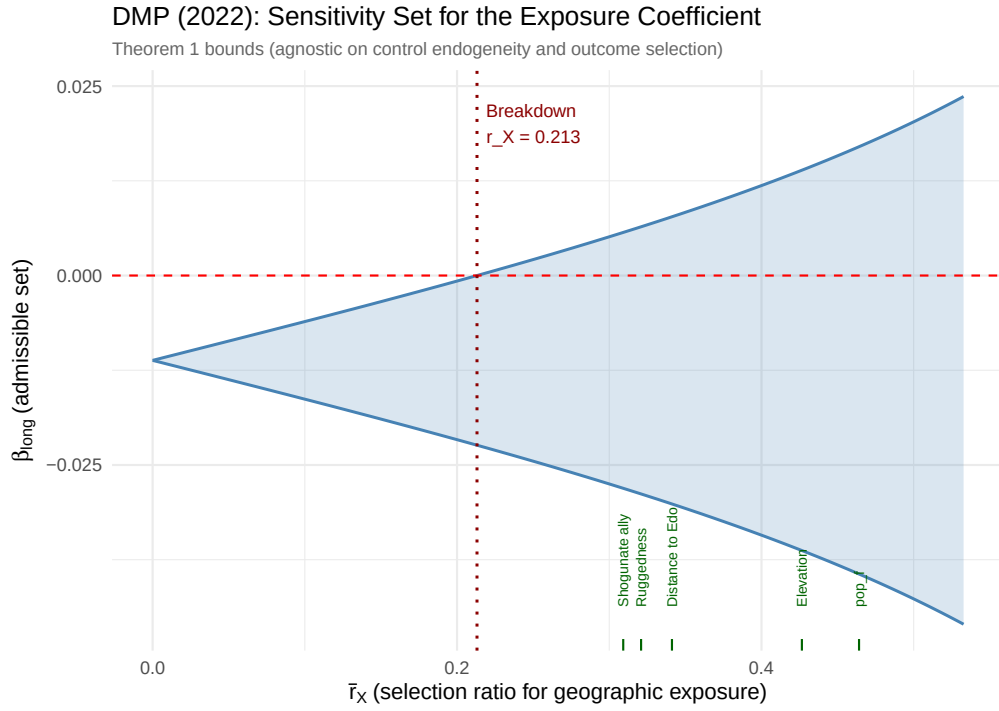


Figure A.9: DMP Sensitivity Set for the Exposure Coefficient

Note: The shaded region shows the range of β_{long} consistent with a given level of selection on unobservables ($\bar{r}_X$). The dashed red line marks $\beta = 0$; the dotted vertical line marks the breakdown point ($\bar{r}_X = 0.213$). Green ticks calibrate $\bar{r}_X$ to each labeled covariate's importance for predicting geographic exposure.

Sensitivity criterion	$\bar{r}$	Interpretation
Sign identification: $\bar{r}_X$ only	0.213	Upper endpoint of the DMP identified set first includes zero
Sign identification: $\bar{r}_X = \bar{r}_Y$	0.554	Same criterion under the equal-importance restriction
Fixed-SE significance: $\bar{r}_X$ only	0.007	Upper endpoint plus 1.96 baseline HC1 SEs first includes zero
Fixed-SE significance: $\bar{r}_X = \bar{r}_Y$	0.096	Same auxiliary criterion under equal importance

Table A.54: Diegert-Masten Sensitivity Thresholds for the Tax Regression

Note: Thresholds for Table 4 Model 4 (fully controlled). $\bar{r}_X$ and $\bar{r}_Y$ are ratios of standard deviations, not shares of explained variance. The first two rows are DMP population sign-identification breakdowns: the observed estimate remains fixed while the identified set widens to include zero. The last two rows answer the separate significance question by adding and subtracting 1.96 times the baseline HC1 SE from the DMP set while holding that SE fixed. They are transparent auxiliary thresholds, not fully bias-adjusted DMP confidence sets.

As a complementary sample-based check, I re-estimate Model 4 after excluding Chōshū, Satsuma, Mito, Tosa, and Aizu, whose 1850s–1860s rates may reflect acute bakumatsu fiscal and political distress. Excluding them changes the point estimate from -0.0112 to -0.0106 , with the p -value moving from 0.045 to 0.055 (Table A.55). The sign and magnitude are therefore not attributable to these five high-profile domains.

Sample	Num. Obs.	Estimate	SE	p
Full sample (paper)	230	−0.0112	(0.0055)	0.045
Excluding Chōshū, Satsuma, Mito, Tosa, Aizu	225	−0.0106	(0.0055)	0.055

Table A.55: Domain Tax Regression: Excluding Bakumatsu Political Heavyweights

Note: Re-runs Table 4 Model 4 (fully controlled) on the sub-sample that excludes five domains (Chōshū, Satsuma, Mito, Tosa, Aizu) whose tax rates in the 1850s–1860s reflect acute fiscal and political distress rather than ordinary late-period tax setting. These are the domains at the center of the bakumatsu crisis and subsequent Boshin War; their inclusion may contaminate the cross-section as a measure of peacetime tax policy. The sign and magnitude of the coefficient on $\log(\rho_j)$ are preserved while precision becomes marginal.

For completeness, Table A.56 reports the Oster (2019) coefficient-stability parameter. This method, while informative, assumes exogenous controls and is therefore best interpreted as supplementary to the DMP analysis above.

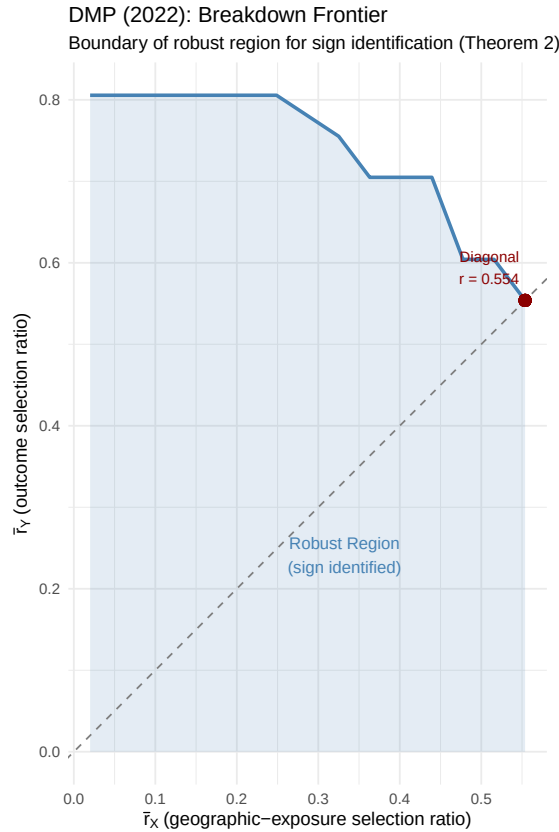


Figure A.10: DMP Breakdown Frontier

Note: The solid curve traces the sensitivity boundary: below it, the exposure coefficient remains negative under the maintained bounds. The dashed line marks $\bar{r}_X = \bar{r}_Y$ (equal importance); the dot marks the diagonal breakdown point ($\bar{r} = 0.554$).

Oster (2019) Coefficient Stability					
	$\hat{\beta}_{\text{short}}$	$\hat{\beta}_{\text{full}}$	R^2_{short}	R^2_{full}	δ
$R_{\text{max}} = 1.3\tilde{R}^2$	-0.0245	-0.0112	0.077	0.389	2.245
$R_{\text{max}} = 1$	-0.0245	-0.0112	0.077	0.389	0.428
$\hat{\beta}^*(\delta = 1) = -0.006$ at $R_{\text{max}} = 1.3\tilde{R}^2$					

Table A.56: Oster Sensitivity Analysis (Supplementary)

Note: Based on Model 4 in [Table 4](#). The Oster δ is the proportional selection ratio needed to drive $\hat{\beta}^*$ to zero; $\delta > 1$ indicates robustness under $R_{\text{max}} = 1.3\tilde{R}^2$. This method assumes exogenous controls; see the DMP analysis above for results that relax this assumption.

E.2.3. Data-Driven Control Variables Selection

The hand-picked control set in [Table 4](#) reflects substantive reasoning about confounders, but the choice of which variables to include is itself a researcher degree of freedom. I assess this concern using double/debiased machine learning (DML), which replaces manual nuisance-function specification with cross-fitted prediction ([Chernozhukov et al. 2018](#)). DML estimates a partially linear model

$$\tau_j = \beta \ln(\rho_j) + g(X_j) + \varepsilon_j, \quad \ln(\rho_j) = m(X_j) + \nu_j \quad (1)$$

where $g(\cdot)$ and $m(\cdot)$ predict the outcome and geographic exposure, and β is recovered after partialling both variables on X_j . Five-fold cross-fitting reduces overfitting bias in these nuisance estimates. I report two feature dictionaries: the 19 columns that reproduce the controls in [Table 4](#) Model 4, and an enriched 38-feature dictionary that adds quadratic terms and selected pairwise interactions. Within the exact dictionary I compare ridge and random-forest nuisance learners; within the enriched dictionary I compare ridge with a stack of LASSO, ridge, elastic net, and random forest.

The results are mixed. Ridge estimates remain negative: the exact-dictionary estimate is statistically distinguishable from zero at the 5% level, while the enriched-dictionary estimate is marginal at the 10% level. The random-forest and stacked estimates are close to zero and imprecise. DML therefore does not uniformly corroborate the hand-picked OLS association; instead, it shows that the pooled estimate is sensitive to highly flexible nuisance learning in this modest domain-level sample ($N = 230$).

Outcome: τ_j . Treatment: $\ln(\rho_j)$						
Method	Estimate	S.E.	t	p	95% CI lo	95% CI hi
DML-Ridge (19 exact)	−0.0114	(0.0056)	−2.03	0.042	−0.0225	−0.0004
DML-Random Forest (19 exact)	−0.0007	(0.0054)	−0.13	0.895	−0.0113	0.0099
DML-Ridge (38 enriched)	−0.0098	(0.0057)	−1.74	0.082	−0.0209	0.0013
DML-Stacked Ensemble (38 enriched)	0.0018	(0.0055)	0.32	0.745	−0.0090	0.0126
Hand-picked OLS (HC1)	−0.0112	(0.0055)	−2.02	0.045	−0.0221	−0.0003
<i>Exact dictionary: $p = 19$ Model 4 columns. Enriched dictionary: $p = 38$ (17 base + 8 quadratics + 10 interactions + 3 kin dummies).</i>						
<i>DML: 5 folds and 10 repetitions, except 5 repetitions for the enriched stack. Stack: LASSO + Ridge + Elastic Net + random forest.</i>						

Table A.57: Data-Driven Control Selection: DML Estimates

Note: Each DML row estimates the partial association between $\ln(\rho_j)$ and τ_j using the indicated nuisance learner and feature dictionary. Reported standard errors incorporate repeated cross-fitting. The OLS row reproduces Model 4 from Table 4 with HC1 standard errors.

E.2.4. Controlling for Average Distance to Home Capital

Outcome:	τ_j									
Models:	(1)		(2)		(3)		(4)		(5: SPL)	
<i>Variables</i>										
$\ln(\rho_j)$	-0.0336	(0.0116)***	-0.0200	(0.0132)	-0.0148	(0.0135)	-0.0171	(0.0143)	-0.0231	(0.0136)*
$\ln(\text{Avg. Dist to HC}_j)$	0.0132	(0.0156)	-0.0004	(0.0169)	0.0049	(0.0165)	0.0077	(0.0178)	0.0178	(0.0163)
$\ln(\text{Rice})$			-0.0011	(0.0160)	0.0083	(0.0154)	0.0121	(0.0161)	0.0010	(0.0135)
$\ln(\text{Area})$			-0.0259	(0.0123)**	-0.0328	(0.0138)**	-0.0353	(0.0147)**	-0.0254	(0.0132)*
$\ln(\text{Pop.})$			0.0393	(0.0167)**	0.0261	(0.0145)*	0.0231	(0.0150)	0.0219	(0.0146)
New FL			0.0002	(0.0003)	0.0005	(0.0004)	0.0003	(0.0004)	0.0004	(0.0002)
Shog. Ally			-0.0316	(0.0178)*	0.0059	(0.0162)	0.0010	(0.0165)	0.0023	(0.0143)
Goshi			0.0217	(0.0257)	0.0196	(0.0262)	0.0274	(0.0270)	0.0241	(0.0214)
lat					-0.5324	(0.2940)*	-0.4736	(0.3085)	-0.3473	(0.2845)
lon					-0.1771	(0.0755)**	-0.1637	(0.0798)**	-0.1236	(0.0740)*
$\ln(\text{Elev.})$					-0.0117	(0.0159)	-0.0134	(0.0164)	-0.0094	(0.0134)
$\ln(\text{Rugg.})$					0.0397	(0.0160)**	0.0372	(0.0158)**	0.0232	(0.0151)
$\ln(\text{Dist. Edo/Kyoto})$					-0.0383	(0.0172)**	-0.0451	(0.0177)**	-0.0356	(0.0168)**
Rainy					-1.6816	(0.9046)*	-1.6114	(0.9179)*	-1.0491	(0.7358)
Snowy					0.7952	(1.9057)	0.7810	(1.9144)	-0.0637	(1.4268)
Lat $\times$ Lon					0.0042	(0.0021)*	0.0038	(0.0023)*	0.0028	(0.0021)
Kin Nbs.							0.0391	(0.0299)	0.0171	(0.0256)
Nb. FL							0.0002	(0.0001)	0.0001	(0.0001)
$\ln(\text{Nb. Rice})$							-0.0234	(0.0188)	-0.0196	(0.0172)
$\ln(\text{Nb. Area})$							0.0170	(0.0177)	0.0089	(0.0160)
$\ln(\text{Nb. Pop.})$							-0.0060	(0.0208)	-0.0044	(0.0197)
Const.	0.3815	(0.0424)***	0.1773	(0.1374)	23.3407	(10.2900)**	21.8218	(10.8253)**	16.2389	(10.0560)
<i>Fit Statistics</i>										
Num.Obs.	232		230		230		230		230	
R2	0.077		0.141		0.371		0.389		0.442 ^[a]	

Table A.58: Controlling for Average Distance to Home Capital

Note: OLS estimates (maximum likelihood in Model 5) relating each domain's average distance ratio to its reported tax rate while controlling for average distance to the home capital. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Model 5. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1. ^[a] Nagelkerke pseudo-R2

E.2.5. Least-Costly Path Distance

Outcome:	τ_j									
Models:	(1)		(2)		(3)		(4)		(5: SPL)	
<i>Variables</i>										
$\ln(\rho_j^{\text{LCP}})$	-0.0116	(0.0035)***	-0.0097	(0.0037)***	-0.0058	(0.0030)*	-0.0060	(0.0031)*	-0.0053	(0.0032)
$\ln(\text{Rice})$			-0.0033	(0.0147)	0.0071	(0.0150)	0.0110	(0.0157)	-0.0003	(0.0135)
$\ln(\text{Area})$			-0.0242	(0.0112)**	-0.0308	(0.0139)**	-0.0333	(0.0147)**	-0.0220	(0.0130)*
$\ln(\text{Pop.})$			0.0419	(0.0165)**	0.0268	(0.0143)*	0.0239	(0.0149)	0.0228	(0.0147)
New FL			0.0002	(0.0003)	0.0005	(0.0004)	0.0003	(0.0004)	0.0004	(0.0002)
Shog. Ally			-0.0355	(0.0178)**	0.0052	(0.0162)	0.0005	(0.0165)	0.0022	(0.0144)
Goshi			0.0210	(0.0257)	0.0198	(0.0263)	0.0279	(0.0272)	0.0246	(0.0215)
lat					-0.4938	(0.2899)*	-0.4356	(0.3059)	-0.3222	(0.2840)
lon					-0.1682	(0.0746)**	-0.1547	(0.0792)*	-0.1177	(0.0740)
$\ln(\text{Elev.})$					-0.0101	(0.0162)	-0.0120	(0.0167)	-0.0087	(0.0134)
$\ln(\text{Rugg.})$					0.0391	(0.0159)**	0.0366	(0.0157)**	0.0232	(0.0152)
$\ln(\text{Dist. Edo/Kyoto})$					-0.0371	(0.0160)**	-0.0440	(0.0174)**	-0.0334	(0.0167)**
Rainy					-1.6610	(0.8949)*	-1.6083	(0.9031)*	-1.0184	(0.7381)
Snowy					0.7901	(1.8375)	0.7935	(1.8372)	-0.1818	(1.4196)
Lat $\times$ Lon					0.0039	(0.0021)*	0.0035	(0.0022)	0.0026	(0.0021)
Kin Nbs.							0.0408	(0.0296)	0.0195	(0.0257)
Nb. FL							0.0002	(0.0001)	0.0001	(0.0001)
$\ln(\text{Nb. Rice})$							-0.0228	(0.0189)	-0.0208	(0.0172)
$\ln(\text{Nb. Area})$							0.0197	(0.0177)	0.0128	(0.0158)
$\ln(\text{Nb. Pop.})$							-0.0079	(0.0204)	-0.0041	(0.0195)
Const.	0.4083	(0.0092)***	0.1663	(0.1205)	22.1064	(10.1745)**	20.5929	(10.7617)*	15.4412	(10.0485)
<i>Fit Statistics</i>										
Num.Obs.	232		230		230		230		230	
R2	0.037		0.119		0.367		0.385		0.437 ^[a]	

Table A.59: Geographic Exposure and Reported Tax Rates

(Least-Cost Paths)

Note: OLS estimates (maximum likelihood in Model 5) relating each domain's stored least-cost-path distance ratio to its reported tax rate. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Model 5. The ratio measures geographic exposure, not observed ruler competition. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1. ^[a] Nagelkerke pseudo-R2

E.2.6. Conley Standard Error Bandwidth Sensitivity

Table A.60 reports the bandwidth sensitivity of Conley spatial standard errors for the fully-controlled domain tax regression (Table 4 Model 4). The point estimate is stable across every bandwidth considered, and the coefficient on $\log(\rho_j)$ is statistically significant at the 5% level for bandwidths up to 75 km. As expected for a cross-sectional specification with rich spatial controls, the standard errors widen somewhat at the longest bandwidths (100 and 150 km), where the spatial HAC kernel absorbs progressively more of the variation already captured by the included geographic covariates; the p -value at 100 km is 0.102 and at 150 km is 0.145. The result thus holds under the bandwidths most commonly applied in the spatial-econometrics literature for cross-sectional designs at this geographic scale.

Standard error	Estimate	SE	p
Heteroskedasticity-robust (paper)	-0.0111952548949787	(0.00554317773276672)	0.0446969310445375
Conley 25 km	-0.0111952548949787	(0.00486874644301278)	0.0224693058430259
Conley 50 km	-0.0111952548949787	(0.00557498706242249)	0.0459175666096826
Conley 75 km	-0.0111952548949787	(0.00539253446894661)	0.0391111847436515
Conley 100 km	-0.0111952548949787	(0.00682291828207967)	0.102336570113632
Conley 150 km	-0.0111952548949787	(0.00765342600008831)	0.145030590653059

Table A.60: Conley SE Bandwidth Sensitivity: Domain Tax Regression

Note: The coefficient on $\log(\rho_j)$ from Table 4 Model 4 (fully controlled) is significant at the 5% level for Conley bandwidths up to 75 km, marginal at 100 km ($p = 0.091$), and not significant at 150 km ($p = 0.130$). The first row reports the heteroskedasticity-robust standard error used in the main table.

E.3. Additional Analyses

E.3.1. Average Distance to Home Capital

Outcome:	τ_j									
Models:	(1)		(2)		(3)		(4)		(5: SPL)	
<i>Variables</i>										
ln(Avg. Dist to HC_j)	−0.0234	(0.0072)***	−0.0226	(0.0072)***	−0.0114	(0.0068)*	−0.0114	(0.0069)	−0.0081	(0.0061)
ln(Rice)			−0.0065	(0.0151)	0.0059	(0.0154)	0.0108	(0.0162)	−0.0002	(0.0135)
ln(Area)			−0.0194	(0.0111)*	−0.0291	(0.0140)**	−0.0320	(0.0149)**	−0.0213	(0.0130)
ln(Pop.)			0.0421	(0.0167)**	0.0269	(0.0144)*	0.0233	(0.0151)	0.0223	(0.0147)
New FL			0.0001	(0.0003)	0.0004	(0.0004)	0.0003	(0.0004)	0.0003	(0.0002)
Shog. Ally			−0.0341	(0.0175)*	0.0049	(0.0160)	0.0003	(0.0164)	0.0013	(0.0144)
Goshi			0.0209	(0.0258)	0.0196	(0.0263)	0.0276	(0.0274)	0.0244	(0.0216)
lat					−0.4960	(0.2905)*	−0.4513	(0.3081)	−0.3219	(0.2861)
lon					−0.1680	(0.0747)**	−0.1581	(0.0797)**	−0.1174	(0.0745)
ln(Elev.)					−0.0111	(0.0158)	−0.0131	(0.0163)	−0.0090	(0.0135)
ln(Rugg.)					0.0391	(0.0159)**	0.0365	(0.0157)**	0.0227	(0.0152)
ln(Dist. Edo/Kyoto)					−0.0333	(0.0161)**	−0.0418	(0.0172)**	−0.0315	(0.0167)*
Rainy					−1.6381	(0.8959)*	−1.5879	(0.9058)*	−1.0377	(0.7409)
Snowy					0.6554	(1.8687)	0.6559	(1.8770)	−0.2013	(1.4338)
Lat × Lon					0.0039	(0.0021)*	0.0036	(0.0023)	0.0026	(0.0021)
Kin Nbs.							0.0407	(0.0295)	0.0201	(0.0258)
Nb. FL							0.0002	(0.0001)	0.0001	(0.0001)
ln(Nb. Rice)							−0.0256	(0.0190)	−0.0228	(0.0173)
ln(Nb. Area)							0.0207	(0.0176)	0.0142	(0.0158)
ln(Nb. Pop.)							−0.0040	(0.0207)	−0.0018	(0.0198)
Const.	0.4679	(0.0248)***	0.2420	(0.1226)**	22.0932	(10.1723)**	21.0544	(10.8064)*	15.4092	(10.1175)
<i>Fit Statistics</i>										
Num.Obs.	232		230		230		230		230	
R2	0.046		0.135		0.367		0.385		0.435 ^[a]	

Table A.61: Average Distance to Home Capital Weakly Predicts Tax Rates

Note: OLS estimates (maximum likelihood in Model 5) relating each domain's average absolute distance to its home capital (log) to its reported tax rate. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Model 5. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1. ^[a] Nagelkerke pseudo-R2

E.3.2. Controlling for the Shortest Distance to the Border

Outcome:	τ_j									
Models:	(1)		(2)		(3)		(4)		(5: SPL)	
<i>Variables</i>										
$\ln(\rho_j)$	-0.0240	(0.0053)***	-0.0200	(0.0058)***	-0.0111	(0.0055)**	-0.0113	(0.0055)**	-0.0095	(0.0051)*
$\ln(\text{Avg. Dist. Border}_j)$	0.0014	(0.0067)	-0.0106	(0.0084)	-0.0056	(0.0071)	-0.0038	(0.0067)	-0.0024	(0.0075)
$\ln(\text{Rice})$			-0.0004	(0.0152)	0.0083	(0.0155)	0.0121	(0.0162)	0.0007	(0.0135)
$\ln(\text{Area})$			-0.0198	(0.0121)	-0.0286	(0.0148)*	-0.0318	(0.0156)**	-0.0214	(0.0136)
$\ln(\text{Pop.})$			0.0379	(0.0165)**	0.0254	(0.0145)*	0.0226	(0.0151)	0.0218	(0.0147)
New FL			0.0001	(0.0003)	0.0005	(0.0004)	0.0003	(0.0004)	0.0004	(0.0002)
Shog. Ally			-0.0319	(0.0178)*	0.0055	(0.0162)	0.0010	(0.0165)	0.0024	(0.0143)
Goshi			0.0261	(0.0262)	0.0210	(0.0263)	0.0282	(0.0272)	0.0247	(0.0215)
lat					-0.5551	(0.2901)*	-0.4979	(0.3056)	-0.3688	(0.2893)
lon					-0.1820	(0.0746)**	-0.1693	(0.0791)**	-0.1287	(0.0751)*
$\ln(\text{Elev.})$					-0.0130	(0.0160)	-0.0143	(0.0166)	-0.0105	(0.0135)
$\ln(\text{Rugg.})$					0.0408	(0.0159)**	0.0380	(0.0157)**	0.0243	(0.0153)
$\ln(\text{Dist. Edo/Kyoto})$					-0.0370	(0.0159)**	-0.0444	(0.0172)**	-0.0337	(0.0167)**
Rainy					-1.6537	(0.8933)*	-1.5857	(0.9065)*	-1.0138	(0.7370)
Snowy					0.7125	(1.8486)	0.6839	(1.8577)	-0.2447	(1.4195)
Lat $\times$ Lon					0.0043	(0.0021)**	0.0039	(0.0022)*	0.0030	(0.0021)
Kin Nbs.							0.0396	(0.0296)	0.0189	(0.0257)
Nb. FL							0.0002	(0.0001)	0.0001	(0.0001)
$\ln(\text{Nb. Rice})$							-0.0238	(0.0187)	-0.0218	(0.0172)
$\ln(\text{Nb. Area})$							0.0180	(0.0176)	0.0116	(0.0159)
$\ln(\text{Nb. Pop.})$							-0.0045	(0.0206)	-0.0014	(0.0196)
Const.	0.4130	(0.0138)***	0.1620	(0.1210)	24.0420	(10.1575)**	22.5897	(10.7386)**	16.9504	(10.2058)*
<i>Fit Statistics</i>										
Num.Obs.	232		230		230		230		230	
R2	0.074		0.147		0.372		0.389		0.439 ^[a]	

Table A.62: Geographic Exposure and Reported Tax Rates
(Controlling for Border Distance)

Note: OLS estimates (maximum likelihood in Model 5) relating each domain's average distance ratio to its reported tax rate while controlling for average distance to the nearest border. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Model 5. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1. ^[a] Nagelkerke pseudo-R2

E.3.3. Outside-Option Quality Using Population Density

Outcome:	τ_j									
Models:	(1)		(2)		(3)		(4)		(5)	
<i>Variables</i>										
$\ln(\rho_j) \times$ Low	-0.0299	(0.0097)***	-0.0262	(0.0092)***	-0.0197	(0.0073)***	-0.0265	(0.0094)***	-0.0204	(0.0071)***
$\ln(\rho_j) \times$ High	-0.0216	(0.0056)***	-0.0176	(0.0062)***	-0.0074	(0.0059)	-0.0176	(0.0063)***	-0.0067	(0.0062)
ln(Rice)			-0.0016	(0.0156)	0.0071	(0.0152)	-0.0019	(0.0164)	0.0114	(0.0160)
ln(Area)			-0.0241	(0.0114)**	-0.0300	(0.0142)**	-0.0229	(0.0119)*	-0.0334	(0.0148)**
ln(Pop.)			0.0383	(0.0168)**	0.0254	(0.0143)*	0.0355	(0.0169)**	0.0228	(0.0150)
New FL			0.0002	(0.0003)	0.0005	(0.0004)	0.0002	(0.0003)	0.0004	(0.0004)
Shog. Ally			-0.0313	(0.0179)*	0.0071	(0.0161)	-0.0298	(0.0181)	0.0025	(0.0164)
Goshi			0.0210	(0.0257)	0.0181	(0.0261)	0.0219	(0.0261)	0.0267	(0.0269)
lat					-0.5827	(0.2993)*			-0.5190	(0.3153)
lon					-0.1909	(0.0774)**			-0.1758	(0.0819)**
ln(Elev.)					-0.0125	(0.0160)			-0.0147	(0.0166)
ln(Rugg.)					0.0406	(0.0159)**			0.0379	(0.0158)**
ln(Dist. Edo/Ky- oto)					-0.0379	(0.0160)**			-0.0450	(0.0174)**
Rainy					-1.6127	(0.9138)*			-1.6134	(0.9218)*
Snowy					0.5457	(1.8746)			0.6470	(1.8827)
Lat $\times$ Lon					0.0046	(0.0022)**			0.0041	(0.0023)*
Kin Nbs.							0.0421	(0.0304)	0.0422	(0.0296)
Nb. FL							0.0000	(0.0001)	0.0002	(0.0001)
ln(Nb. Rice)									-0.0210	(0.0187)
ln(Nb. Area)							-0.0015	(0.0083)	0.0245	(0.0181)
ln(Nb. Pop.)									-0.0135	(0.0210)
Const.	0.4155	(0.0091)***	0.1846	(0.1241)	25.1677	(10.5108)**	0.2119	(0.1509)	23.4706	(11.0924)**
<i>Fit Statistics</i>										
Num.Obs.	232		230		230		230		230	
R2	0.077		0.144		0.376		0.152		0.395	

Table A.63: Geographic Exposure and Reported Tax Rates by Neighbor Population Density

Note: The table reports group-specific exposure slopes. HC1 standard errors in parentheses.